

XDPE1A2G5C Digital Multi-phase Controller

16-phase Dual Loop Voltage Regulator

Features

- Digital controller assisted with high performance analog front ends with a fully programmable ARM® Cortex™ –M0 Processor
- PMBus compliant with rev 1.4 with AVSBus serial interface
 - Output voltage control
 - Query voltage, current, temperature faults
 - Fault Response
 - Maximum supported bus speed 1 MHz
 - Supports 1.8/3.3 V operation
- AVSBus
 - Supports 1.2V/1.8V operation
 - Supports clock frequencies up to 50MHz
- Output voltage regulation range
 - 0.05 V to 3.1 V (1 mV/step)
- Digitally controlled phasing for full flexibility in both loop configuration and phase firing order
- Configurable autonomous phase add/drop
- Automatic phase detection at start-up
- Supports start-up into pre-bias voltage
- Integrated power stage current sense
- Differential output voltage sense
- Fast current balancing with current-mode control
- Digitally programmable PID (Proportional, Integral, Differential) loop compensation
- Digitally programmable load line slope and offset
- Digital temperature compensation
- Extensive fault detection and protection capability
 - IUVP, OUV & OOV (fixed and tracking)
 - OCP instantaneous & averaged (total current)
 - Multiple OTP thresholds
 - Peak phase current pulse limiting
 - Negative current limit protection
- Internal non-volatile memory (NVM) to store multiple custom configurations
- RoHS compliant and Halogen free 56-lead QFN plastic package

Potential applications

Potential applications

- CORE power regulator for AI based Microprocessors
 - Telecom applications
 - ASIC POWER

Product validation

Qualified for industrial applications according to the relevant tests of JEDEC47/20/22

Description

The XDPE1A2G5C digital dual loop 16 phase controller provides power for AI servers, workstation, and telecom applications that use AVS BUS. Voltage on either the first loop or the second loop can be provided by a multi-phase buck converter with up to 16 synchronous-rectified channels. Typical configurations of 16+0, 15+1, and up to 8+8 are supported.

Command and monitoring functions are controlled through the PMBus and AVS interfaces which supports dynamic voltage identification with 1 mV/step, output range up to 3.1 V, offset and trim resolution of 625 μ V and accuracy better than 0.5%.

The XDPE1A2G5C controller utilizes digital technology to implement all control functions, providing the ultimate system solution in terms of flexibility and stability. Advanced control loop features, such as current mode control, variable frequency operation, Active Transient Response (ATR/FATR), fast DVID response, automatic phase shedding and Pulse Frequency Modulation (PFM) enable optimal response to a highly dynamic load with fast di/dt load transients across a wide range of load current.

The XDPE1A2G5C controller supports multi-vendor pin compatible integrated power stages with trivalent PWM interfaces, fast response, integrated current sense, integrated temperature sense, and advanced fault detection capability, interface and signaling.

In addition to supporting power stage fault protection features, the XDPE1A2G5C controller also includes a set of sophisticated over-voltage, under-voltage, over-temperature, over-current, and phase fault protections. These attributes provide a complete and advanced protection feature set for microprocessor and power systems.

Infineon strongly recommends pairing Infineon's power stages with our Digital XDP™ family of controllers to ensure correct interoperability. Interoperability when pairing with other vendor power stages/ discrete power components cannot be guaranteed by Infineon and requires thorough evaluation and characterization by the power stage/ discrete power component vendor.

Table 1 Part number and package summary

Part Number	Package
XDPE1A2G5C	56-lead 7 mm x 7 mm QFN PG-VQFN-56

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1 Ordering information

Table 2 Ordering information

Base Part Number	Package Type	Standard Pack Form and Quantity		Orderable Part Number
XDPE1A2G5C	QFN 7 mm x 7 mm	Tape and Reel	3000	XDPE1A2G5C0000XUMA1 Note 1
				XDPE1A2G5CxyzXUMA1 Note 2

Note:

1. Standard part number with default configuration
2. Customer Specific Configuration File, where x = Firmware ID and yzz = Custom Configuration File ID (Codes assigned by Product Marketing).

Sample Programming

The customer can program the parts to their specific system requirements using software/hardware available from Infineon or through other controller programming facilities (contact Infineon for recommendations). Infineon Field Application Engineers are available to assist with system and configuration file optimization and programming of the controllers. Alternatively, samples can be ordered with a customer specific custom configuration pre-programmed at the factory, but lead times for these types of samples are significantly longer than for non-configured samples. The generic part numbering format is shown below:

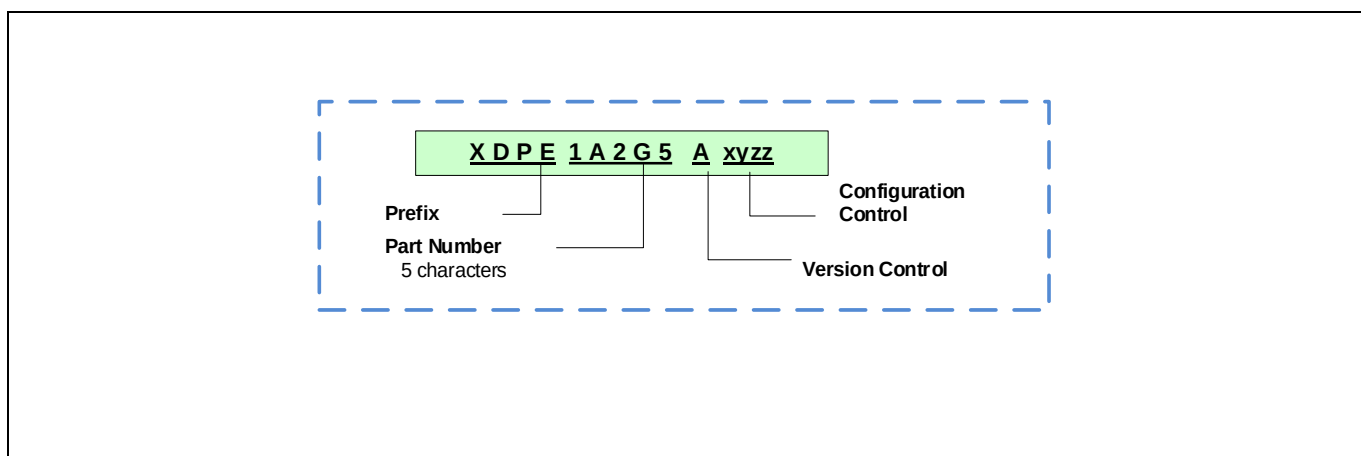
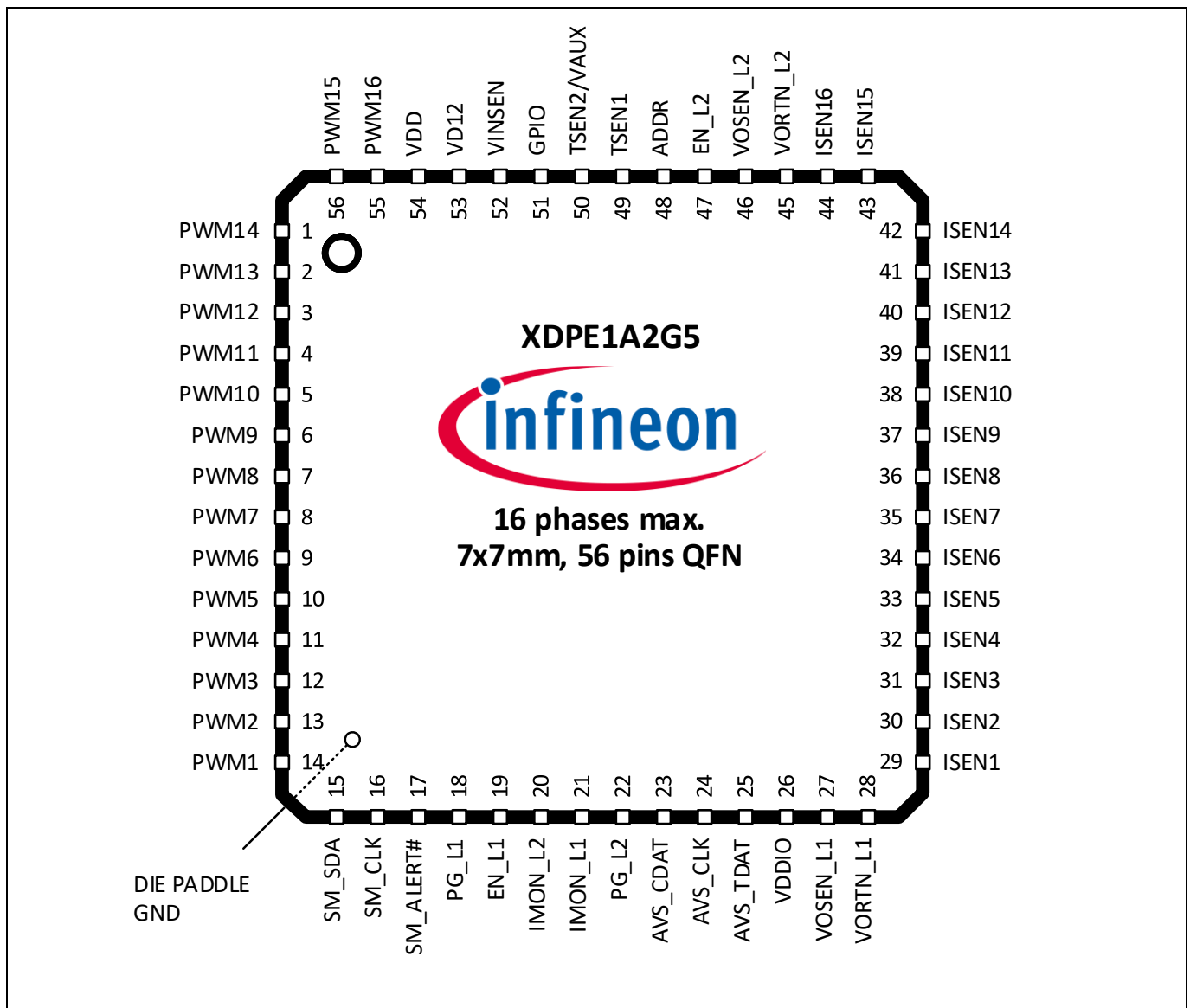


Figure 1 Part number decoding

**Figure 2 Package top view**

2 Functional block diagram

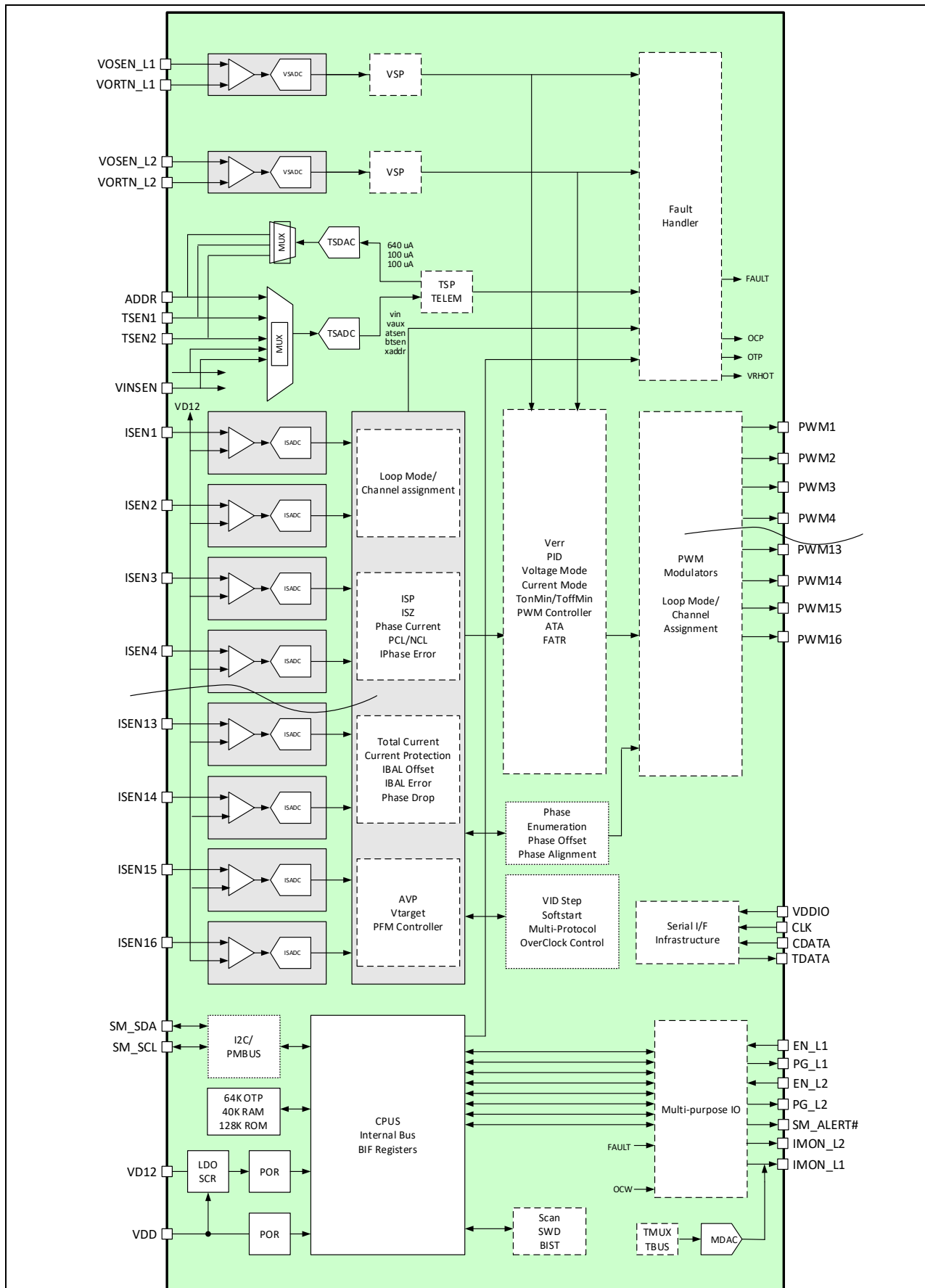


Figure 3 Block diagram

3 Typical application drawings

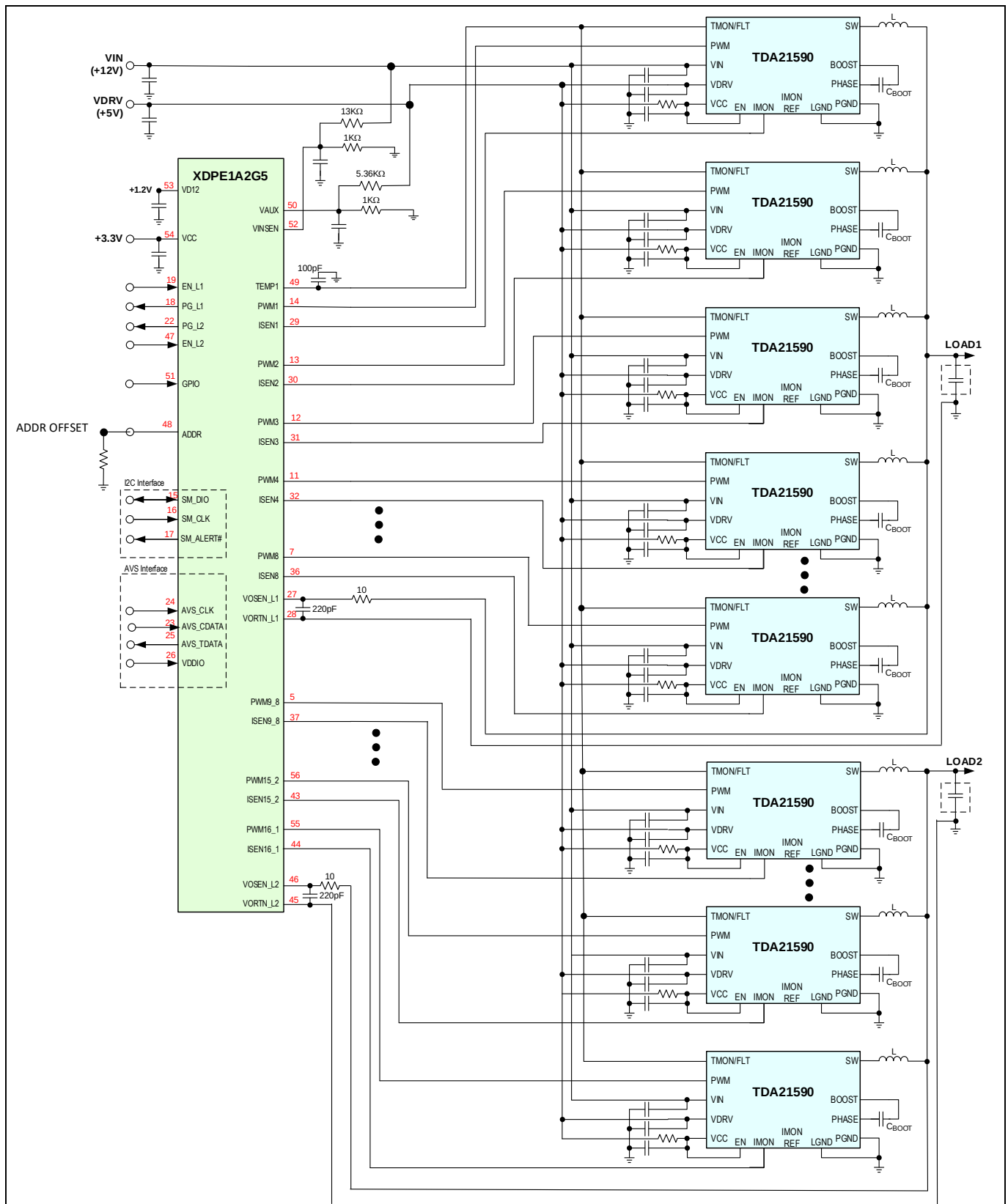


Figure 4 Dual loop VR using XDPE1A2G5C Controller and TDA21590 Power Stage in 8+8 phase

Pin descriptions

4 Pin descriptions

Table 3 Pin descriptions

Key: I = Input / O = Output / S = Supply

Pin#	Pin name	I/O	Type	Pin description
1	PWM14	O	CMOS	PWM output. Refer to PWM1 description.
2	PWM13	O	CMOS	PWM output. Refer to PWM1 description.
3	PWM12	O	CMOS	PWM output. Refer to PWM1 description.
4	PWM11	O	CMOS	PWM output. Refer to PWM1 description.
5	PWM10	O	CMOS	PWM output. Refer to PWM1 description.
6	PWM9	O	CMOS	PWM output. Refer to PWM1 description.
7	PWM8	O	CMOS	PWM output. Refer to PWM1 description.
8	PWM7	O	CMOS	PWM output. Refer to PWM1 description.
9	PWM6	O	CMOS	PWM output. Refer to PWM1 description.
10	PWM5	O	CMOS	PWM output. Refer to PWM1 description.
11	PWM4	O	CMOS	PWM output. Refer to PWM1 description.
12	PWM3	O	CMOS	PWM output. Refer to PWM1 description.
13	PWM2	O	CMOS	PWM output. Refer to PWM1 description.
14	PWM1	O	CMOS	Pulse Width Modulation (PWM) output. This signal is used to drive the PWM input of the FET driver IC. Any PWM pin can be assigned any phase angle on either loop. See Note 3 below
15	SM_SDA	I/O	Open Drain	SMBus/I2C bi-directional serial data interface.
16	SM_SCL	I	Open Drain	SMBus/I2C serial clock interface.
17	SM_ALERT#	O	Open Drain/ CMOS	Multipurpose pin configurable as SM_ALERT# signal for a PMBus device, Low-power Mode (LPM#) enable, or FAULT#
18	PG_L1	O	Open Drain	Voltage regulator “Ready” output signal for Loop 1. The PG_L1 indicator will be asserted when the controller is ready to accept AVS commands. This open-drain output requires an external pull-up resistor (1 K Ω recommended). PG_L1 will be pulled low when the enable is de-asserted or a shutdown fault occurs.
19	EN_L1	I	CMOS	Active high Output Enable input. Asserting the EN_L1 pin will activate loop 1, pending status of the internal power-on-reset circuit and any existing fault states. De-asserting then asserting EN_L1 after a latched-fault based shutdown will cause the controller to enter the soft-start state. Faults will be cleared when EN_L1 is reasserted. This pin can also be configured to be a common enable for both Loop 1 and Loop 2.
20	IMON_L2	O	Analog	IMON output that will provide a current that is analogous to the total output current of Loop 2
21	IMON_L1	O	Analog	IMON output that will provide a current that is analogous to the total output current of Loop 1

Pin descriptions

Pin#	Pin name	I/O	Type	Pin description
22	PG_L2	O	Open Drain	Voltage regulator “Ready” output signal for Loop 2. See PG_L1 description for more detail.
23	AVS_CDAT	I	CMOS	Bus line that is driven by the AVSBus controller (ASIC, FPGA, or Processor) and carries data to the target (VR Controller). Float or connect to ground if not used. Note 3
24	AVS_CLK	I	CMOS	Bus clock that is driven by the controller and clocks data for both the AVS_CDAT and AVS_TDAT lines. Note 3
25	AVS_TDAT	O	CMOS	Bus line that is driven by an AVSBus Target (VR Controller) and carries data to the AVSBus controller (ASIC, FPGA, or Processor). Note 3
26	VDDIO	S	External Supply	Common voltage for the controller and target I/O circuitry.
27	VOSEN_L1	I	Analog	Loop 1 voltage sense inputs. VOSEN_L1 and VORTN_L1 are inputs to the precision differential remote sense amplifier and should be connected to the sense pins of the remote load. The sense amplifier converts the differential voltage of the remote output sense to a single-ended voltage referenced to chip ground. It is recommended to add a 10 Ohm, 220 pF filter for VOSEN_L1
28	VORTN_L1	I	Analog	Loop 1 voltage sense input reference.
29	ISEN1	I	Analog	Current sense input used with PWM1 output. See Note 5
30	ISEN2	I	Analog	Current sense input used with PWM2 output.
31	ISEN3	I	Analog	Current sense input used with PWM3 output.
32	ISEN4	I	Analog	Current sense input used with PWM4 output.
33	ISEN5	I	Analog	Current sense input used with PWM5 output.
34	ISEN6	I	Analog	Current sense input used with PWM6 output.
35	ISEN7	I	Analog	Current sense input used with PWM7 output.
36	ISEN8	I	Analog	Current sense input used with PWM8 output.
37	ISEN9	I	Analog	Current sense input used with PWM9 output.
38	ISEN10	I	Analog	Current sense input used with PWM10 output.
39	ISEN11	I	Analog	Current sense input used with PWM11 output.
40	ISEN12	I	Analog	Current sense input used with PWM12 output.
41	ISEN13	I	Analog	Current sense input used with PWM13 output.
42	ISEN14	I	Analog	Current sense input used with PWM14 output.
43	ISEN15	I	Analog	Current sense input used with PWM15 output.
44	ISEN16	I	Analog	Current sense input used with PWM16 output.
45	VORTN_L2	I	Analog	Loop B voltage sense input reference.
46	VOSEN_L2	I	Analog	Loop 2 voltage sense inputs. Refer to VOSEN_L1 and VORTN_L1 description. It is recommended to tie these pins to GND in single rail applications.
47	EN_L2	I	CMOS	Active high Output Enable input. Asserting the EN_L2 pin will activate loop 2, pending status of the internal power-on-reset circuit and any existing fault states. De-asserting then asserting EN_L2 after a latched-fault based shutdown will cause the controller to enter the soft-start state. Faults will be cleared

Pin descriptions

Pin#	Pin name	I/O	Type	Pin description
				when EN_L2 is reasserted.
48	ADDR	I	Analog	Used to set the PMBus address of the controller along with the address in NVM by tying an external resistor between this pin and GND.
49	TSEN1	I	Analog	Loop 1 or Loop 2 external temperature sense input. Connects to the integrated power stage temperature sensor output.
50	TSEN2	I	Analog	Loop 1 or Loop 2 external temperature sense input. Connects to the integrated power stage temperature sensor output.
	VAUX	I	Analog	Auxiliary voltage monitor to measure power stage supply voltage through an external resistor divider.
51	GPIO	I/O	CMOS	General Purpose IO
52	VINSEN	I	Analog	Monitor VIN with an external resistor divider
53	VD12	S	Internal Supply	Internally generated 1.2 V voltage reference used to power digital core logic and to provide a reference voltage for external Power Stages. The pin is provided for attaching external decoupling capacitors. Decouple using high quality 1.0 μ F + 0.1 μ F ceramic capacitors. Do not apply voltage to or ground this pin.
54	VDD	S	External Supply	3.3 V power supply input to the controller. This pin should be connected to the system +3.3 V supply and decoupled using high quality 1.0 μ F + 0.1 μ F ceramic capacitors.
55	PWM16	O	CMOS	PWM output. Refer to PWM1 description.
56	PWM15	O	CMOS	PWM output. Refer to PWM1 description.
Die paddle		S	Ground	Exposed Ground pad beneath the device must be soldered to the PCB ground for proper operation.

Note:

3. If the AVS interface is not enabled, the AVS interface pins can be left floating or grounded.
4. Leave unused PMW output pins floating
5. Leave unused ISEN input pins floating

5 Absolute maximum ratings

Table 4 Absolute maximum ratings

Symbol	Description	Min to Max	Conditions
VDD	Supply voltage	GND-0.3 V to 4.0 V	See Note 6 below
VOSEN_L1/L2		GND-0.3 V to 3.7 V	Do not exceed VDD + 0.2 V
VORTN_L1/L2		GND-0.3 V to 0.5 V	-
ISENx		GND-0.3 V to 3.7 V	See Note 6 below
VDDIO, AVS_CLK, AVD_CDAT, AVS_TDAT		GND-0.3 V to 2.0 V	-
VD12		-15 mA to +1 mA	Do not drive this pin
	All other pins	GND-0.3 V to 4.0 V*	See Note 6 below, and Note 7
T _J	Junction temperature	-40~150 °C	-
T _{STG}	Storage temperature	-65~150 °C	-

Note:

6. The absolute maximum VDD supply voltage is 4.0 V with the conditions that the junction temperature range is maintained between $-40\text{ °C} \leq T_J \leq +125\text{ °C}$ and the product is not operated at the absolute maximum VDD supply voltage for more than 24 hours cumulatively over the lifetime of the product.
7. *The lesser of 4 V or VDD+0.2 V; outputs in Hi-Z state

Attention: Stresses beyond those listed under 'Absolute maximum ratings may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications are not implied.

6 Electrical specifications

Table 5 Recommended operating conditions for reliable operation with margin

Operating Junction Temperature Range T_j	-40 °C to 125 °C
Supply Voltage Range VDD	+2.97 V to +3.63 V

Table 6 Temperature information

Symbol	Description	7 mm x 7 mm QFN
$\theta_{JA(0)}$	Junction-to-ambient thermal resistance at 0 lfm	27.1 °C/W
$\theta_{JA(200)}$	Junction-to-ambient thermal resistance at 200 lfm	23.1 °C/W
$\theta_{JA(500)}$	Junction-to-ambient thermal resistance at 500 lfm	21.7 °C/W
θ_{JC}	Junction-to-case thermal resistance	2.8 °C/W

The electrical characteristics table lists the spread of values guaranteed within the recommended operating conditions. Typical values represent the median values, which are related to 25 °C.

Table 7 Electrical characteristics

Parameter	Test conditions	Min	Typ	Max	Units
VDD supply					
Nominal supply current	Phase count maximum	–	80	–	mA
	VREN = low	–	55	–	mA
PWM[x:1]					
Output low voltage (V_{OL})	$I_{LOAD} = +5$ mA	–	–	0.4	V
Output high voltage (V_{OH})	$I_{LOAD} = -5$ mA	2.6	–	–	V
Output current in Hi-Z state	pin driven to 2.3 V	-1	–	1	μA
PWM low threshold	Configured as inputs prior to power-up when used to detect shorted or open PWM	-	0.8	-	V
SM_SDA, SM_SCL					
Input low voltage (V_{IL})	Configured as 3.3 V buffer	–	–	1.0	V
Input high voltage (V_{IH})	Configured as 3.3 V buffer	2.1	–	–	V
Input low voltage (V_{IL})	Configured as 1.8 V buffer	–	–	0.8	V
Input high voltage (V_{IH})	Configured as 1.8 V buffer	1.35	–	–	V
Input current	pin driven to 0.0 V	-1	–	1	μA
	pin driven to 3.4 V	-1	–	1	μA
Output low voltage (V_{OL})	open drain output, $I_{LOAD} = +5$ mA	–	–	0.4	V
Pin capacitance	See Note 9	–	–	4	pF
AVS_CDAT, AVS_CLK					
Input low voltage (V_{IL})	CPU I/O Voltage (VDDIO) = 1.2 V	–	–	35% VDDIO	V

Electrical specifications

Parameter	Test conditions	Min	Typ	Max	Units
Input high voltage (VIH)	CPU I/O Voltage (VDDIO) = 1.2 V	65% VDDIO	–	–	V
Hysteresis voltage (VHYS)	CPU I/O Voltage (VDDIO) = 1.2 V See Note 9	0.1	–	–	V
Input leakage current (IL)	pin driven to VDDIO	-10	–	10	μA
Pin capacitance	See Note 9	–	–	2	pF

AVS_TDAT

Output low voltage (VOL)	CMOS driven, ILOAD = +4 mA	–	–	0.22	V
Output high voltage (VOH)	CMOS driven, ILOAD = -4 mA	VDDIO - 0.22	–	–	V

EN_L1, EN_L2

Input low voltage (VIL)		–	–	0.6	V
Input high voltage (VIH)	See Note 8	0.7	–	–	V
Input current	pin driven to 0.0 V	-1	–	1	μA
	pin driven to 3.4 V	-1	–	1	μA

PG_L1, PG_L2

Output low voltage (VOL)	open drain output, ILOAD = +5 mA	–	–	0.4	V
Output current in Hi-Z state	pin driven to +3.4 V	-1	–	1	μA

SM_ALERT#

Output low voltage (VOL)	CMOS driven, ILOAD = +5 mA	–	–	0.4	V
Output high voltage (VOH)	CMOS driven, ILOAD = -5 mA	2.6	–	–	V
Output low voltage (VOL)	open drain output, ILOAD = +5 mA	–	–	0.4	V
Output current in Hi-Z state	pin driven to +3.4 V	-1	–	1	μA

IMON_L1/L2

Output voltage range	R _{IMON} < 1.6 kΩ, See note 9		0 to 1.05		V
Output current range			0 to 640		μA
Output current resolution			0.625		μA
IMON IDAC accuracy	Tested at 100uA	-3		3	%
Update rate	See note 9		25		MHz
Code zero offset			200		nA

VOSEN_L1/L2 (Voltage Sense Amplifier)

Input voltage range		–	0 to 3.5	–	V
Input current	VOSEN_L1/L2 = 2.5 V	70	–	300	μA

VORTN_L1/L2

Input voltage range		–	-0.3 to 0.3	–	V
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Electrical specifications

Input current	VORTN_L1/L2 = 0 V	-120	-	-10	μA
ISEN[x:1] (Integrated current sense)					
Input voltage range	Internal 1 kΩ resistor enabled See note 14	-	-200 to +550	-	mV
	Internal 500 Ω resistor enabled See note 14		-100 to +450		mV
	External 1 kΩ resistor with Vmode enabled, See note 14	-	-200 to +650	-	mV
ISEN Input current	Vmode enabled $V_{ISEN} = 1.85$ V, See Note 10	-	<1	-	μA
ISEN Input resistance	Internal 1 kΩ resistor enabled $V_{ISEN} = 1.85$ V, See Note 10	-	1	-	kΩ
	Internal 500 Ω resistor enabled $V_{ISEN} = 1.85$ V, See Note 10	-	500	-	Ω
	Vmode $V_{ISEN} = 1.85$ V, See Note 10		>1	-	MΩ
ISEN accuracy	Internal 1 kΩ resistor enabled Calibrated; 60A load $-5^{\circ}\text{C} \leq T_A \leq 85^{\circ}\text{C}$ See notes 10 , 11 , and 12	-	±1.9	-	%
	Internal 1 kΩ resistor enabled Calibrated; 60A load $-40^{\circ}\text{C} \leq T_A \leq 125^{\circ}\text{C}$ See notes 10 , 11 , and 12	-	±2.1	-	%
	Internal 500 Ω resistor enabled Calibrated; 144A load $-5^{\circ}\text{C} \leq T_A \leq 85^{\circ}\text{C}$ See notes 10 , 11 , and 12	-	±1.2	-	%
	Internal 500 Ω resistor enabled Calibrated; 144A load $-40^{\circ}\text{C} \leq T_A \leq 125^{\circ}\text{C}$ See notes 10 , 11 , and 12	-	±1.6	-	%
	Internal 333 Ω resistor enabled Calibrated; 180 A load $-5^{\circ}\text{C} \leq T_A \leq 85^{\circ}\text{C}$ See Note 10 , 11 , and 12	-	±1.9	-	%
	Internal 333 Ω resistor enabled Calibrated; 180 A load $-40^{\circ}\text{C} \leq T_A \leq 125^{\circ}\text{C}$ See Note 10 , 11 , and 12	-	±2.3	-	%
	External 1 kΩ resistor with Vmode Calibrated; 60A load $-5^{\circ}\text{C} \leq T_A \leq 85^{\circ}\text{C}$ See notes 10 , 11 , and 12	-	±2.0	-	%

Electrical specifications

	External 1 kΩ resistor with Vmode Calibrated; 60A load -40 °C ≤ TA ≤ 125 °C See notes 10, 11, and 12	-	±2.2	-	%
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VAUX, VINSEN

Input resistance	See note 9	-	>1	-	MΩ
Input voltage range		-	0 to 1.2	-	V

ADDR

Measured voltage	See note 9		0 to 1.2		V
Current Measurement			0 to 640		μA

TSEN1, TSEN2 (for power stages with Integrated temperature sense)

Input Voltage Range		-	0 to 2.2	-	V
Valid Temperature Range		-	-40 to 200	-	°C
Sensor Reference	Sensor Voltage at 25°C	-	0.8	-	V
Sensor Scale		-	8	-	mV/°C
Bias Current	TSEN1, TSEN2 = 1.2 V	-	-12	-	μA
Driver Fault Threshold		2.2	2.4	2.6	V

Switching Frequency

Frequency range	See note 9	250	-	3000	kHz
Frequency accuracy	TA = 25°C, VDD = 3.3V	-2	-	2	%
	-5 ≤ TA ≤ 85°C, 2.97 V ≤ VDD ≤ 3.63 V; See note 9	-4	-	4	%
	-40 ≤ TA ≤ 120°C, 2.97 V ≤ VDD ≤ 3.63 V; See note 9	-7	-	7	%

System Set-point Accuracy

0.25 V ≤ VID ≤ 0.5 V	See notes 9 and 13	-	±8	-	mV
0.5 V < VID ≤ 0.75 V	See notes 9 and 13	-	±5	-	mV
0.75 V < VID ≤ 3.1 V	-5 °C ≤ TA ≤ 85 °C; 3.0 V ≤ VDD ≤ 3.6 V, VDD > VID + 200 mV, Typ = 3 σ; see notes 9 and 13	-	±0.5	-	%
0.75 V < VID ≤ 3.1 V	-40 °C ≤ TA ≤ 125 °C; 3.0 V ≤ VDD ≤ 3.6 V, VDD > VID + 200 mV, Typ = 3 σ; see notes 9 and 13	-	±0.8	-	%

Fault Monitors

OUVP threshold accuracy	See note 9	-	±20	-	mV
Fixed OOV threshold accuracy		-	±20	-	mV

Electrical specifications

Tracking OOV threshold accuracy		–	±20	–	mV
OT threshold accuracy	TSENx = 100 °C (1.4 V) See note 9	–	±3	–	°C
VDD UVLO threshold	VDD Rising	–	2.78	2.95	V
	VDD Falling	2.5	2.7	–	V
VDD UVLO hysteresis		–	80	–	mV

Startup Timing

3.3V stable to controller ready	See note 9 and 16	–	10	–	ms
EN_Lx high to loopx start	See note 9	–		300	µs
xVREN low to Loopx shutdown	See note 9	–		300	µs
Common EN high to loopx start	See notes 9 and 15			600	µs

Note:

8. These pins should not be left open. In most applications they should be driven by an open drain driver with an external pull-up resistor.
9. Not subject to production test – verified by design and/or characterization.
10. Current sense reference voltage is 1.2V through the internal supply
11. Runtime offset calibration enabled, over $-40 \leq TA \leq 85$ °C, $2.97 \text{ V} \leq VDD \leq 3.63 \text{ V}$; typical=3 sigma
12. External 1 kΩ tolerance is not included in the accuracy spec; internal 1 kΩ resistor error is included
13. Production test limits verified at 0.85 V, 1.1 V, 1.85 V. Full range not subject to production test.
14. For Isense input voltage ranges above +450 mV, the current sense DAC offset must be enabled
15. VR start is sequential with a common EN pin or when both EN_Lx pins are asserted simultaneously. Loop1 will start 300us after EN and loop2 will start 300us after loop1
16. For standard production parts with default firmware patch

7 Theory of operation

7.1 Startup configuration

The XDPE1A2G5 is a digital power management controller with a firmware-based state-machine (See figure below). Operation is controlled by application specific firmware and configuration settings stored in ROM and NVM, then loaded into RAM and control registers during initialization. Configuration and firmware updates are supported, but the number of reprogramming cycles is limited. See Section 8.17 for more details on Configuration Management.

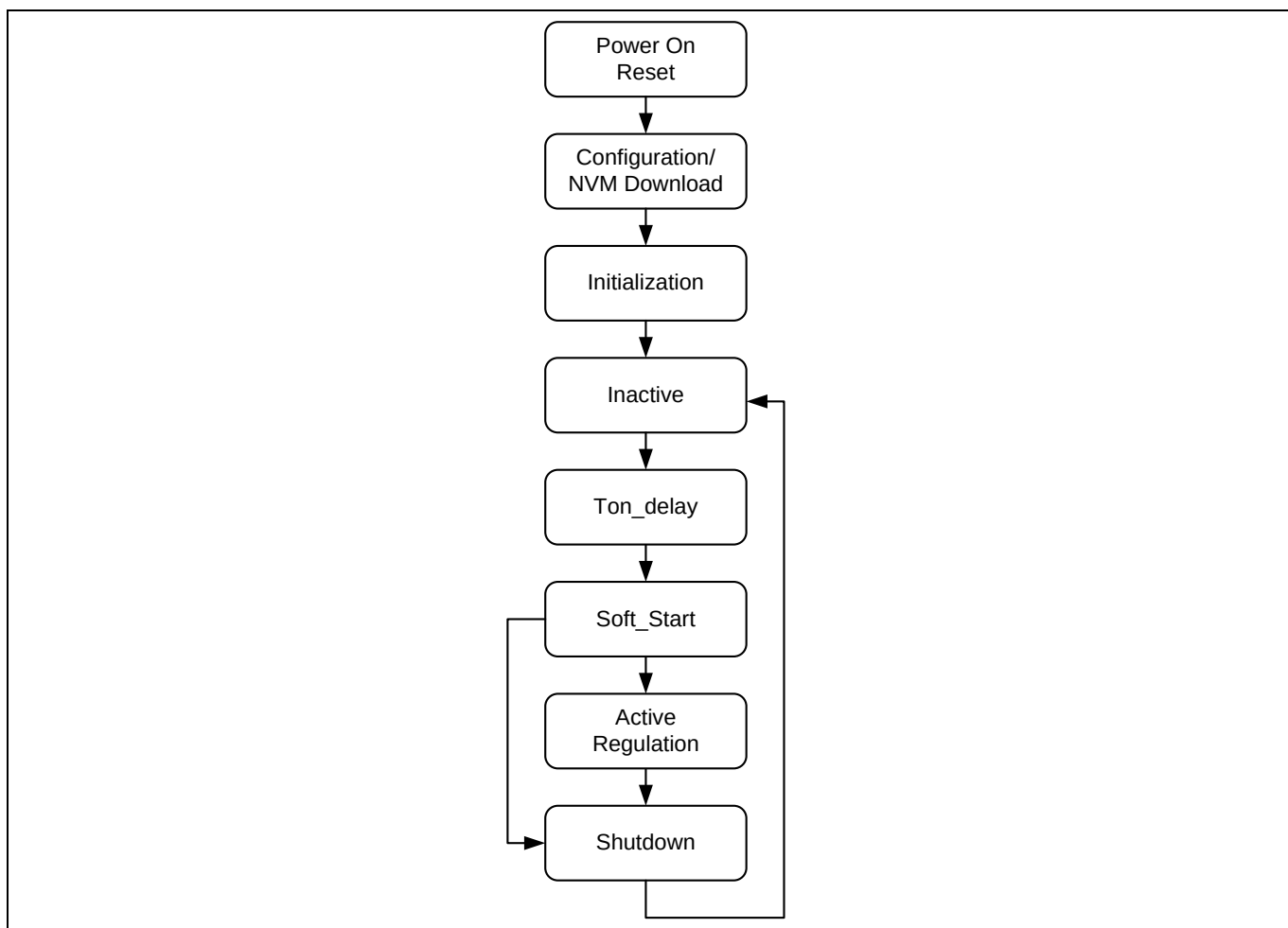


Figure 5 State machine

7.2 Power-up configuration

Operating from a single +3.3 V (VDD) supply to the controller, an on-chip regulator generates an internal +1.2 V voltage. Controller operation is initialized by an internal threshold-based power-on reset circuit. During controller configuration, the contents of the NVM are downloaded into the control registers. During this period, the PWM outputs are held in high impedance (Hi-Z) state, allowing board pull-up or pull-down resistors to set the correct default levels for static input signals. To ensure no sensitive analog measurements such as XValent resistor decoding is executed prior to 3.3 V supply is settled, the controller provides a delay which extends the “Configuration/NVM Download” state between power-on reset and entering the Initialization state. This delay is used to ensure the internal analog circuitry settles before making any precision measurements that are performed in the next state.

7.3 Initialization

During the Initialization state, the controller measures the external temperatures, input voltage, and executes the various calibration routines within the IC. Prior to exiting the Initialization state, the controller performs the external resistor pin set measurement that determines the PMBus (see Table 36) address. To properly set the device address, a resistor with at least a 1% tolerance must be connected from the ADDR pin to ground.

7.4 Inactive state and TON_DELAY

Upon completion of the Initialization process, the controller will enter the Inactive State. Before the system can be started, the controller will verify that the following conditions are satisfied:

1. $V_{DD} > 2.95\text{ V}$ (start threshold)
2. No shutdown faults are asserted that are defined in the programmed shutdown mask.
3. $V_{IN} > V_{IN_ON}$ threshold
4. $V_{AUX} > V_{AUX_ON}$ threshold (if pin is configured for Vaux sensing)
5. Assertion of EN_Lx: It is recommended that EN_Lx be asserted only after VDD, VIN and power supplies for the power stages are ready. EN_Lx is generally pulled up with an external resistor to the 3.3 V controller supply, but it is also compatible with a low voltage pull-up (e.g. 1.8 V) to allow for flexibility in system-level sequencing.
6. Optionally, the XDPE1A2G5 controller may also be configured to wait for TSEN inputs to rise above 0.4 V (the valid temperature range) and/or the PWM voltage to rise above 1.0 V (indicating the trivalent bias resistor pull-up supply is valid) on all phases. This allows the power stage driver to indicate its readiness via its TOUT or PWM I/O. Alternatively, the XDPE1A2G5 controller may be configured to auto detect populated phases, where the PWM voltage indicates the presence of populated phases and the controller may start up with just the phases that were detected.

Once the above startup conditions are satisfied, the controller will wait for a programmable period of time (TON_DELAY) before ramping up the output voltage.

7.5 Soft-start

Prior to entering the Active Regulation state, the controller performs a controlled, monotonic Soft-Start ramp of the voltage output. At the onset of Soft-Start, the controller will perform a pre-bias condition measurement of the output voltage. The controller will set the initial startup ramp voltage at the appropriate level so that it will not sink current from the pre-biased output. Soft-Start is performed by actively regulating the output voltage while digitally ramping up a reference voltage from the measured pre-biased voltage to its final target value.

For the case when Vboot is a nonzero value, the output voltage will ramp to Vboot and assert the PG_Lx pin when the voltage ramps above the POWER_GOOD_ON threshold. The voltage will ramp at the rate set by the TON_RISE setting.

7.6 Active regulation

Each rail of the controller implements a variable switching frequency, digital PID current mode voltage error control loop and inner digital PI current control loop, input voltage feedforward and dual-edge PWM architecture. During operation, the output voltage is sensed differentially and digitized by a precision analog-to-digital converter (ADC). The resultant digital error signal is then fed into a digital PID compensator.

The locations of the poles and zeroes are determined by the digital loop coefficients Kp, Ki, Kd, Kfp1 and Kfp2, which are the PID (proportional, integral, and derivative) and low pass filter pole terms, respectively.

Theory of operation

The output of the PID compensator is scaled by $1 / N_{ph}$ and becomes the target current for each phase. The target current is compared against the measured sense current, and this error current feeds a PI compensator for each phase to generate a delta pulse width value that is combined with the feedforward duty cycle computation to determine the cycle width.

The cycle width for each phase current loop is converted to a PWM pulse using a digital dual-edge pulse width modulator, with programmable maximum duty cycle limit and phase current limit capability. In addition to using this dual edge modulation scheme, the XDP™ controller also includes non-linear control mechanisms such as variable frequency and diode body braking high speed transient control, and frequency active transient response (FATR), to minimize voltage excursions during transient and repetitive transient events.

7.7 Shutdown

The shut-down state can be entered from either soft-start or active regulation states through user intervention (de-asserting EN_Lx) or through a detected fault including OTP, OCP, IUVP, OOVp and OUVP conditions. The controller's two rails may be programmed to shut down together or independently in case of a fault condition.

The XDPE1A2G5 controller supports two user selectable shutdown options in response to de-assertion of EN_Lx. The first option is a closed-loop shutdown where the controller ramps down the output voltage at a user defined slew rate. Once the output voltage has ramped to below 200 mV, the controller would switch off the output power stage FETs to avoid undershoot below GND due to residual inductor current. The second option is a hard-shutdown (Hi-Z) response, where the output stage power FETs are immediately switched off. In either case, the EN_Lx inputs are deglitched using the TOFF_DELAY setting to provide immunity to errant shutdowns due to noise on these pins. If immediate shut-down is required, TOFF_DELAY should be set to 0 msec. For cases where the shutdown is caused by a fault, the resultant shutdown response is always Hi-Z except for OOVp, in which case the low side FETs are turned on under this condition.

7.8 PWM output configuration

The switching frequencies for the rails are independently selectable and configurable by setting the appropriate register values for each rail output. PWM outputs are 3.3 V compatible trivalent signals which are set to high impedance during reset, configuration, initialization, and inactive states. PWM outputs can be assigned to any phase channel and to either loop. The multi-phase PWM output signals are flexible and can be configured to fire in any order that is required to take advantage of input and output ripple cancellation by interleaving adjacent phases, or to facilitate PCB layout.

Table 8 PWM Channel Selection for typical Loop Configurations

Loop Configuration	Loop A Phase Map Setting	Loop A Phase Channels	Loop B Phase Map Setting	Loop B Phase Channels
1+0	0x0001	1	0x0000	na
2+0	0x0003	1, 2	0x0000	na
...				
16+0	0xFFFF	1 - 16	0x0000	na
1+1	0x0001	1	0x8000	16
2+1	0x0003	1, 2	0x8000	16
...				
15+1	0x7FFF	1 - 15	0x8000	16
14+2	0x3FFF	1 - 14	0xC000	15, 16
12+4	0x0FFF	1 - 12	0xF000	13, 14, 15, 16

Theory of operation

Loop Configuration	Loop A Phase Map Setting	Loop A Phase Channels	Loop B Phase Map Setting	Loop B Phase Channels
8+8	0x00FF	1 - 8	0xFF00	9 - 16

8 Application information

8.1 Input power sense

The XDPE1A2G5 controller provides the capability to estimate the input power consumption of the individual voltage regulator. The input power supply voltage is monitored via VINSEN for feedforward control, telemetry, power sequencing, and fault detection. Input power is obtained by multiplying the measured input voltage by the estimated the input current.

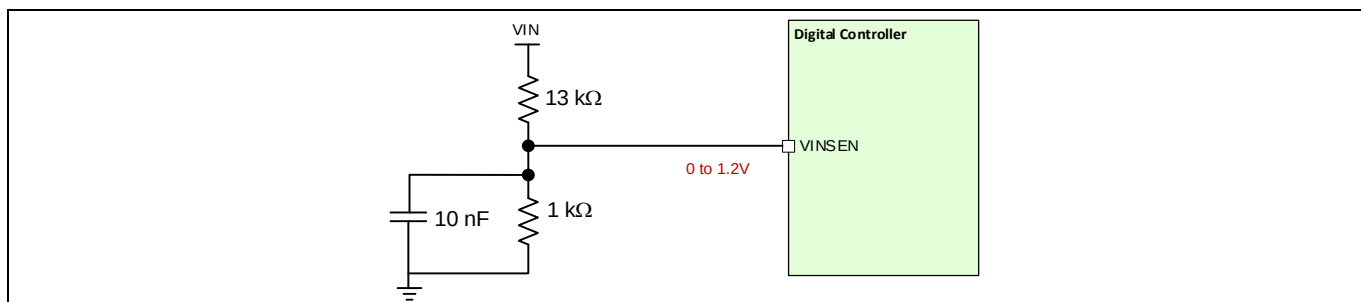


Figure 6 VINSEN monitoring

8.2 Differential output voltage sense (VOSEN_Lx / VORTN_Lx)

The differentially sensed remote output voltage of each rail (VOSEN_Lx/VORTN_Lx) is used for PID loop compensation, over voltage fault protection, and telemetry. The output voltages of all rails are sensed differentially and converted to a digital representation over a range of 0 to 3.1 V using a high speed, precision analog-to-digital converter (ADC). An on-chip factory trimmed temperature compensated bandgap voltage reference ensures precise set-point accuracy.

It is advantageous to add a higher bandwidth external RC filter network to increase the attenuation of high frequency noise that may couple onto the voltage sense lines. This additional filter provides minimal phase margin impact on the feedback loop. If such an additional filter is used, it is recommended that values of $R_{ext} = 10\ \Omega$ and $C_{ext} = 220\ \text{pF}$ be used.

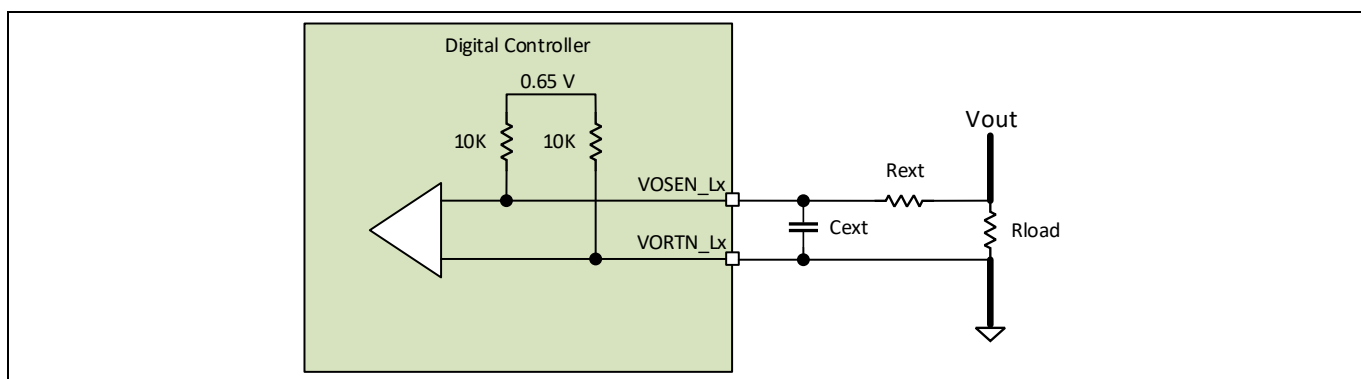


Figure 7 Output voltage sense internal impedance and external filtering

8.3 Output phase current sense (ISENx)

The phase current sense circuitry converts the differential signal of the remote current sense connected to the ISENx pins and the VD12 pin as a reference into a digital representation of the instantaneous phase current.

The phase currents are used for current-mode control, computing total output current, current monitoring, telemetry, adaptive voltage positioning (AVP), phase current balance, phase error detection, positive and negative current limit, output over-current protection, and load adaptive response (auto phase shedding). The phase currents can be temperature compensated based on the internal die temperature of the controller. This insures that the integrated 1 k Ω resistors at each ISENx input are well compensated over the full temperature range. Current sense gain and offset parameters are trimmed at ATE which ensures a high degree of accuracy. Offset parameters are also continuously calibrated during controller operation to ensure good accuracy with varying operating conditions.

Integrated current sense is a scheme using a power stage which incorporates a current or voltage output, IMON, proportional to the output current through the power stage high-side and low-side transistors. Some of Infineon's integrated power stages vary their IMON gain as a function of output voltage (Duty Cycle), VDRV voltage, and current. The XDPE1A2G5 controller has implemented controls to compensate the varying IMON gain and produce exceptional phase current measuring accuracy.

Figure 8 shows the recommended method to connect the integrated current sense signals between the XDPE1A2G5 controller and an integrated power stage. The controller supports an input signal range for integrated current sense of -200 mV to +650 mV (+550mV with the internal 1 k Ω) with 1750 μ V resolution. This provides a phase current range of -40 A to +130 A (110 A with the internal 1 k Ω) with 0.35 A resolution. The integrated power stage IMON pin output is a transconductance output of 5 μ A/A.

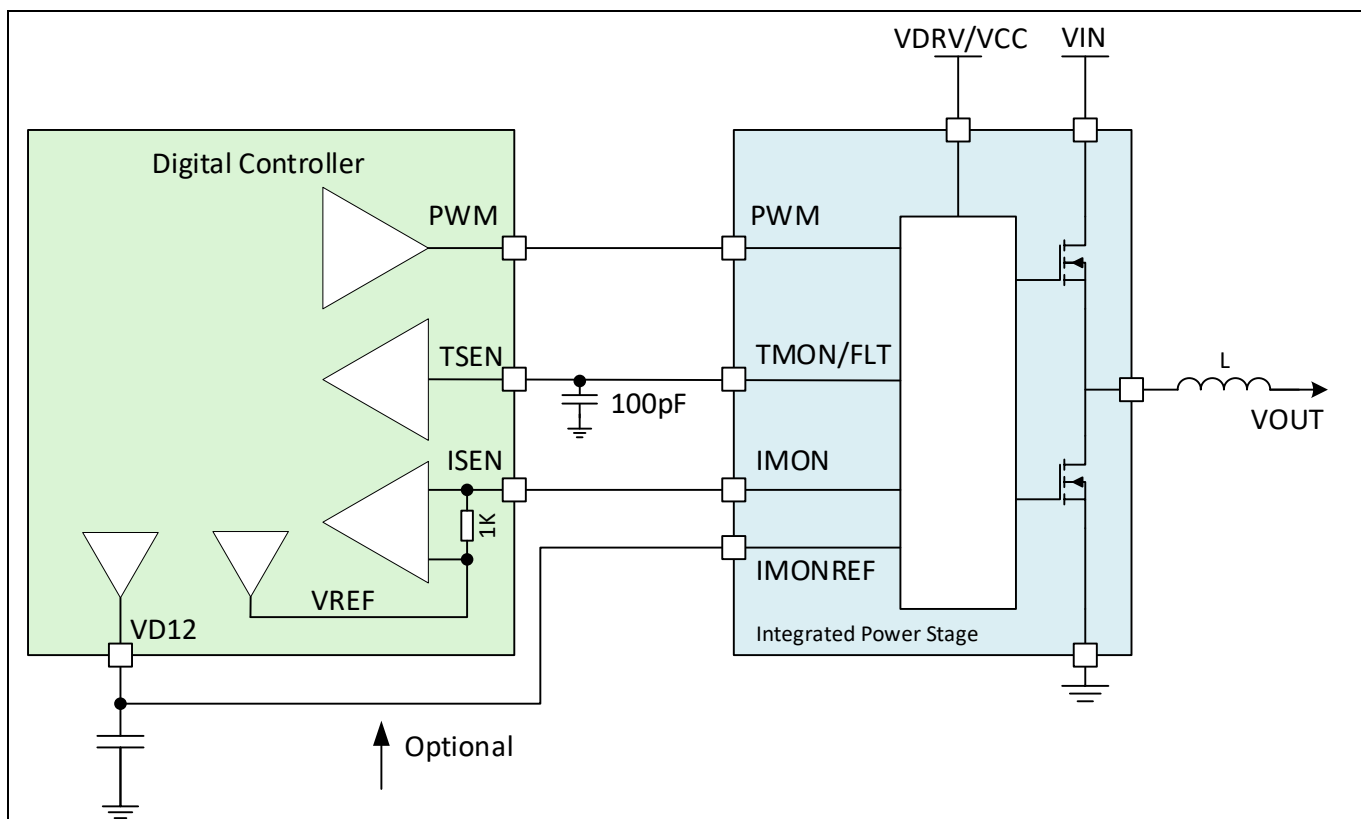


Figure 8 Recommended connections between integrated power stage and controller

The XDPE1A2G5C allows the user to parallel as many as four power stages on each PWM pin. The max phase current shown in Tables 9 and 10 is the total phase current for all paralleled power stages.

Table 9 Maximum Positive Phase Current

Paralleled Power stages	Resistor Location	Resistor (Ω)	Max Sense Voltage (mV)	Max Phase Current (A)	Max Output Current (A)	Internal Vref Isink (mA)	VD12 pin Isink (mA)
3	Internal	333	333	200	3200	-16	NA
2	Internal	500	450	180	2880	-14.4	NA
1	Internal	1000	550	110	1760	-10.4	NA
4	External	250	650	520	8320	NA	-40

Table 10 Maximum Negative Phase Current

Paralleled Power stages	Resistor Location	Resistor (Ω)	Min Sense Voltage (mV)	Min Phase Current (A)	Max Sink Current (A)	Internal Vref Isource (mA)	VD12 pin Isource (mA)
3	Internal	333	-66	-40	-640	3.2	NA
2	Internal	500	-100	-40	-640	3.2	NA
1	Internal	1000	-200	-40	-640	3.2	NA
4	External	250	-200	-40	-200	NA	1

8.3.1 Support for output current greater than 2048A

The XDPE1A2G5C has firmware and hardware updates that allow the controller to be used in applications with output currents greater than 2048A. Register `t1m()_lp_iout_scale` sets the max output current. The max supported loop output current is shown in Table 11

Table 11 Max Output Current per loop

<code>T1m()_lp_iout_scale</code> (hex)	Loop A (<code>t1m0_lp_iout_scale</code>)	Loop B (<code>t1m1_lp_iout_scale</code>)
0	2048	1024
1	4096	2048
2	8192	4096
3	reserved	reserved

The following registers are not scaled by the `t1m_lp_iout_scale` register and need to be scaled external

Table 12 Registers requiring external scaling

Section	Access	Register Name	Register Address	Required Action
PMBUS	Read/write	FW_CONFIG_REGULATION. MFR_RDROOP_IDELTA_SEG3	0xC5 [111:96]	Write a value that is 1/(1+tlm_lp_iout_scale) the required value
		FW_CONFIG_REGULATION. MFR_RDROOP_ITHR_SEG2	0xC5 [95:80]	
config		Mod0_phase2_delta	0x70000C0C [9:5]	
		Mod1_phase2_delta	0x7000100C [9:5]	
		Mod0_phase3_delta	0x70000C0C [14:10]	

Application information

Section	Access	Register Name	Register Address	Required Action
		Mod1_phase3_delta	0x7000100C [14:10]	
		Mod0_phase4_delta	0x70000C0C [19_15]	
		Mod1_phase4_delta	0x7000100C [19_15]	
		Mod0_phase5_delta	0x70000C0C [24:20]	
		Mod1_phase5_delta	0x7000100C [24:20]	
		Mod0_phase6_delta	0x70000C0C [29:25]	
		Mod1_phase6_delta	0x7000100C [29:25]	
		Mod0_phase7_delta	0x70000C10 [4:0]	
		Mod1_phase7_delta	0x70001010 [4:0]	
		Mod0_phase8_delta	0x70000C10 [9:5]	
		Mod1_phase8_delta	0x70001010 [9:5]	
		Tlm_imon_l1_max	0x70002044 [12:2]	
		Tlm_imon_l2_max	0x70002048 [12:2]	
		Mod0_ce_ni_thresh	0x70000C40 [17:12]	
		Mod1_ce_ni_thresh	0x70001040 [17:12]	
	Read only	lp1_adc_lpf	0x70007464 [13:0]	Multiply register reading by (1+tlm_lp_iout_scale) for correct phase current
		lp2_adc_lpf	0x70007464 [27:14]	
		lp3_adc_lpf	0x70007468 [13:0]	
		lp4_adc_lpf	0x70007468 [27:14]	
		lp5_adc_lpf	0x7000746C [13:0]	
		lp6_adc_lpf	0x7000746C [27:14]	
		lp7_adc_lpf	0x70007470 [13:0]	
		lp8_adc_lpf	0x70007470 [27:14]	
		lp9_adc_lpf	0x70007474 [13:0]	
		lp10_adc_lpf	0x70007474 [27:14]	
		lp11_adc_lpf	0x70007478 [13:0]	
		lp12_adc_lpf	0x70007478 [27:14]	
		lp13_adc_lpf	0x7000747C [13:0]	
		lp14_adc_lpf	0x7000747C [27:14]	
		lp15_adc_lpf	0x70007480 [13:0]	
		lp16_adc_lpf	0x70007480 [27:14]	

The controller produces an internal synthesized current based on the externally sensed phase current from the power stages and the internally created emulated current. The controls, as shown in Figure 9, are available to configure the device to produce accurate synthesized current based on the inductors and power stage drivers used. Controls for both Standard and TLVR inductor designs are available.

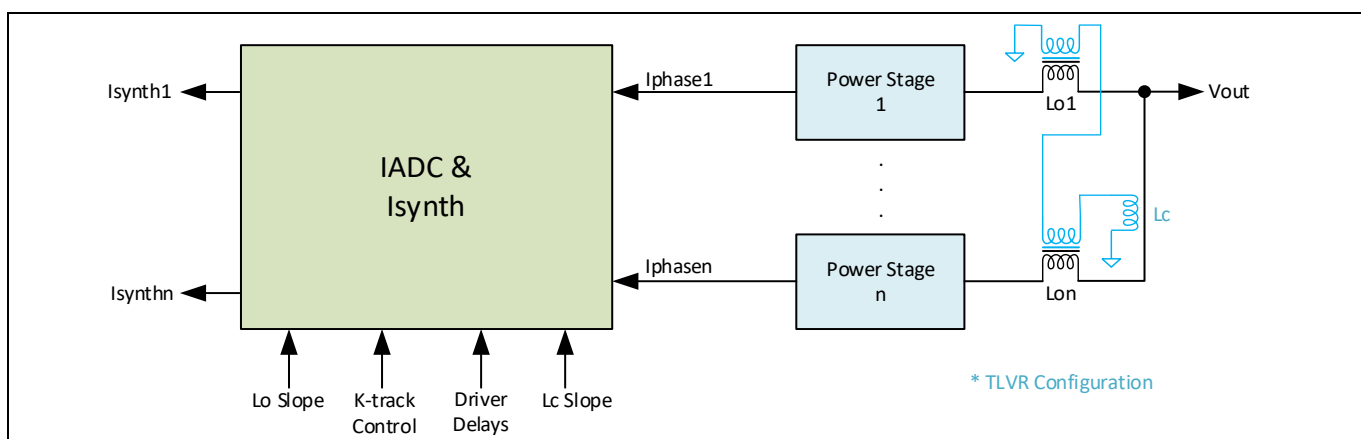


Figure 9 Synthesized current controls

Table 13 Isynth controls

Controls	Description
Lo Slope	Sets the value of the output inductor so that the expected slope of the inductor current can be used to accurately create the emulated current
K-track Control	Sets the gain terms to control how quickly the emulated current is modified based on the error of the sensed current versus the present emulated current. Gain terms are available for PWM ON, OFF, and Tri-state conditions.
Driver Delays	Sets the positive, negative, and tri-state delays based on the power stage driver used. This compensates for the PWM to Switch Node delays and any IMON delays of the power stage.
Lc Slope	Sets the value of the Lc Inductor used in a TLVR design so that the expected slope of the contributing inductor current can be used to accurately create the emulated current

8.4 Temperature sense (TSENx, Internal)

XDPE1A2G5 controllers provide the capability to measure the temperature of up to two external temperature sensors. The temperature sensors are connected through the TSENx pins from the TMON output of an integrated power stage. In the case of TDA21570 or equivalent power stages, this voltage is at a nominal value of 0.8 V @ 25 °C and a temperature slope of 8 mV/°C. The TMON (temperature output) pins of multiple power stages are OR-tied together so that the power stage with the hottest temperature (highest TMON voltage) will drive the signal line to the controller TSENx pin.

Additionally, the power stage is able to provide catastrophic over temperature protection within the power stage itself to quickly trigger VR shutdown in case of temperatures exceeding the allowed junction limit. This is accomplished by the power stage driving its TMON/FLT to 3.3 V. The controller will respond to this when the TSENx voltage goes greater than 2.4 V typically.

In addition to the two external temperature sense inputs, the XDPE1A2G5 incorporates an internal die temperature sensor (TSEN_INT). This is useful for local temperature monitoring, and for using the die temperature in device characterization and ATE trimming.

Application information

Each temperature sensor input can be configured as the Fault or Telemetry source for Loop A or Loop B as shown in Table 14. This is configured through the PMBus command MFR_SELECT_TEMPERATURE_SENSOR (CCh).

Table 14 Temperature Sensor Select

Function	Loop A or B Source Select
FAULT	0 – TSEN1
	1 – TSEN2
	2 – TSEN_INT
READ_TEMPERATURE_1, READ_TEMPERATURE_2	0 – TSEN1, TSEN2
	1 – TSEN1, TSEN_INT
	2 – TSEN2, TSEN1
	3 – TSEN2, TSEN_INT
	4 – TSEN_INT, TSEN1
	5 – TSEN_INT, TSEN2

8.5 Auxiliary voltage sense (VAUX)

The XDPE1A2G5 supports monitoring of an auxiliary supply rail (typically used to monitor Power Stage VDRV/VCC to control sequencing). The supply voltage on the VAUX pin is compared against a programmable threshold. Once the rising VAUX voltage crosses the turn-on threshold (Vaux ON) and ENx is asserted, all PWM outputs become active when there are no faults in the VR. The VAUX supply voltage is valid until it falls below the programmed turn-off level (Vaux UV).

A 6.4:1 attenuation network is connected to the VAUX pin as shown in Figure 10. Recommended values for a 5V system are R1=5.36 kΩ and R2=1 kΩ, with a 1% tolerance or better. C1 is required to have a minimum 1nF for noise suppression, with a maximum value of 10nF.

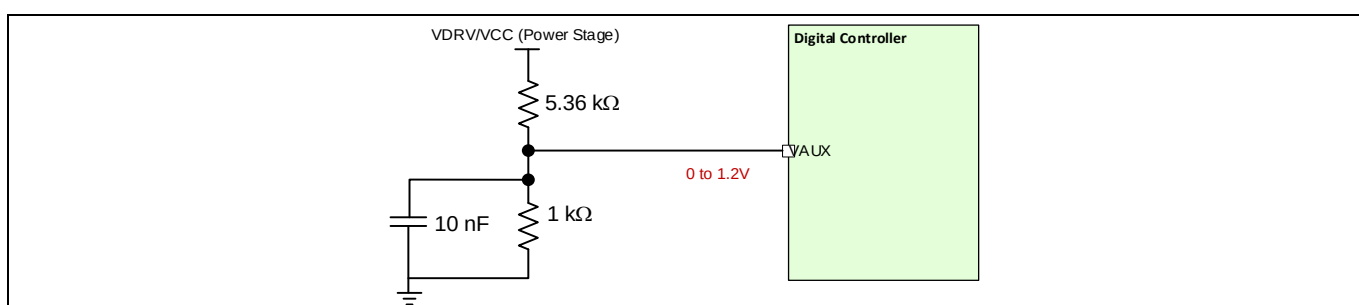


Figure 10 VAUX sense resistor divider network

8.6 AVSBus

8.6.1 Interface

The XDPE1A2G5 implements a fully compliant AVSBus interface with an AVS clock frequency up to 25MHz. See section 8.6.3.1 for details for operation with AVS clock frequencies up to 50MHz. This is typically a three-wire interface between a processor and a VR that consists of clock, controller data and target data signals. (Figure 11) The XDPE1A2G5 implements all the required AVSBus commands per PMBus Rev 1.4 Part III. A list of supported commands is shown in Table 17.

Application information

Alternatively, a 2-wire interface (Figure 12) can be implemented making the target data signal optional. This renders the system incapable of sending data back to the controller, but may be useful in simplifying the bus interface.

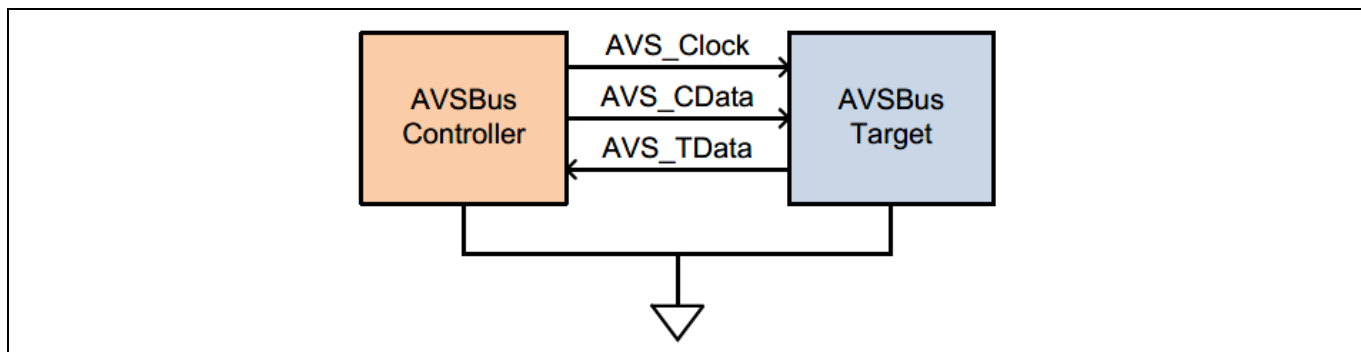


Figure 11 AVSBus 3-wire Interface

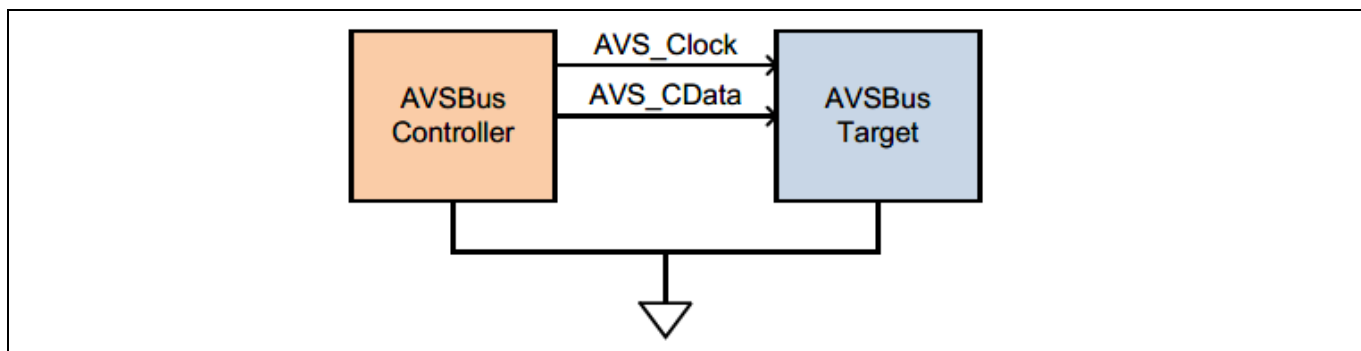


Figure 12 AVSBus 2-wire Interface

8.6.2 Operation

The interface operation consists of complementary functions in the controller and the target devices. The AVSBus Controller must initiate all data transfers and it must guarantee that AVS_CData is held at a logic value '1' when the clock is not running. This must occur during initialization prior to any bus transfers, as well as during idling between frames. AVS_CData is not allowed to be at a logic value '0' when the clock starts.

The AVSBus Target listens for and may respond to controller commands but cannot initiate a transfer under any circumstances. In general, the target must guarantee that AVS_TData is held at a logic value '1' when the clock is not running. This must occur during initialization prior to any bus transfers, between frames, as well as at any time during a transfer when the target is not sending data to the controller. An exception to this rule is that the AVSBus Target will set AVS_TData to a logic value '0' when it needs to alert the AVSBus Controller to start a frame so that it can send a "Status Response Frame"

The XDPE1A2G5 implements mechanisms that allow recovery of the bus in case noise or some other artifact causes the bus to enter an incorrect state. One mechanism is target resynchronization. After receiving 34 clocks while the AVS_Cdata is held high, the device will resynchronize its bus interface and wait for the next start code. The second mechanism is a Bus Timeout. When the device determines that an ongoing transfer has been effectively aborted because the clock stopped running for a user programmable amount of time (timeout), the device will reset the bus interface and again wait for the next start code.

Application information

8.6.3 Electrical Interface

The XDPE1A2G5 will accept clock and data signals from the controller driven to the VDDIO level and will also provide its data to the controller, also driven to the VDDIO level. This is shown in Figure 14.

8.6.3.1 Timing

Both the controller and target launch data from the rising edge of the clock. The XDPE1A2G5C (target) captures data on the clock falling edge. The AVS controller can capture data at any time between the clock falling edge and the rising edge of the following clock.

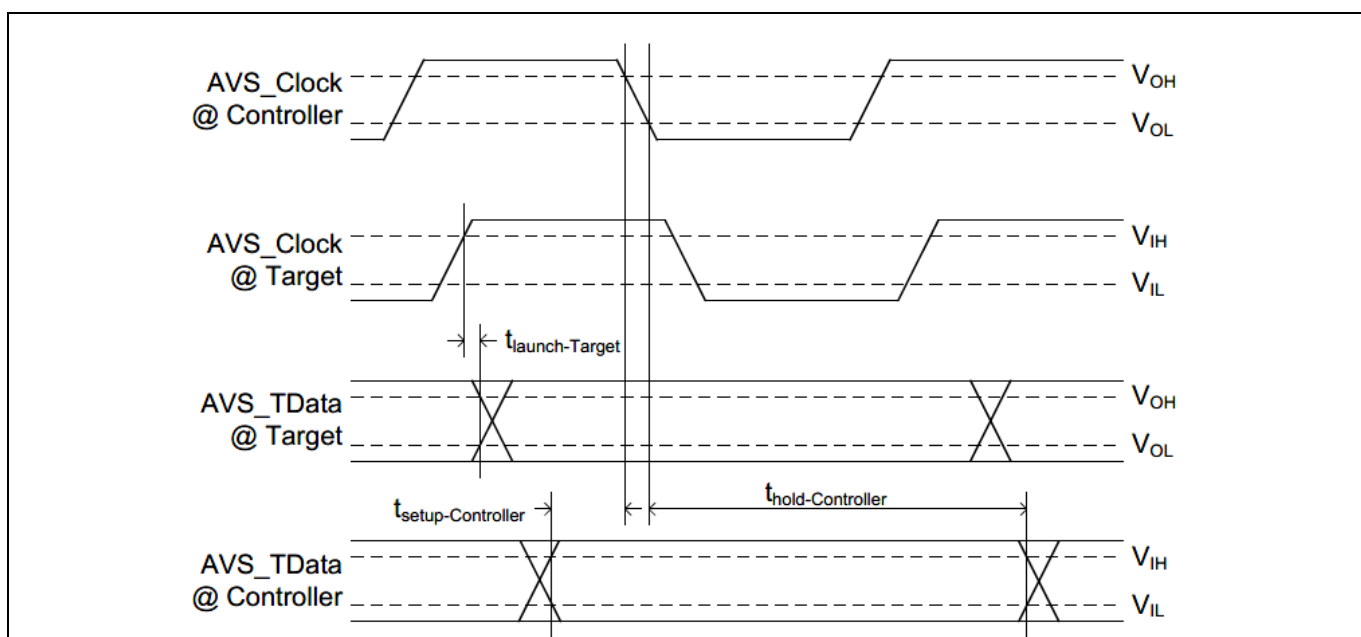


Figure 13 Timing diagram for AVS_TData

The timing diagram shows a skew in the clock and data lines, t_{delay} , which corresponds to the transmission delays on the board. A comparable delay affects the data lines. Setup and hold times are measured at the pads of the controller or target, as pertinent.

Table 15

Parameter	Description	Min.	Typ.	Max.	Units
t_p	Period for active clock	20		200	ns
t_{high}	Duration of the high-phase of clock	40			% t_p
t_{low}	Duration of the low-phase of clock	30			% t_p
Target to Controller data path					
$t_{launch-target}$	Time from rising clock edge in target to data-out transition at target's data-out port		6	11	ns
$t_{setup-controller}$	Time from data-out edge in target to capture clock edge in controller	0			% t_p
$t_{hold-controller}$	Time from capture clock edge in controller to data-out edge in target (for next bit)	20			% t_p

Due to transmission delays, clock and data rise and fall times, and $t_{launch-target}$ delays, 50MHz operation requires that the target data be clocked into the controller at some point in time after the falling edge of the clock pulse.

Application information

The target holds the data on the TDATA line until the next clock rising edge (plus $t_{\text{launch-target}}$). So, the controller can clock in the target data at any time after the clock falling edge up to and including the rising edge of the following clock pulse

8.6.3.2 Electrical Drive Levels

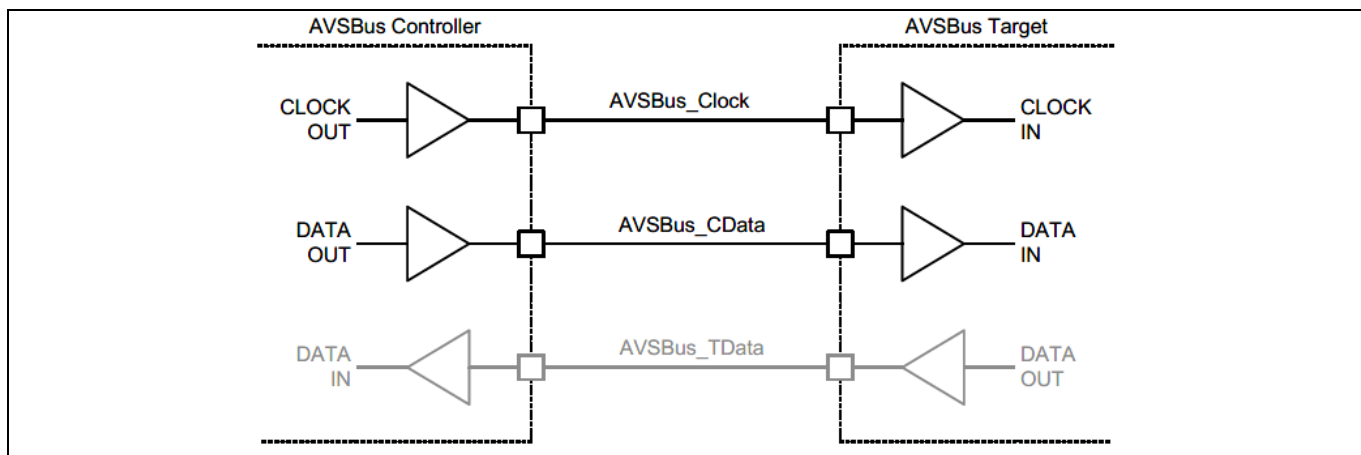


Figure 14 Electrical Interface

The threshold levels for the incoming clock and controller data lines are set by the VDDIO voltage level. The nominal levels are shown in Table 16

Table 16 Nominal Clock and Data Threshold Levels

Nominal VDDIO	V _{il}	V _{il}
1 V	0.35 V	0.65 V
1.2 V	0.42 V	0.78 V
1.5 V	0.53 V	0.98 V
1.8 V	0.63 V	1.17 V

8.6.4 AVSBus Status Alert Mask

The XDPE1A2G5 Target will alert the Controller that its status has changed by asserting the AVS_TDATA low. This will happen when the following Warnings or Fault bits are set.

- OCW = Output Over-current Warning (STATUS_IOUT, bit 5)
- UVW = Output Under-voltage Warning (STATUS_VOUT, bit 5)
- OTW = Over-temperature Warning (STATUS_TEMP, bit 6)
- OPW = Output Over-power Warning (STATUS_IOUT, bit 0)
- Faults or Warnings in MFR_SPECIFIC (STATUS_MFR_SPECIFIC, bits 7:0)

The XDPE1A2G5 has a Mask Register that can mask any or all chosen warnings or faults listed above. Setting the associated bit to a '1' will mask that warning or fault from asserting the AVS_TDATA low. The Status Alert and specific warning bits will still be set and can be seen when an AVS Status Read command is performed.

8.6.5 AVSBus Command Group

Table 17 AVSBus Command Group 0b

Data Types	Command Name	Transaction Type	Range/Resolution	Description
0000b	Voltage	Read/Write	16-bit unsigned integer field with 1 LSB = 1 mV	This type allows the AVSBus Controller to read or write a new voltage target for a rail on an AVSBus Target.
0001b	Vout Transition Rate	Read/Write	8-bit unsigned integer field with 1 LSB = 1 mV/μs	This type allows the AVSBus Controller to read or write a new transition rate for a rail on an AVSBus Target.
0010b	Current	Read	16-bit unsigned integer field with 1 LSB = see Table 19	This type allows the AVSBus Controller to read the current for a rail on an AVSBus Target. Writes are not supported.
0011b	Temperature	Read	16-bit signed integer field with 1 LSB = 0.1 °C	This type allows the AVSBus Controller to read the temperature for a rail on an AVSBus Target. Writes are not supported.
0100b	Voltage Reset	Write	All 0's	This type allows the AVSBus Controller to force a "Predetermined Value" for an AVSBus Target rail, particularly while handling exceptions that could otherwise cause damage. The "Predetermined Value" is set by PMBus VOUT_COMMAND.
0101b	Power Mode	Read/Write	000b – 111b	This type allows the AVSBus Controller to set the mode of operation for the output of an AVSBus Target rail. The Power Mode settings are the same as the PMBus Power Mode descriptions.
0110b	Reserved for Future Use			
0111b	Reserved for Future Use			
1000b	Reserved for Future Use			
1001b	Reserved for Future Use			
1010b	Reserved for Future Use			
1011b	Reserved for Future Use			

Application information

Data Types	Command Name	Transaction Type	Range/Resolution	Description
1100b	Reserved for Future Use			
1101b	Reserved for Future Use			
1110b	AVSBus Status	Read/Write	<VDONE> <OCW> <UVW> <OTW> <OPW> <RSV_3> <MFR_SPEC_8>	This type allows the AVSBus Controller to read the <AVSBus_Status> of an AVSBus Target, or to clear the target's <AVSBus_Status> by a write. When writing to this data type, all Status bits written with a 1(binary) are cleared. All others remain unchanged. Persistent faults will be immediately re-asserted. MFR_SPEC_8 = PMBus STATUS_MFR_SPECIFIC Register.
1111b	AVSBus Version	Read	0001b	This type allows the AVSBus Controller to read the version of AVSBus implemented by an AVSBus Target.

Table 18 MFR Specific Commands. Command Group 1b

CmdGroup	CmdDataType	Description	Comments
1	0x0	Read Input Current	See Table 20 for resolution setting
1	0x1	Read Input Voltage	See Table 23 for resolution setting
1	0x2	Read Actual Output Voltage	See Table 24 for resolution setting
1	0x3	Read Output Power	See Table 21 for resolution setting
1	0x4	Read Input Power	See Table 22 for resolution setting
1	0x5	Read Aux Voltage	125mV per lsb
1	0x8	Read and Write load line	Linear 16 format; 0.25 $\mu\Omega$ lsb. See Table 34 for actual load line resolution settings. The power on reset (POR) load line is set by the PMBus VOUT_DROOP command. The load line can be updated with the VOUT_DROOP command until an AVS load line command is written. After an AVS load line command is sent, the VOUT_DROOP command is ignored until the next POR
1	0x9	Read minimum output current	Same resolution as READ IOUT
1	0xA	Read maximum output current	Same resolution as READ IOUT. Reading this command clears min and max IOUT
1	0xB	Read minimum output voltage	Lsb resolution is 0.122mV
1	0xC	Read maximum output voltage	Lsb resolution is 0.122mV. Reading this command clears min and max VOUT
1	0xD	Read MFR_REV	Returns value in PMBus command 0x9B

Application information

CmdGroup	CmdDataType	Description	Comments
1	0xE	Read MFR_MODEL	Returns value in PMBus command 0x9A
1	0xF	Read MFR_ID	Returns value in PMBus command 0x99

The MFR read commands (Table 18) in the XDPE1A2G5C have a resolution that can be set by the user. Table 19 to Table 24 below show the telemetry resolution and the register settings

Table 19 Read IOUT

LSB	tlm_iout_hs_telem_gain_exp	tlm_iout_hs_telem_gain_man
10 mA	0x37	0xC80
20 mA	0x36	0xC80
40 mA	0x35	0xC80
50 mA	0x35	0xA00
100 mA	0x34	0xA00

Table 20 Read IIN

LSB	tlm_iin_hs_telem_gain_exp	tlm_iin_hs_telem_gain_man
10 mA	0x37	0xC80
20 mA	0x36	0xC80
40 mA	0x35	0xC80
50 mA	0x35	0xA00
100 mA	0x34	0xA00

Table 21 Read POUT

LSB	tlm_pout_hs_telem_gain_exp	tlm_pout_hs_telem_gain_man
10 mW	0x37	0xC80
20 mW	0x36	0xC80
40 mW	0x35	0xC80
50 mW	0x35	0xA00
100 mW	0x34	0xA00

Table 22 Read PIN

LSB	tlm_pin_hs_telem_gain_exp	tlm_pin_hs_telem_gain_man
10 mW	0x37	0xC80
20 mW	0x36	0xC80
40 mW	0x35	0xC80
50 mW	0x35	0xA00
100 mW	0x34	0xA00

Table 23 Read VIN

LSB	tlm_vin_hs_telem_gain_exp	tlm_vin_hs_telem_gain_man
1 mV	0x3A	0xFA0
10 mV	0x37	0xC80

Application information

Table 24 Read VOUT

LSB	tlm_vout_hs_telem_gain_exp	tlm_vout_hs_telem_gain_man
1 mV	0x3A	0xFA0
10 mV	0x37	0xC80

8.6.6 AVSBus Addressing

The XDPE1A2G5 can control two loops with one device. This requires addressing control to differentiate between the two loops that use the same AVSBus interface. This is accomplished by address registers 0x70004004 [3:0] for Loop1 and 0x70004404 [3:0] for Loop2 that can be set to a value of 0 – 14 for each loop. Note that address 15 should not be used since this is the all call address on the AVSBus.

Table 25 AVS Address Registers

Register	Description	Value
0x70004004[3:0]	AVSBus Address Loop1	0 to 14
0x70004404[3:0]	AVSBus Address Loop2	0 to 14

8.6.7 VOUT transition rate

The XDPE1A2G5 supports dynamic VID changes by writing to the AVS ‘VOLTAGE’ command. The DVID slew rate is set by writing to the AVS ‘VOUT TRANSITION RATE’ command. The slew rate can be set as low as 1 mV/μs up to 50 mV/μs. The practical limit of the slew rate will be determined by the number of phases, per phase inductance, and the total output capacitance.

During dynamic VID changes to a higher value, the tracking output over voltage protection (OOVP) is temporarily disabled. However, the fixed OOVP remains enabled.

8.7 Phase Add/Drop: Autonomous

The XDPE1A2G5 controller provides a selectable autonomous phase add/drop feature. By comparing the total measured current with a set of user programmable thresholds, the controller will autonomously add or drop phases. Phases are added or dropped one at a time based on the current level.

Two types of user programmable parameters are provided for current based autonomous phase control:

1. The threshold parameter determines the current level at which a phase is added.
2. The hysteresis parameter, when subtracted from the threshold, determines the current level at which a phase is dropped. Independent threshold parameters are provided for each phase count transition. There is a single hysteresis parameter (6 A or 10 A) for all phase count transitions.

In addition to the phase add method described above, the XDPE1A2G5 controller provides the following methods for immediate phase adding which helps to improve transient performance:

1. When error voltage exceeds a programmable threshold (ATRH Entry Threshold).
2. During DVID events.

Application information

Due to the different output configurations of the power stage when in a reduced phase count state, the Pole filters are automatically scaled based on the active phase count, and there are an independent set of PID and Pole filter settings for single phase count for optimal transient performance.

8.8 Linear loop compensation (PID / Poles / PI)

The XDPE1A2G5 uses digital voltage and current feedback loops to maintain a tightly regulated output voltage with inherent phase current balance. During operation, the output voltage is sensed differentially and digitized by a precision analog-to-digital converter (ADC). The digitized voltage is compared to a target voltage whose error is fed into a digital compensator that consists of PID parameters, K_p (proportional coefficient), K_i (integral coefficient), and K_d (derivative coefficient), as well as the low-pass filter pole locations, K_{fp1} and K_{fp2} . Various PID coefficient and Poles settings are available for optimizing loop response during soft-start, steady state, load transient, and power saving states.

The output of each loop's digital compensator is converted into a control current that is divided and fanned-out equally to all active phases. The phase current is sensed and compared to a target emulation current. The error is digitized and is used to adjust the emulation current creating a synthesized phase current output. The control current going to the phases is compared to the synthesized phase current of each phase to create a phase current error that is feed into a digital compensator that consists of PI parameters. This is the inner phase current feedback loop whose output is converted into PWM pulses for each phase. The PI coefficients settings are available to optimize the inner loop response based on the output inductors used in the design.

Figure 15 shows a simplified diagram of the compensation loops use in the linear current-mode feedback control.

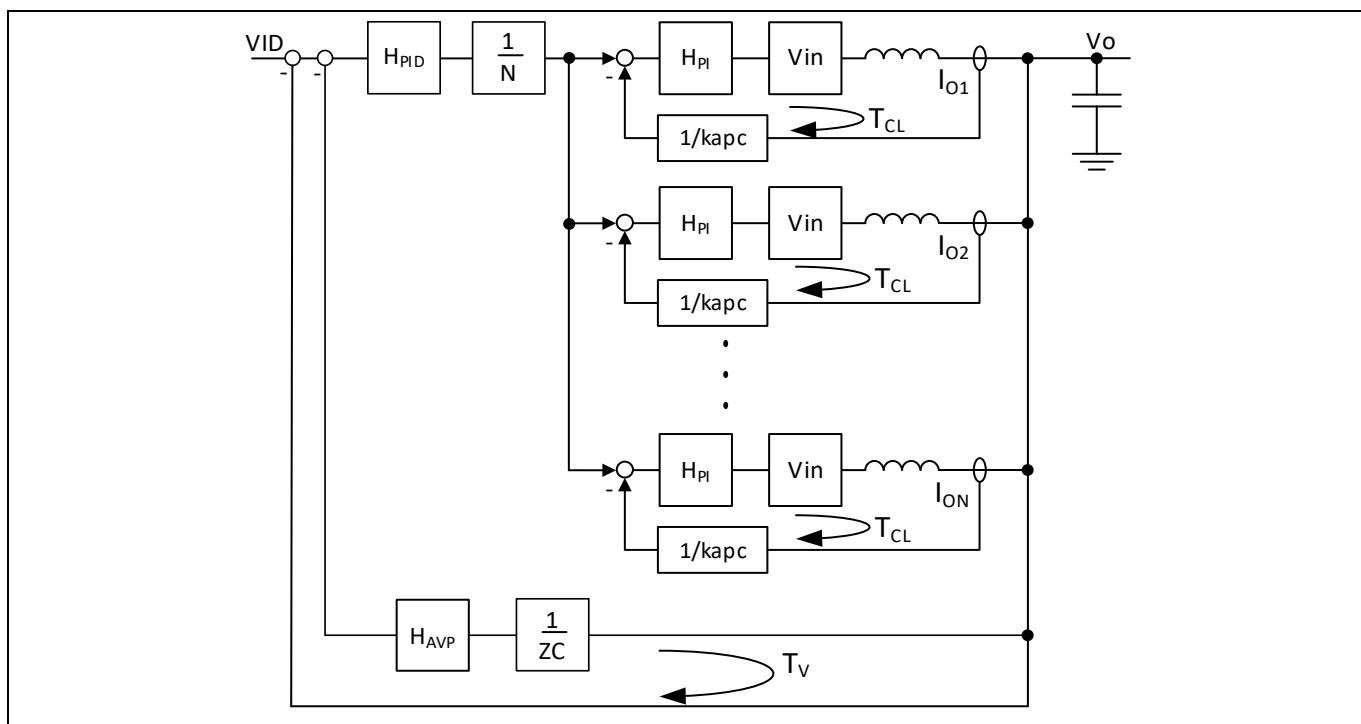


Figure 15 Voltage and current compensation loops

8.9 Active transient response (ATR / FATR)

The XDPE1A2G5 controller supports an Active Transient Response function which allows for improved transient regulation with reduced output capacitance. Error voltage tracking comparators with programmable

Application information

thresholds (magnitude and slope) continuously monitor the output voltage. The ATR High (ATRH) function will increase the switching frequency (PFM) and will boost the control current (Increased Pulse Width) during a load step to reduce output undershoot. The ATR Low (ATRL) function will blank PWM output pulses and can also enter Diode Braking (PWM Hi-Z) during a load release to reduce output overshoot. Frequency Active Transient Response (FATR) allows control of the ATR non-linear gains, PID, Diode Braking, Vout Offset, and High-speed Phase Balance based on load oscillation frequency. The load oscillation frequency can be detected by the control engine and thresholds set to enable/disable functions or change gains to the linear and non-linear controls.

8.9.1 ATRH and ATRL

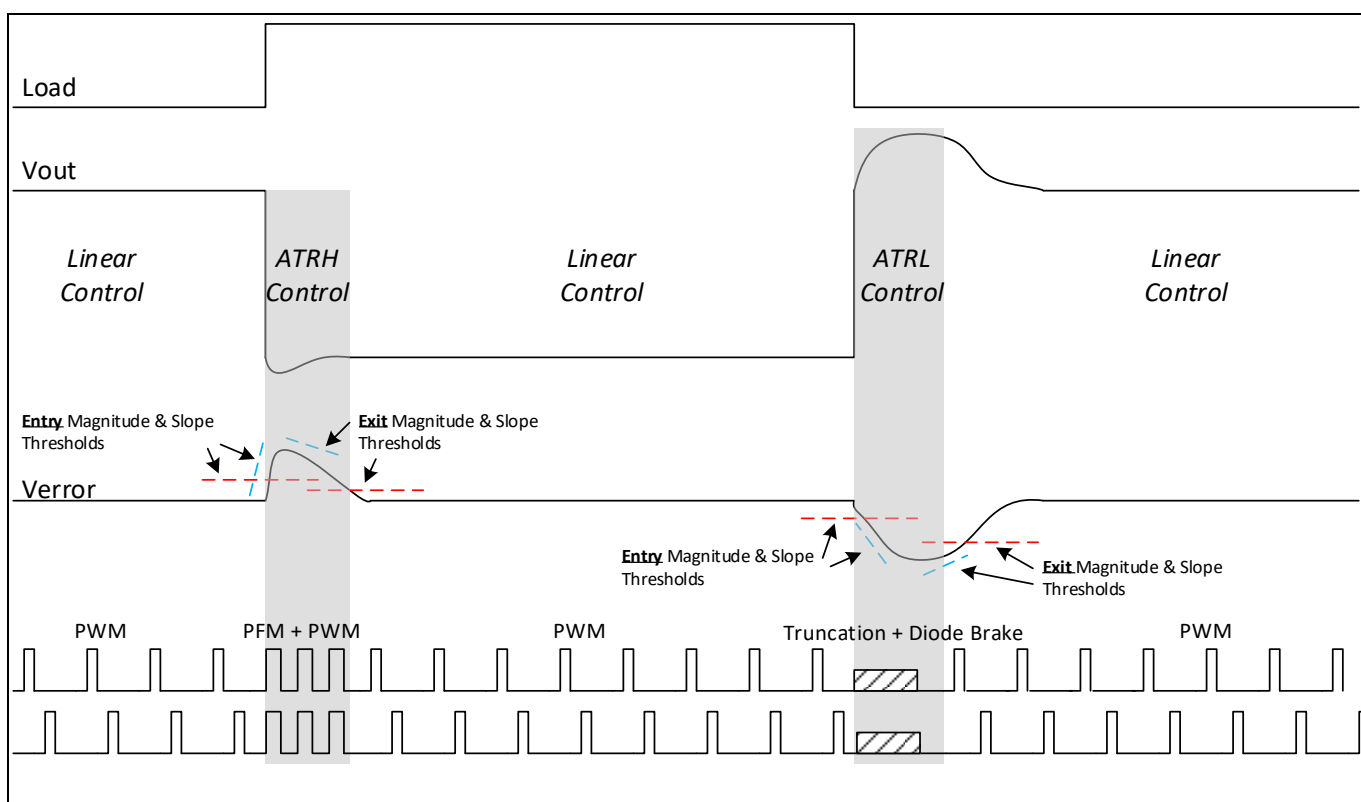


Figure 16 ATRH & ATRL Transient control

Table 26 ATRH & ATRL transient controls

Control	Description
ATRH Entry & Exit Thresholds	Error magnitude and slope thresholds to control entry into and exit out of ATRH control. Magnitude: 0 – 318.75 mV, 1.25 mV. Slope: 0 – 0.492 mV/ nsec, 7.8 μV/nsec
Frequency Boost	Gain control to adjust the Pulse Frequency Modulation (PFM) during ATRH control. $\Delta T_{step} = K_p \times \text{error} + K_d \times \text{error slope}$
Control Current Boost	Gain control to adjust control current to each phase during ATRH control. $\Delta I_{pid} = K_p \times \text{error} + K_d \times \text{error slope}$
ATRH Kp	Proportional Gain term applied to the control current during ATRH control.
ATRH Ki	Integral Gain term applied to the control current during ATRH control. Only reacts

Application information

Control	Description
	to positive error
ATRH Kd	Differential Gain term applied to the control current during ATRH control.
ATRH Lift	Positive voltage offset applied during ATRH control.
Frequency Boost Increase	Frequency Boost strength control versus load oscillation frequency. Strength increase: 1/8 – 7/8 every 1/8 switching cycle, 0 disables control.
Control Current Decrease	Frequency Boost strength control versus load oscillation frequency. Strength decrease: 1/8 – 7/8 every 1/8 switching cycle, 0 disables control.
ATRL Entry & Exit Thresholds	Error magnitude and slope thresholds to control entry into and exit out of ATRL control. Magnitude: 0 – 318.75 mV, 1.25 mV. Slope: 0 – 0.492 mV/ nsec, 7.8 μV/nsec
Diode Braking Max Tri-state Time	Maximum allowable tri-state time during diode braking. 320 nsec – 2240 nsec, 320 nsec increments.
Diode Braking Frequency Threshold	Load oscillation frequency above which diode braking is disabled. 49 kHz – 12.5 MHz, 200 Hz, 0 disables body braking

8.9.2 FATR

The load oscillation frequency is measured by the controller using the FATR detectors. The detectors measure the time between each excursion of the output voltage error beyond a programmable OV and UV threshold. Two sets of detectors and a frequency threshold are available to allow for robust detection of higher load oscillation frequencies where the error voltage starts to become very small.

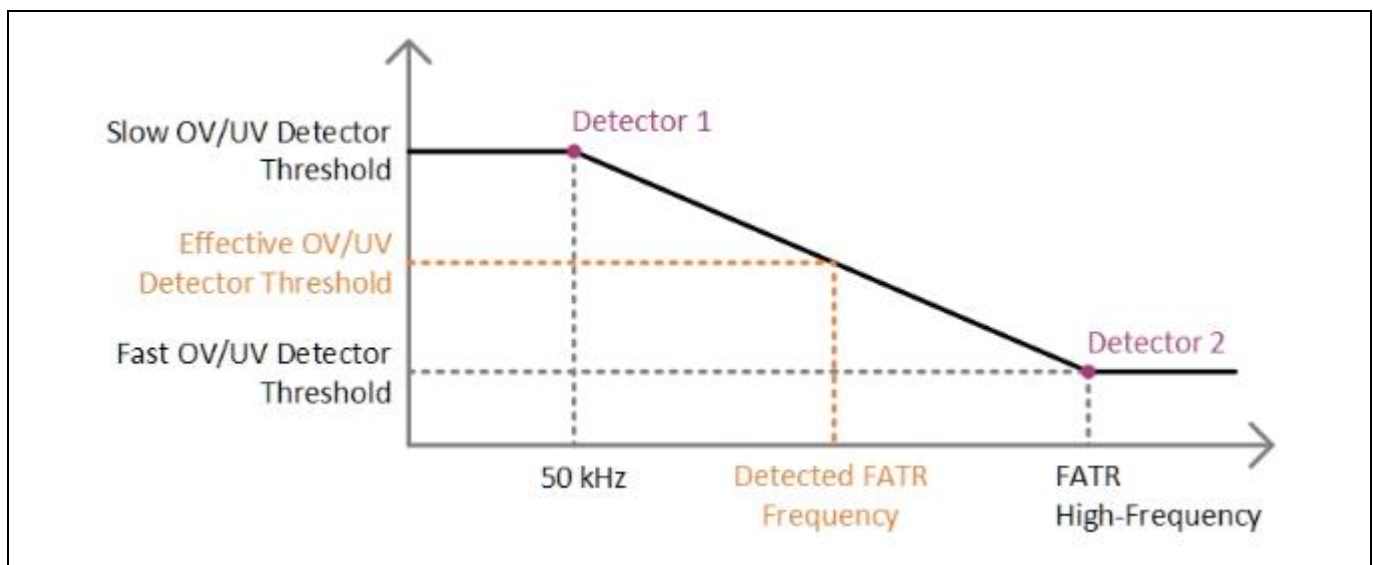


Figure 17 Load oscillation frequency detector

Table 27 FATR controls

Control	Description
OV/UV Detector1 Threshold	OV/UV error voltage thresholds for load oscillation frequencies below 50kHz
FATR High Frequency Threshold	Frequency threshold above which OV/UV Detector2 Thresholds will be used.
OV/UV Detector2 Threshold	OV/UV error voltage thresholds for load oscillation frequencies above High-Frequency Threshold

Application information

Control	Description
Body Braking Disable Threshold	Frequency threshold above which body braking (ATRL Hi-Z) is disabled
ATRL Disable Threshold	Frequency threshold above which ATRL is disabled
Auto Phasing Disable Threshold	Frequency threshold above which auto phase drop is disabled
HSPB Enable Threshold	Frequency threshold above which High-speed Phase Balance (HSPB) is enabled
FATR PID Enable Threshold	Frequency threshold above which FATR PID settings are used
FATR Vout Offset Enable Threshold	Frequency threshold above which FATR Vout Offset is enabled
FATR Vout Offset	Voltage offset used when load oscillation frequency above the FATR Vout Offset Enable Threshold

8.10 High-speed phase current balance (HSPB)

In addition to the inherent phase current balance in current-mode control, the controller features High-speed Phase Balance control which is particularly valuable at load oscillation frequencies around the voltage regulator switching frequency. This control reduces “Beat Frequencies” that occur when the load oscillation frequency is equal to the switching frequency.

The HSPB control monitors the peak current of each phase and compares it to the average current of all the active phases. When the peak current of one phase is greater than the current imbalance threshold for a programmable number of switching cycles, and the load oscillation frequency is greater than the HSPB Enable Threshold, the switching frequency will move to an alternate, and typically higher switching frequency. When switching at the alternate frequency and an imbalance occurs the switching frequency will revert back to the original frequency setting. It will also revert back when the load oscillation frequency goes below the HSPB Enable Threshold.

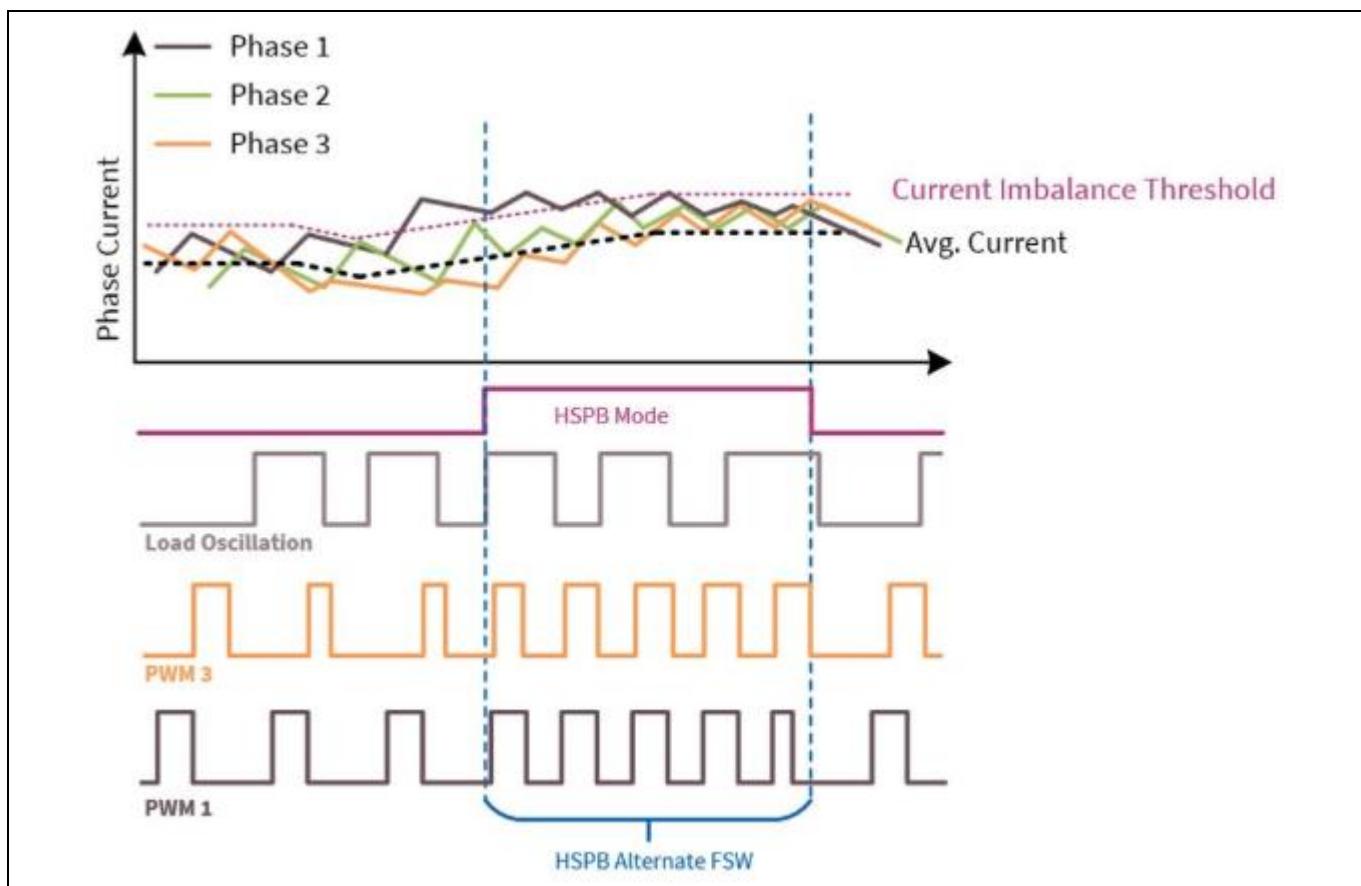


Figure 18 HSPB response

8.11 Telemetry

The XDPE1A2G5 can provide real-time accurate measurement of input voltage, input current, output voltage, output current, phase current, and temperature, and provides this information over PMBus.

8.11.1 Telemetry sample rate, filter bandwidth, and range/resolution

Table 28

Telemetry	Register	Sensing/ Calculation	Sample Rate (Typical)	Filter Bandwidth ¹⁷ (Typical)	Range/Resolution
Input Voltage (Local)	PMBus: READ_VIN	VINSEN	125 kHz	1.9 kHz	0 – 63.75 V/ 0.25 V
Input Current (Loop)	PMBus: READ_IIN	IIN	125 kHz	15 kHz	0 - 127.875 A/ 0.125 A
Input Power (Loop)	PMBus: READ_PIN	VIN x IIN	125 kHz	15 kHz	0 – 8184 W/ 1, 2, 4, 8 W
Output Voltage	PMBus: READ_VOUT	VOSEN_Lx – VORTN_Lx	50 MHz	Bypass	VOUT_MAX/ VOUT_MODE
Output Current	PMBus: READ_IOUT	Sum of ISENx	25 MHz	4.86 kHz	0 – 8184 A/ 1, 2, 4, 8 A Note 18
Output	PMBus: READ_POUT	VOUT x IOUT	25 MHz	5.8 kHz	0 – 4092 W/

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Telemetry	Register	Sensing/ Calculation	Sample Rate (Typical)	Filter Bandwidth ¹⁷ (Typical)	Range/Resolution
Power					1, 2, 4 W
Phase Current	lphx_adc_lpf	ISENx	25 MHz	486 Hz	-43.4 A – 179.2 A/ 0.022 A
Temperature (External)	PMBus: READ_TEMPERATURE_1	TSEN1/TSEN2	125 kHz	478 Hz	-40 C – 215 C/ 1 C
Temperature (Internal)	PMBus: READ_TEMPERATURE_2	Die Temperature	125 kHz	1.2 Hz	-40 C – 215 C/ 1 C
Auxiliary Voltage	Vaux_supply	VAUX	125 kHz	478 Hz	0 – 8.125 V/ 0.125 V

Note:

17. Filter Rates are programmable. Range is a function of the sample rate. See Section 8.11.2

18. Output Current range/resolution is user settable with register tlm_lp_iout_scale

8.11.2 Programmable filter bandwidth

The filter bandwidth for each telemetry is fully programmable. The bandwidth is a function of the update rate, Table 29 shows the list of telemetry and their filter bandwidth programming range. The low-pass filter can be bypassed for any telemetry input by setting the filter index to 0.

Table 29 Programmable filter bandwidth range

Telemetry	Filter Bandwidth Range
Input Voltage	2.4 Hz – 11.78 kHz
Input Current (Loop)	2.4 Hz – 11.78 kHz
Input Current (System)	2.4 Hz – 11.78 kHz
Input Power (Loop)	2.4 Hz – 11.78 kHz
Input Power (System)	2.4 Hz – 11.78 kHz
Output Voltage	1.942 kHz – 9.42 MHz
Output Current	486 Hz – 2.355 MHz
Output Power	1.942 kHz – 9.42 MHz
Phase Current	486 Hz – 2.355 MHz
Temperature (External)	1.2 Hz – 5.89 kHz
Temperature	1.2 Hz – 5.89 kHz

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Telemetry	Filter Bandwidth Range
(Internal)	
Auxiliary Voltage	19 Hz – 94.2 kHz

8.12 Fault handler

The XDPE1A2G5 controller provides a comprehensive fault detection system. Reporting and handling features include:

- Controller VDD under-voltage lock out (VDD UVLO)
- Driver PWM faults with Infineon integrated power stages
- Driver not ready – Tmon fault with Infineon integrated power stages
- Driver fault with Infineon integrated power stages
- Input under/over-voltage protection (IUVP/IOVP)
- Input over-current warning and fault (IOCP)
- Input over-power warning (IOPP)
- Output under/over-voltage protection (OUVP/OOVP)
- Output Over-current protection (OOCPP)
- Pulse-by-pulse peak current limit protection (P2CL)
- Output over-power warning (OOPP)
- Over temperature protection (OTP)
- Temperature Warning (OT_WARN)
- Configuration (CRC) failure
- Communication failure (CML)

All individual faults are latched and can be reported over the PMBus. A configurable fault signal, FAULT1, is provided for fault reporting. The fault conditions available for reporting on this pin are listed in Table 30. Each fault output signal supports individual masking of any fault, allowing a subset of faults to be reported over that signal. The fault signals can also be configured as active high or active low polarity, and CMOS level driven or open drain outputs.

Table 30

Protection	PMBus Command/Register	Range/Resolution	Fault Reporting	VR Fault Response
VIN UVLO	VIN_OFF	0 – 63.75 V/ 0.25 V	STATUS_INPUT[3] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
VIN UV Warn	VIN_UV_WARN_LIMIT	0 – 63.93 V/ 0.0625 V	STATUS_INPUT[4] SMBALERT#/FAULT1	Asserts alert pins
VIN OV Fault	VIN_OV_FAULT_LIMIT VIN_OV_FAULT_RESPONSE	0 – 63.93 V/ 0.0625 V	STATUS_INPUT[7] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
VIN OV Warn	VIN_OV_WARN_LIMIT	0 – 63.93 V/ 0.0625 V	STATUS_INPUT[6] SMBALERT#/FAULT1	Asserts alert pins
IIN Fault	IIN_OC_FAULT_LIMIT IIN_OC_FAULT_RESPONSE	0 – 127.875 A/ 0.125 A	STATUS_INPUT[2] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
IIN Warn	IIN_OC_WARN_LIMIT	0 – 127.875 A/ 0.125 A	STATUS_INPUT[1] SMBALERT#/FAULT1	Asserts alert pins

Application information

Protection	PMBus Command/Register	Range/Resolution	Fault Reporting	VR Fault Response
Loop PIN Warn	PIN_OP_WARN_LIMIT	0 – 8184 W/ 1, 2, 4, 8 W	STATUS_INPUT[0] SMBALERT#/FAULT1	Asserts alert pins
VOUT OV fixed	VOUT_OV_FAULT_LIMIT VOUT_OV_FAULT_RESPONSE VOUT_OV_WARN_LIMIT	0 – VOUT_MAX/ VOUT_MODE	STATUS_VOUT [7:6] SMBALERT#/FAULT1	Fault: shutdown with LS FET ON Warn: Asserts alert pins
VOUT OV Tracking	MFR_OVP_RELATIVE_THRESH VOUT_OV_FAULT_RESPONSE	50 – 500mV/ 50mV Above VID	STATUS_VOUT [7] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
VOUT UV fixed	VOUT_UV_FAULT_LIMIT VOUT_UV_FAULT_RESPONSE VOUT_UV_WARN_LIMIT	0 – VOUT_MAX/ VOUT_MODE	STATUS_VOUT [5:4] SMBALERT#/FAULT1	Fault: Hi-Z Shutdown Warn: Asserts alert pins
VOUT UV Tracking	MFR_UVP_RELATIVE_THRESH VOUT_UV_FAULT_RESPONSE	50 – 500mV/ 50mV Below VID	STATUS_VOUT [4] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
IOUT OC Fault	IOUT_OC_FAULT_LIMIT IOUT_OC_FAULT_RESPONSE	0 – 2046 A/ 1, 2 A	STATUS_IOUT [7] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
IOUT OC Warn	IOUT_OC_WARN_LIMIT	0 – 2046 A/ 1, 2 A	STATUS_VOUT [5] SMBALERT#/FAULT1 VALRT#	Asserts alert pins
P2CL	p2cl_limit	0 – 178.85 A/ 0.35 A	STATUS_MFR_SPECIFIC [5] MFR_COMMON_FAULT_ STATUS1 [6] SMBALERT#/FAULT1	Truncates PWM until current is below the limit Asserts alert pins
NCL	ncl_limit	0 to -43.4 A/ 1.4 A	STATUS_MFR_SPECIFIC [5] MFR_COMMON_FAULT_ STATUS1 [6] SMBALERT#/FAULT1	Hi-Z PWM until current is above the limit Asserts alert pins
POUT Warn	POUT_OP_WARN_LIMIT	0 – 4092 W/ 1, 2, 4 W	STATUS_VOUT [0] SMBALERT#/FAULT1	Asserts alert pins
OT Fault	OT_FAULT_LIMIT OT_FAULT_RESPONSE	64 – 255 C/ 1 C	STATUS_TEMP [7] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
OT Warn	OT_WARN_LIMIT	64 – 255 C/ 1 C	STATUS_TEMP [6] SMBALERT#/FAULT1 VALRT#	Asserts alert pins
PS Fault	NA	NA	STATUS_MFR_SPECIFIC [5] MFR_COMMON_FAULT_ STATUS1 [7] SMBALERT#/FAULT1	Hi-Z Shutdown Asserts alert pins
CRC Fault	NA	NA	STATUS_CML [4] SMBALERT#/FAULT1	Asserts alert pins
Com Fault	NA	NA	STATUS_CML [1] SMBALERT#/FAULT1	Asserts alert pins

8.12.1 VDD under-voltage lockout (VDD UVLO)

An analog comparator continuously monitors the +3.3 V (VDD) supply. When the supply falls below a fixed threshold of approximately 2.7 V, the VDD UVLO fault is asserted. A VDD UVLO assertion will trigger Hi-Z shutdown mode which immediately turns off both the high and low side MOSFETs of the power stage.

8.12.2 Driver PWM Faults

Driver PWM fault is designed to detect when a MOSFET driver/power stage is missing or not powered up. Prior to regulation, the controller will check the power stage's PWM pin status. When the Infineon driver is powered up, it will drive the PWM voltage to a tri-state level of approximately 1.6 V. If the PWM pin is shorted or open, the controller will not send PWM signals out and it will report Driver PWM related faults as follows:

- PWM shorted to GND or VDRV UVLO
- PWM open

8.12.3 Driver not ready – TMON low

In addition to Driver PWM faults the controller also monitors the power stage's TMON signal. Infineon's integrated power stage TMON signal indicates whether the driver is ready and able to respond to PWM signals correctly. The controller will delay soft-start until the TMON voltage is greater than 0.28V (-40 C), ensuring that the driver has sufficient supply voltage to operate properly.

8.12.4 Input voltage fault protection (IUVP/IOVP)

The XDPE1A2G5 controller monitors the VIN (+12 V) supply with the VINSEN pin. The signal is low-pass filtered and compared against a set of user defined programmable thresholds to resolve VIN status conditions. Typical placement of the input under voltage (IUVP), the turn-on (VIN_ON), and input over voltage (IOVP) thresholds are shown in Figure 19.

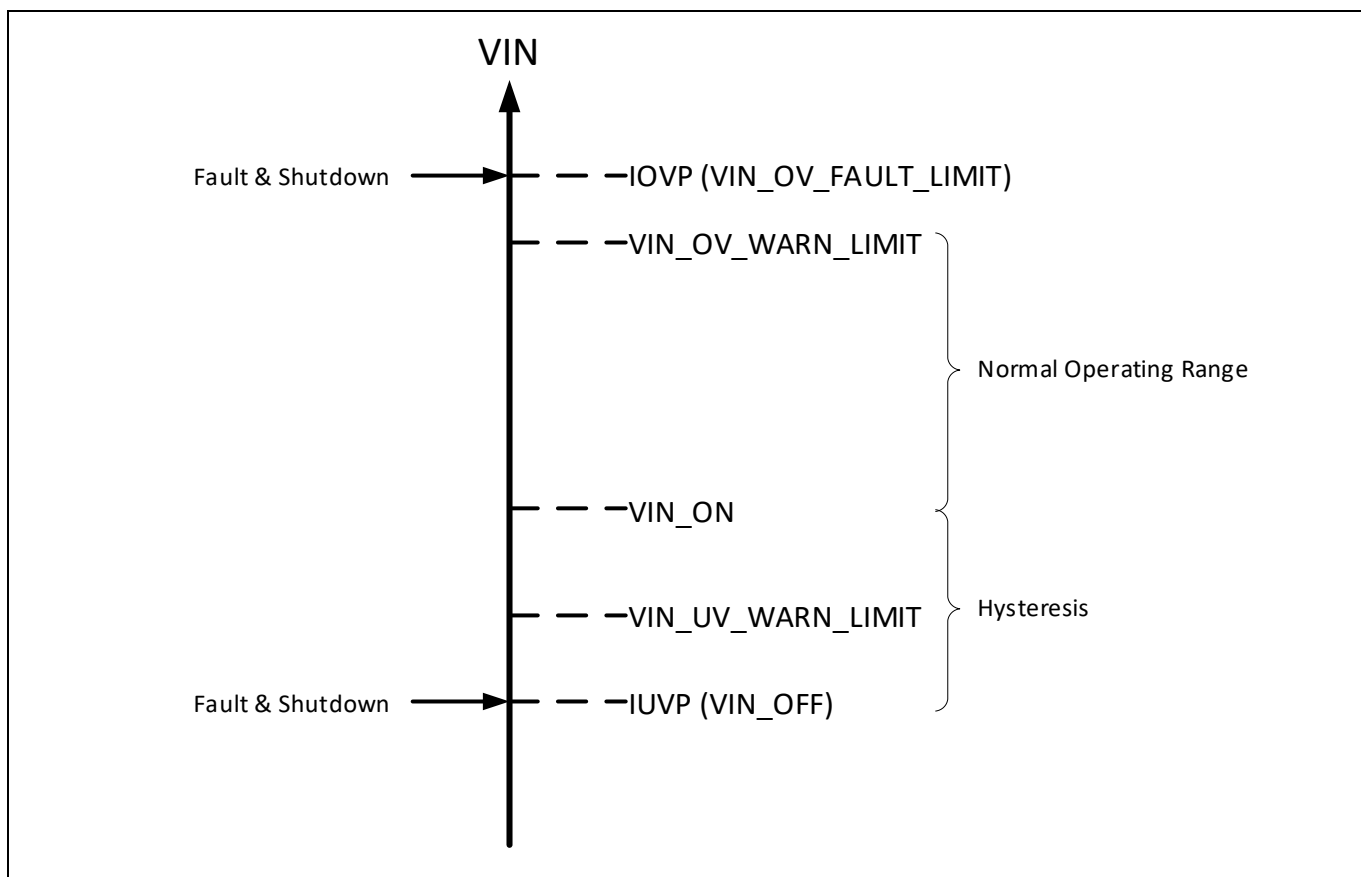


Figure 19 VIN threshold settings

VIN_ON

VIN must be greater than the VIN_ON (35h) threshold to start the system. If the controller has reached the Active Regulation state, it does not require VIN to remain above the VIN_ON threshold to maintain regulation.

Input Under Voltage Protection (IUVP)

When the controller is in the Active Regulation state, VIN dropping below IUVP threshold will assert an IUVP fault and shut down the system. IUVP detection is enabled once VIN ramps above VIN_ON threshold after the Initialization state. VIN OFF (36h) PMBus command sets the IUVP threshold.

Input Over Voltage Protection (IOVP)

When VIN is above the IOVP threshold, the controller can be programmed to shut down the system. The maximum IOVP threshold that can be set is 16.8 V.

In addition to the fault thresholds, warning thresholds can be set with VIN_OV_WARN_LIMIT (42h) and VIN_UV_WARN_LIMIT (43h). These can be used to alert of over-voltage or under-voltage conditions occurring close to the fault shutdown limits.

8.12.5 Input current protection (IOCP)

The controller can be configured to estimate each loop input current. Each loop input current can be set to be:

- Estimated Loop A current
- Estimated Loop B current

Application information

- Estimated sum of Loop A and Loop B current

An IIN over-current fault is triggered when the estimated loop current exceeds the fault threshold set by the IIN_OC_FAULT_LIMIT (5Dh). If the response to the fault is set to shut down the PWMs will go Hi-Z. A warning threshold, IIN_OC_WARN_LIMIT (58h), can be used to flag that an over-current condition is occurring but not shutdown voltage regulator. This is useful for communicating with the system to reduce its load so that the maximum input power is not exceeded.

8.12.6 Output over & under-voltage protection (OVP/OUVP)

Output over and under-voltage protection can be set to both a tracking and a fixed threshold as shown in Figure 20. A tracking feature allows thresholds to be set relative to the target voltage. The programmable thresholds are between 50 mV to 800 mV in 50 mV steps above and below the target voltage. Tracking thresholds are enabled during the soft-start and active regulation states. PWMs will go Hi-Z when either tracking threshold are exceeded.

The fixed thresholds are an absolute voltage with selectable values between 0 V to VOUT_MAX. Fixed OVP detection is active after the initialization state. When a fixed OV condition is detected, the PWMs are forced low to turn on the Low-side FETs to clamp the output voltage through the inductors. Fixed OUVP is active during the active regulation state. PWMs will go Hi-Z when Vout drops below the set threshold.

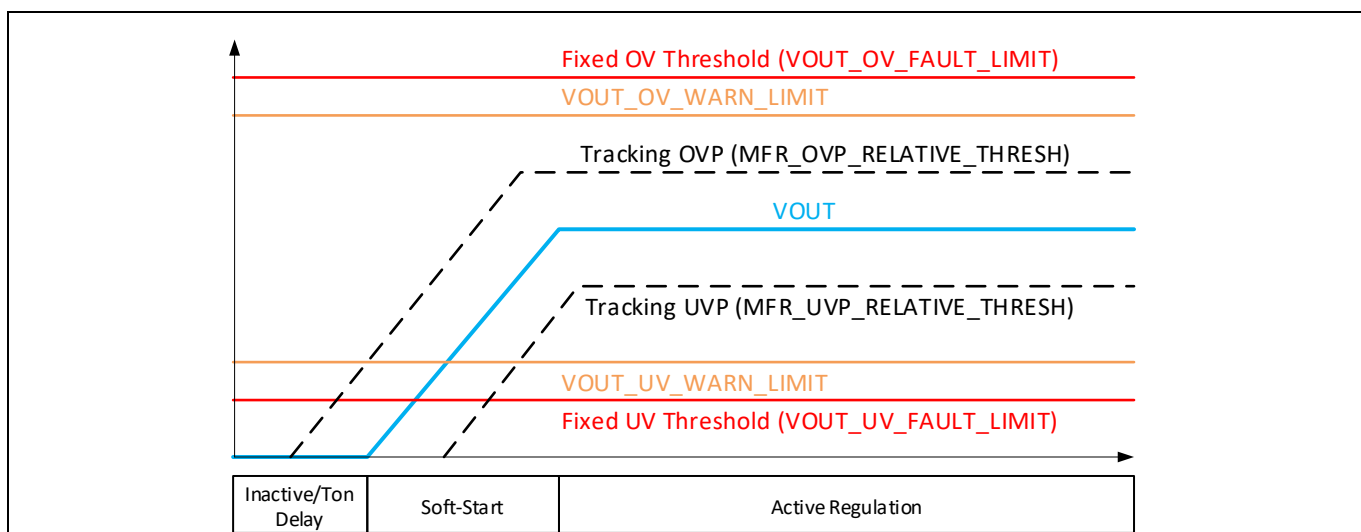


Figure 20 Tracking and Fixed OV and UV Protections

8.12.7 Output over-current protection (OOCF)

The controller calculates the total output current by summing the sampled phase currents from all active phases. The total current signal used for over-current detection is filtered through a programmable filter that is settable to four selections (63, 210, 380, 532 kHz). This provides the user the ability to tailor the required response time to both a fault shutdown and the assertion of a warning signal. Additional control of the response time is controlled via a fault and warning counter that is programmable from 0 to 31 1MHz clock cycles in 1 cycle increments. Typically, the fault limit, IOUT_OC_FAULT_LIMIT (46h), is set 120% higher than the expected maximum current, which is defined by the application/user. The warning limit is defined by IOUT_OC_WARN_LIMIT (4Ah).

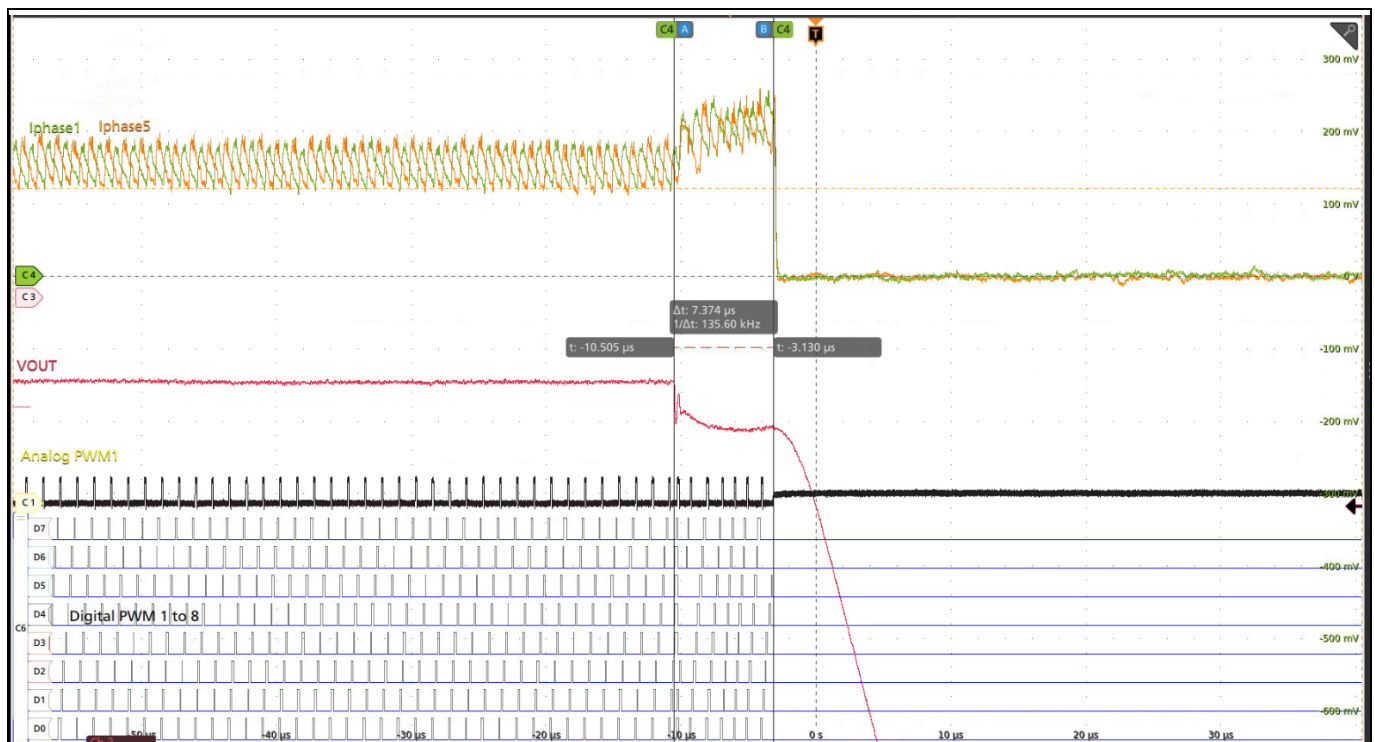


Figure 21 Shutdown response to over-current fault (63 kHz Filter Index, Delay Count = 0)

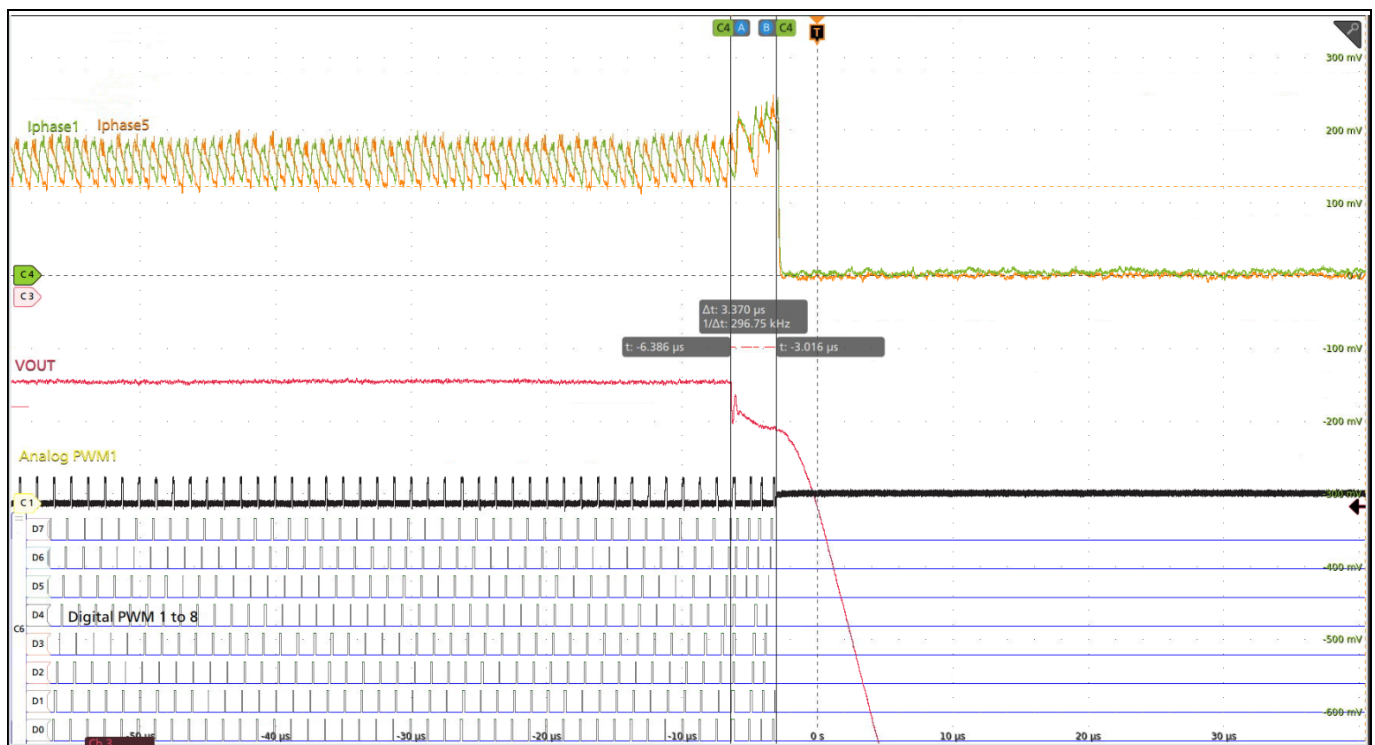


Figure 22 Shutdown response to over-current fault (532 kHz Filter Index, Delay Count = 0)

8.12.8 Per phase current protection (P2CL, NCL)

P2CL

Per phase pulse-by-pulse current limiting prevents inductor saturation by preventing the individual phase currents from exceeding a user-programmed threshold. The threshold can be set up to 178 A in 0.35 A increments. The width limit of the PWM pulse is calculated based on the synthesized inductor current. Once the inductor current exceeds the maximum threshold limit, the PWM pulse is truncated to turn on the low-side FET for the rest of the PWM cycle. The P2CL fault monitors each phase independently.

P2CL can be used to clamp the output current to insulate devices at the input side from delivering excessive power. The controller has been designed to ride thru peak current events. The VR will assert the P2CL flag when the peak is 10A below the set value. The VR will increase the FSW 1.625x as current continues to increase. From 500K up to 800K for the event depicted in Figure 23.

This allows a system manager to receive a P2CL flag and throttle operation accordingly before VOUT droops to a dangerous level

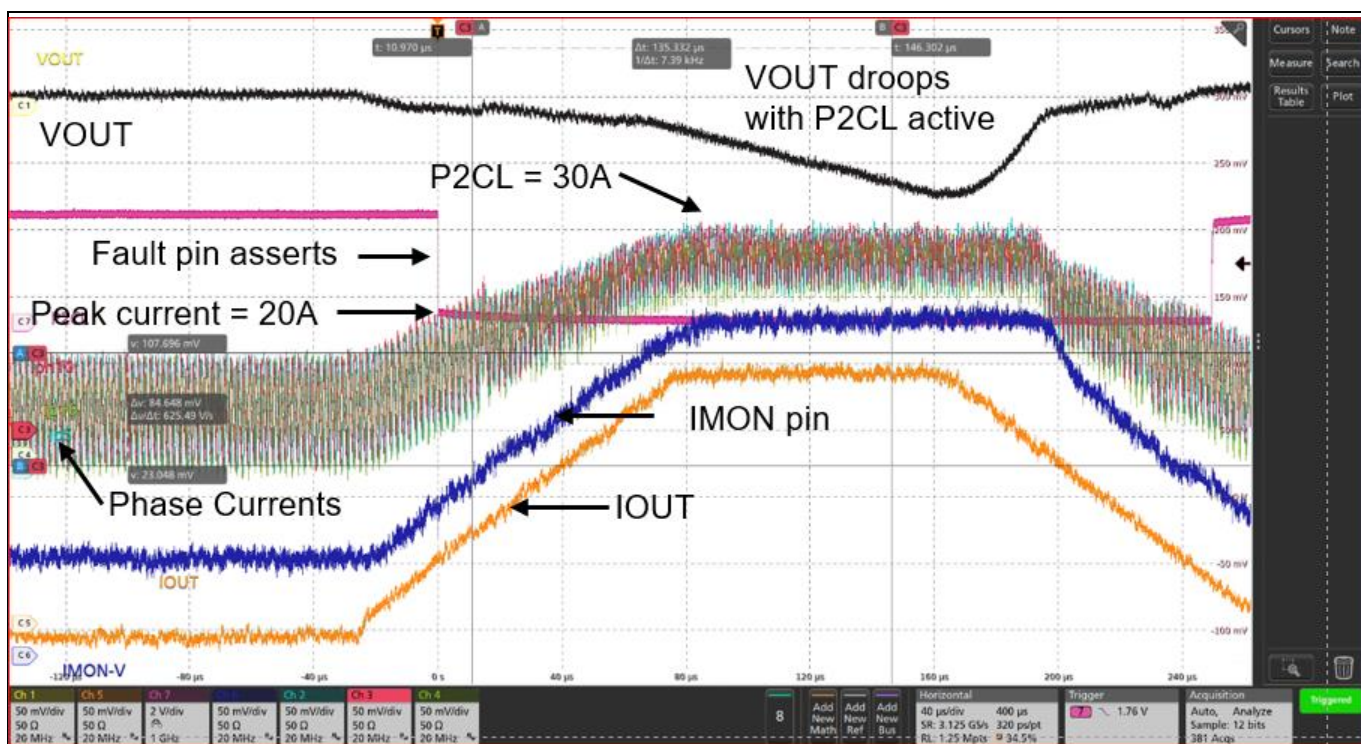


Figure 23 **P2CL Response**

NCL

The controller constantly monitors the inductor ripple current of each phase. Upon detection of negative current in the inductor exceeding a user-programmed threshold (up to -43.4 A in 1.4 A increments), the controller will set the PWM output of that phase to Hi-Z in order to reduce the negative current. The prevention of very high negative currents avoids potential damage to the high-side FET body diode during switching cycles.

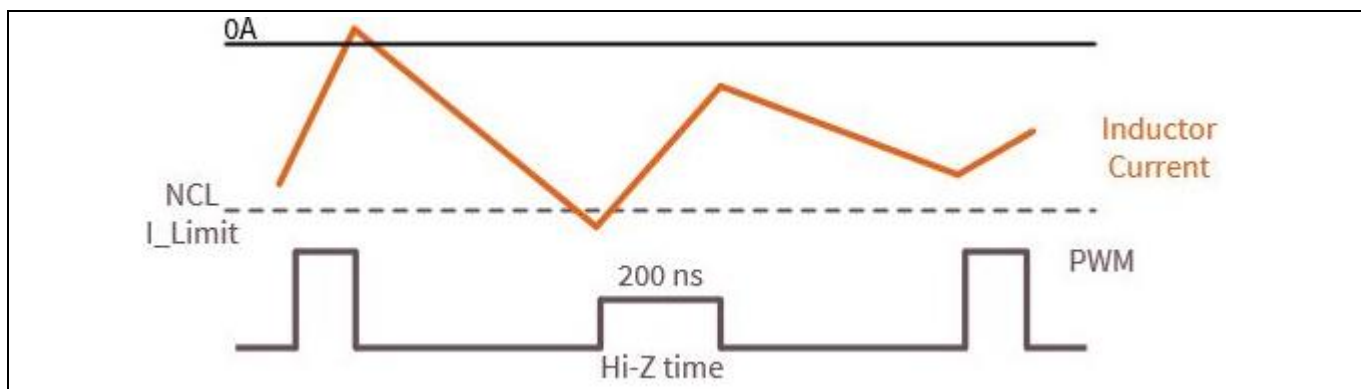


Figure 24 NCL Response

8.12.9 Output over-power protection (OOPP)

The controller monitors both the output current and the output voltage and uses this information to calculate the output power. An Output Over-power Warning will be issued when the power exceeds the user programmable threshold, POUT_OP_WARNING_LIMIT (6Ah). Exceeding this limit does not cause a shutdown. Only the warning bit in VOUT_STATUS bit 0 and any configured warning outputs will be asserted.

8.12.10 Over-temperature protection (OTP)

Over Temperature Protection (OTP) is triggered when the operating temperature reported on TSEN1/2 exceeds the user-programmed shutdown threshold, OT_FAULT_LIMIT (4Fh).

When selected for shutdown, an OTP fault will trigger a Hi-Z shutdown. Temperature protection is enabled after the Initialization state.

8.12.11 Power stage fault detection

In addition to directly interfacing with Infineon Power Stage to monitor temperature, the TSENx input is used to monitor a Power Stage fault condition. Any fault within the Power Stage is communicated to the controller by pulling its temperature sense output (TMON) to 3.3V. A 3.3V at the controller's TSENx pin indicates one of the following has occurred in the Power Stage:

- 1) A power stage High-side FET Short (HSS)
- 2) An over temperature fault (OT)
- 3) A persistent overcurrent fault (OC)
- 4) High-side driver voltage UV fault (Boot UVLO)

Additionally, a Power Stage fault condition, VDRV UV fault, can be communicated to the controller by pulling its current sense output (IMON) to ground.

The controller can shut down or continue operating and assert the SMB_ALERT# and/or FAULT1 pin upon receiving the power stage fault.

The controller can be configured to accommodate future Power Stage Fault ID implementations. Consult with local Infineon representative for availability and enablement.

Application information

8.12.12 CRC failure

The integrity of downloading the configuration from the NVM to the controller's control registers is assured by CRC check. A configuration download CRC failure prevents the controller from leaving the Inactive state.

8.12.13 Communication fault

When receiving unimplemented commands or improperly structured I2C protocol, such as PEC (packet error checking) faults, the controller will report communication failures.

8.12.14 Fault response control

The controller can configure the response to the detected shutdown faults. Depending on the detected fault, these responses can be set to; Ignore, Shut Down (latch), Shut Down (Retry), and Shut Down & Resume. Additional control is available to delay the shutdown as well as delay the time the output is restarted. Table 31 shows a list of faults and their configurable response

Table 31 Configurable fault response

Fault	Configurable Response
Vout Over-voltage	Ignore, Shutdown & Retry (No retry, 1 to 6 times, continuously) , Shutdown & Resume when OK
Vout Under-voltage	Ignore, Shutdown & No Retry
Iout Over-current	Shutdown & Retry (No retry, 1 to 6 times, continuously)
Over-temperature	Ignore, Shutdown & Retry (No retry, 1 to 6 times, continuously), Shutdown & Resume when OK
Vin Over-voltage	Ignore, Shutdown & Retry (No retry, 1 to 6 times, continuously), Shutdown & Resume when OK
Iin Over-current	Ignore, Shutdown & Retry (No retry, 1 to 6 times, continuously), Shutdown & Resume when OK
Ton Max	Ignore, Shutdown & Retry (No retry, 1 to 6 times, continuously), Shutdown & Resume when OK
Power Stage	Ignore, Shutdown & No Retry (Set through Fault Enable Mask)

8.12.15 Fault pin mapping

The controller has a Fault Path that can be mapped to any Loop or Common fault. Pin 17 can be configured as the FAULT# output. Each fault or warning can be masked for the Fault Path in order to control the assertion of the Fault Output.

Table 32 FAULT Common Faults

Fault	
VDD UVLO Fault	POR VDDA Fault

Table 33 FAULT Loop Faults

Fault	
Vout OV Fault	Vout OV Warn
Vout UV Fault	Vout UV Warn

Application information

Fault	
Vin OV Fault	Vin OV Warn
P2CL Limit	Vin UV Warn
Iout OC Fault	Iout OC Warn
Vout UV Tracking Fault	Vout OV Tracking Fault
Iin OC Fault	Iin OC Warn
Over-temperature Fault	Over-temperature Warning
Pin OP Warn	Pout OP Warn
Vout Min/Max Warning	TMON High
IMON low	

8.13 Active voltage positioning (AVP)

The controller provides programmability of a linear DC load line slope via VOUT_DROOP (28h) register. The linear DC load line slope is programmable from 0 to 15.98 mΩ linear 11 mode with least significant bits (LSB) in steps ranging from 15.6 μΩ to 2.4 μΩ steps. The actual load line LSB is set in the vc_vavp_lsb_sel register. The load line LSB can be set to 19.5 μΩ, 9.8 μΩ, 4.8 μΩ, or 2.4 μΩ.

Additionally, a non-linear DC load line slope can be programmed. The slope of the non-linear load line can be broken into two additional segments. The load line slope of each segment is independently programmable 0 to 15.98 mΩ in steps set by the vc_vavp_lsb_sel register and each segment slope will become effective when the load current exceeds the programmable current thresholds as shown in Figure 25.

Linear DC load line is used to reduce peak overshoot voltages during load transient and helps reduce the required output capacitance. The non-linear load line shape shown in Figure 25 is typically used in applications whose transient load currents normally operate within the segment2 load line slope region. This allows for added overshoot margin during a load release typically seen in traditional DC load line applications, but it also limits both the maximum and minimum output voltage over the entire load range.

Table 34

Vc_vavp_lsb_sel	Resolution	Comments
0	5/256 mΩ	19.53125 μΩ
1	5/512 mΩ	9.765625 μΩ
2	5/1024 mΩ	4.8828125 μΩ
3	5/2048 mΩ	2.44140625 μΩ

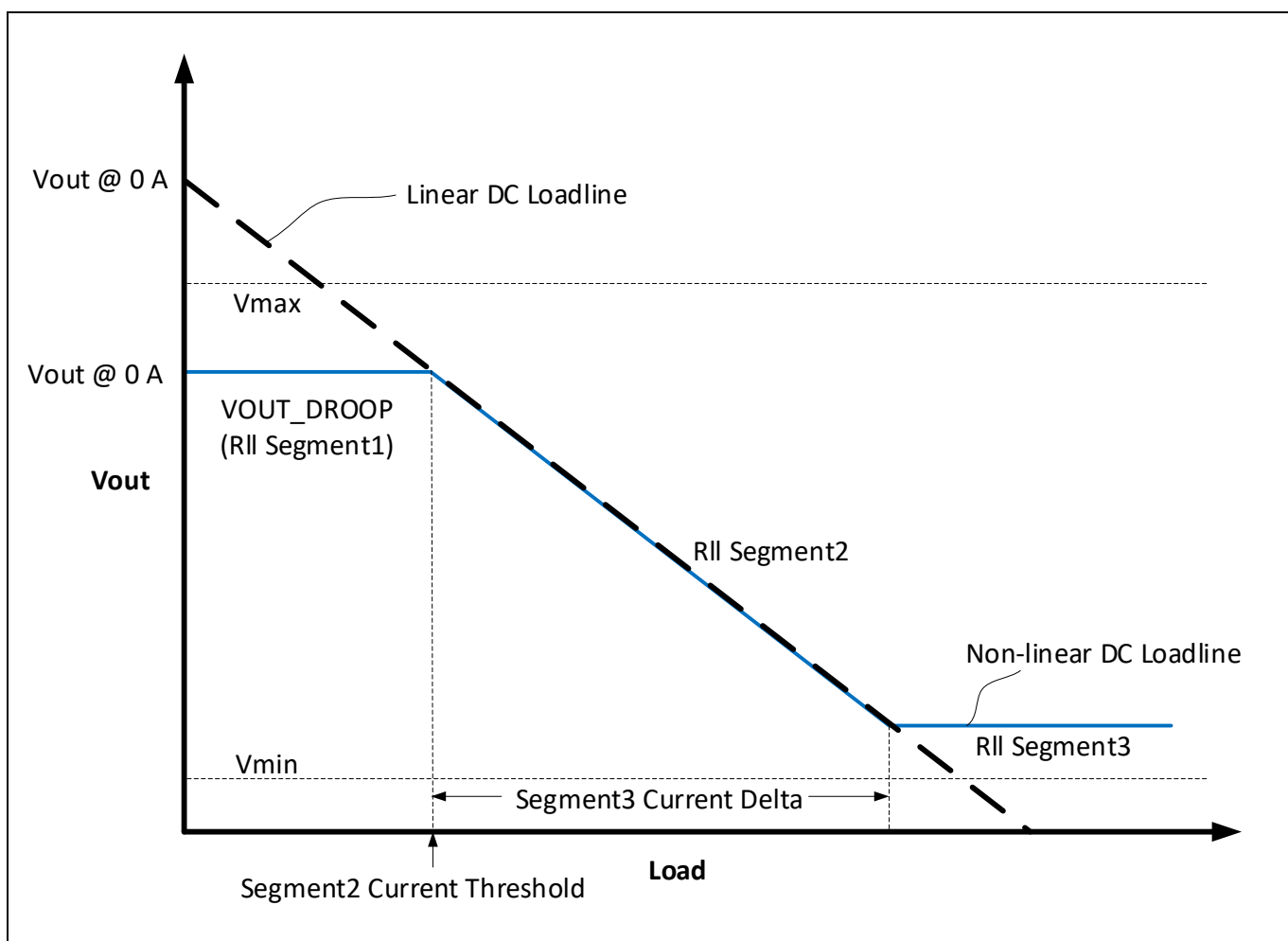


Figure 25 Linear and Non-linear DC Load line slopes

The total current is filtered with a programmable single pole filter to set the AVP bandwidth. This bandwidth can be independently set between 7.8 kHz and 2.2 MHz in 8.3 kHz steps for the load step and the load release.

For applications which cannot tolerate the DC voltage drop introduced by the load line at heavy loads, the controller can still provide some load line benefits with its AC load line feature. Instead of dropping the DC target based on the load current, AC load line temporarily adds this load line offset through a programmable band pass filter. At either low or very high load transient frequencies, the load line is blocked by the filter. At near midband, a non-zero load line is in effect. Figure 26 illustrates the responses to a load step/release sequence with different load line settings.

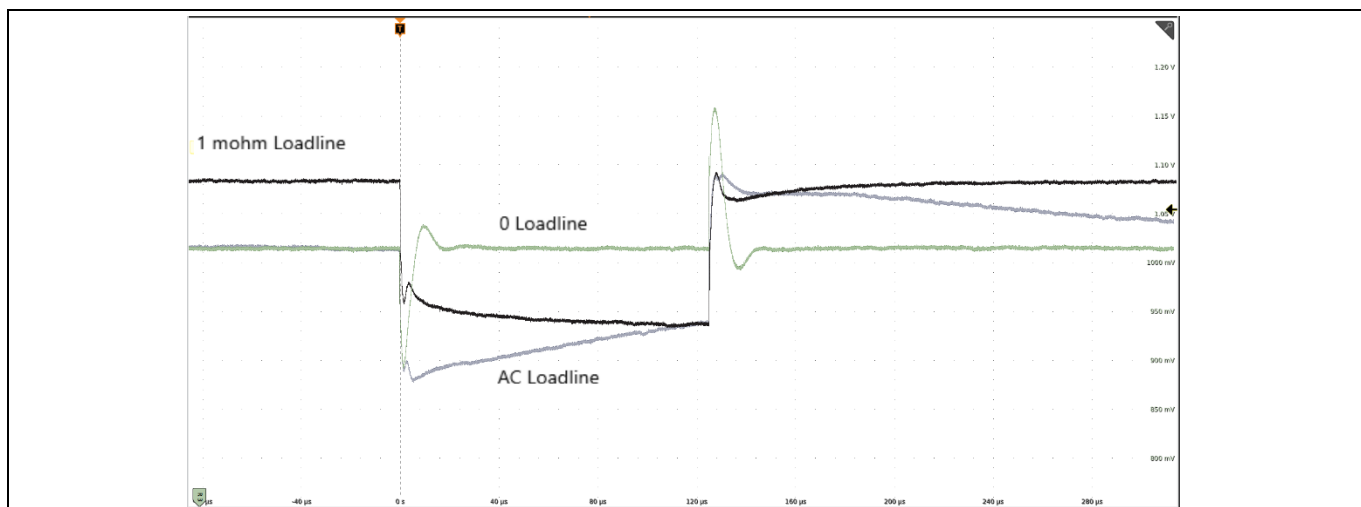


Figure 26 No load line vs DC load line vs AC load line

When using the non-linear load line feature, each segment could have different enough load lines that it could affect the required loop compensation coefficients that make for a stable loop. Separate Kp and Kd terms are available to allow the user to optimize the loop compensation for each segment. Drastic differences in load line, i.e. $R_{ll} = 0 \text{ m}\Omega$ versus $R_{ll} = 1 \text{ m}\Omega$, will typically require different Kd terms, where Kd is much less for the 1 m Ω load line case. In contrast, shallow load lines, i.e. $R_{ll} = 0.25 \text{ m}\Omega$, will typically have Kp and Kd terms set to be the same as for a 0 m Ω load line case.

8.14 Multi-function pins

The controller provides various digital IO pins that can be configurable to support several controller interface input and output signals. In addition to the input/output functionality being configurable, the output type (driven vs. open drain) and polarity are also user-programmable. Table 35 shows the available multi-functional pins and their described functions.

Table 35 Multi-functional pin descriptions

Pin Name/Function	Description
SMBALERT# (pin 17)	
SMBALERT#	Alert output for PMBus fault and warning flags
LPM#	Low Power Mode output. Used to de-assert power stage enable to place VR in low power state when VR_EN is low or in lower power state
FAULT#	Mapped indicator for generic fault. See Table 32 and Table 33
EN_L1 (pin 19)	
EN_L1	VR enable for loopA
Common Enable	Common VR enable for loopA and loopB
TSEN2 (pin 50)	
TSEN2	Second temperature sense
VAUX	Auxiliary voltage sense
Not used	No connection
GPIO (pin51)	

Application information

Pin Name/Function	Description
GPIO	Loop A LPM# Loop B LPM# Loop B Power Good
Not used	no connection

8.15 PMBus interface

All operating parameters in the XDPE1A2G5 controller are configurable via the SMBus/I2C serial interface (SDA, SCL). The digital interface also allows for monitoring of fault status, as well as voltage, current, power, and temperature telemetry. The XDPE1A2G5 controller supports I2C bus speeds of 1 MHz and logic levels of 1.8 V or 3.3 V. The default serial address of the XDPE1A2G5 controller I2C port is resistor programmable through XADDR pin as shown in Table 36. Other serial address options are available and can be programmed into device.

Table 36 Device PMBus Target Address (ADDR pin 48 Resistor 1% -to-GND)

R _{XADDR}	Device address <7-bit Addr + 0> (default +2 stride)	Device address <7-bit Addr + 0> +1 stride	Device address <7-bit Addr> (default +2 stride)	Device address <7-bit Addr + 0> +1 stride
≥ 49.9 kΩ	0xB0	0xB0	0x58	0x58
38.3 kΩ	0xB4	0xB2	0x5A	0x59
29.4 kΩ	0xB8	0xB4	0x5C	0x5A
22.6 kΩ	0xBC	0xB6	0x5E	0x5B
17.4 kΩ	0xC0	0xB8	0x60	0x5C
13.3 kΩ	0xC4	0xBA	0x62	0x5D
10.2 kΩ	0xC8	0xBC	0x64	0x5E
7.87 kΩ	0xCC	0xBE	0x66	0x5F
6.04 kΩ	0xD0	0xC0	0x68	0x60
4.64 kΩ	0xD4	0xC2	0x6A	0x61
3.57 kΩ	0xD8	0xC4	0x6C	0x62
2.67 kΩ	0xDC	0xC6	0x6E	0x63
2 kΩ	0xE0	0xC8	0x70	0x64
1.47 kΩ	0xE4	0xCA	0x72	0x65
1.07 kΩ	0xE8	0xCC	0x74	0x66
≤ 680 Ω	0xEC	0xCE	0x76	0x67

The serial interface supports the following protocols as defined in the System Management Bus (SMBus) Specification (version 2.0 August 3, 2000).

- Write byte
- Read byte
- Send byte
- Write word
- Read word
- Read block
- Write block

Application information

8.16 PMBus protocol

Infineon's XDP™ controllers are compliant to PMBus™ Power System Management Protocol Specification, revision 1.4.

To access the device's configuration and monitoring registers, 4 different protocols are required:

- the PMBus Send Byte protocol with/without PEC (for CLEAR_FAULTS)
- the PMBus Read/Write Byte/Word protocol with/without PEC
- the PMBus Block Read and Block Write protocols with Byte Count = 2, 4, 6, 8, 14, 16, and 31
- the PMBus Block Read Process call (for SMBALERT Mask)

An explanation of which command codes and protocols are required to access them is given in Table 37.

In addition, the XDPE1A2G5 supports:

- Alert Response Address (ARA)
- Bus timeout (30ms)

Note: Clock stretching may occur during a fault

S: Start Condition
A: Acknowledge (0')
N: Not Acknowledge (1')
Sr: Repeated Start Condition
P: Stop Condition
R: Read(1')
W: Write(0')
PEC: Calculated from Command
*****: Data is optional
☐ : Master to Slave
☒ : Slave to Master

Figure 27 PMBus Protocol Legend

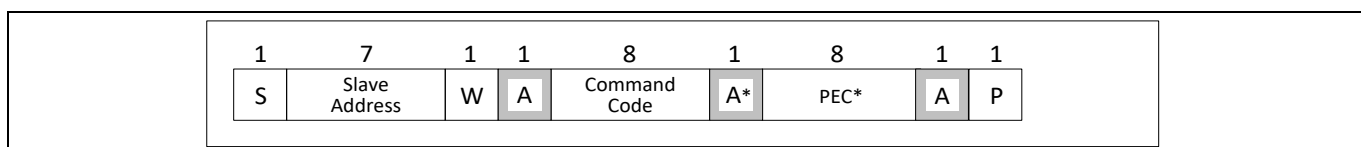


Figure 28 PMBus Send Bytes

Application information

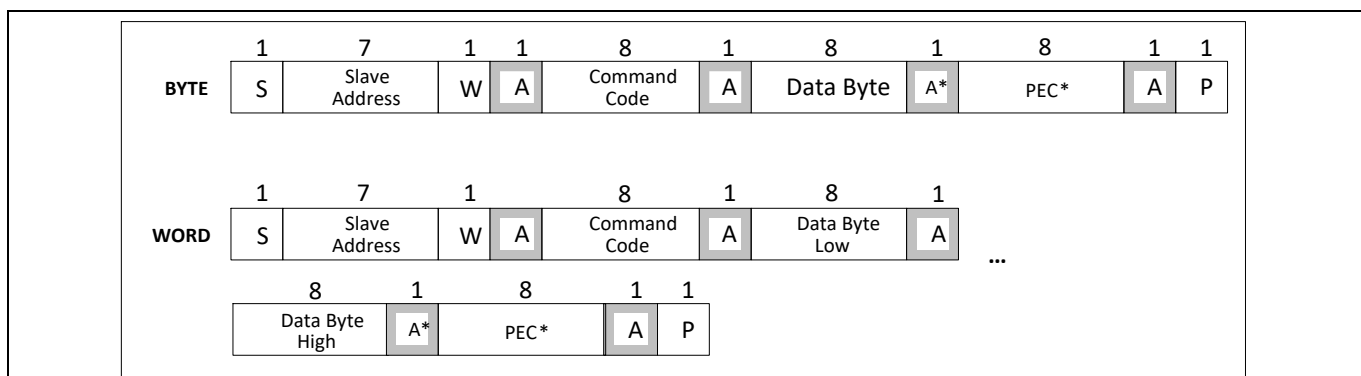


Figure 29 PMBus Write Byte/Word

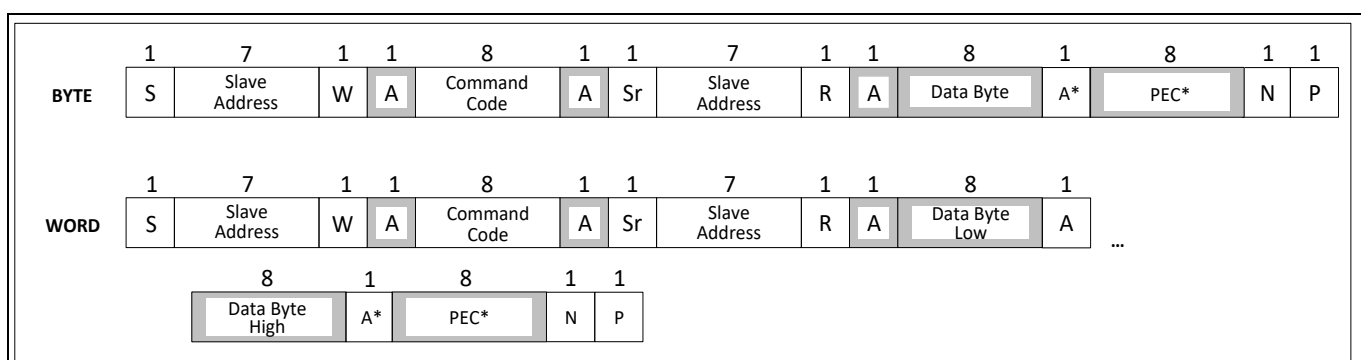


Figure 30 PMBus Read Byte/Word

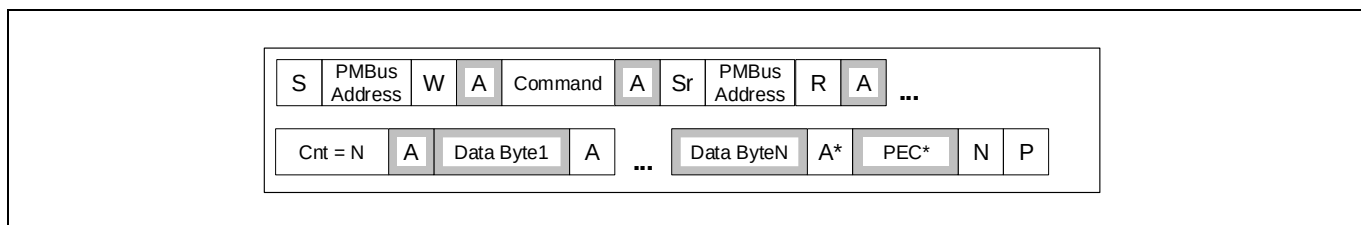


Figure 31 PMBus Block Read with Byte Count = N

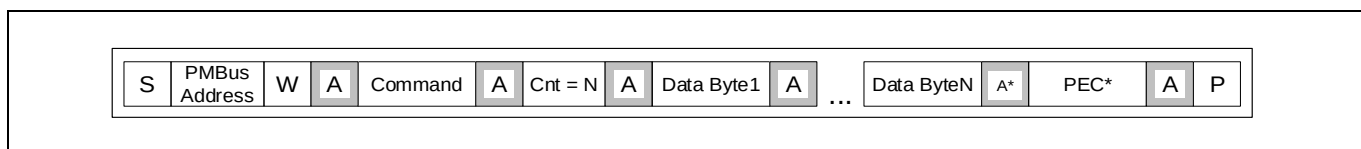


Figure 32 PMBus Block Write with Byte Count = N

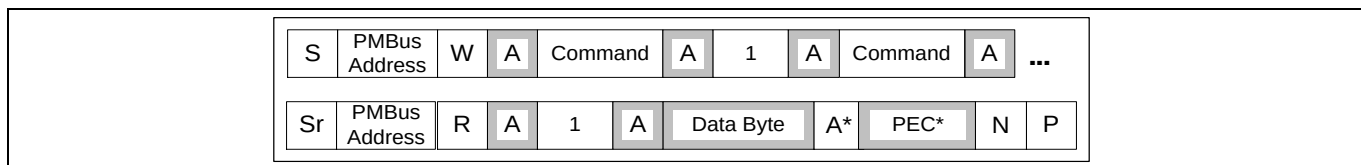


Figure 33 Block Read Process Call

Application information

8.16.1 PMBus command set

Table 37 PMBus command set

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
PAGE	Read/Write Byte	00h	Allows access of each loop via paging. Page 0 = LoopA, Page 1 = LoopB, Page FF = all pages
OPERATION	Read/Write Byte	01h	Enables or disables the output and controls margining. Ignores OVP on Margin High, UVP on Margin Low. Controls if VOUT_COMMAND or High-speed Interface controls the output voltage.
ON_OFF_CONFIG	Read/Write Byte	02h	Configures the combination of CONTROL pin and OPERATION command needed to turn the unit on and off.
CLEAR FAULTS	Send Byte	03h	Clear contents of Fault registers
WRITE_PROTECT	Read/Write Byte	10h	Provides protection from accidental changes
STORE_DEFAULT_ALL	Send Byte	11h	Copy the entire contents of the Operating Memory to the matching locations in the non-volatile Default Store memory.
RESTORE_DEFAULT_ALL	Send Byte	12h	Copy the entire contents of the non-volatile Default Store memory to the matching locations in the Operating Memory.
STORE_USER_ALL	Send Byte	15h	Copy the entire contents of the Operating Memory to the matching locations in the non-volatile User Store memory.
RESTORE_USER_ALL	Send Byte	16h	Copy the entire contents of the non-volatile User Store memory to the matching locations in the Operating Memory.
STORE_USER_CODE	Write Byte	17h	Copy the parameter whose Command Code matches the value in the data byte from the Operating Memory to the matching location in the non-volatile User Store memory.
RESTORE_USER_CODE	Write Byte	18h	Copy the parameter whose Command Code matches the value in the data byte from the nonvolatile User Store memory to the matching location in the Operating Memory. The value in the Operating Memory is overwritten by the value retrieved from the User Store.
CAPABILITY	Read Byte	19h	Returns 0xD0 to indicate Packet Error Checking is supported, Maximum bus speed is 1000 kHz, and the device does have a SMBALERT# pin and does support the SMBus Alert Response protocol
SMBALERT_MASK	Block Write/ Block Read Process Call	1Bh	Set to prevent warning or fault conditions from asserting the SMBALERT# signal. Write command code for STATUS register to be masked in the low byte, the bit to be masked in the High byte.

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
VOUT_MODE	Read/Write Byte	20h	Sets the resolution for VOUT related commands.
VOUT_COMMAND	Read/Write Word	21h	Sets the voltage to which the device should set the output. Format according to VOUT_MODE.
VOUT_TRIM	Read/Write Word	22h	Applies a fixed offset to the output voltage command value. Typically used by the end user. Format according to VOUT_MODE.
VOUT_CAL_OFFSET	Read/Write Word	23h	Applies a fixed offset to the output voltage command value. Typically used by the PMBus device manufacturer to calibrate a device in the factory. Format according to VOUT_MODE.
VOUT_MAX	Read/Write Word	24h	Sets an upper limit on the output voltage the unit can command. Commands above this level are clamped to the setting. Format according to VOUT_MODE.
VOUT_MARGIN_HIGH	Read/Write Word	25h	Sets the margin high voltage when commanded by OPERATION. Must be in format determined by VOUT_MODE.
VOUT_MARGIN_LOW	Read/Write Word	26h	Sets the margin low voltage when commanded by OPERATION. Must be in format determined by VOUT_MODE.
VOUT_TRANSITION_RATE	Read/Write Word	27h	Sets the rate at which the output changes voltage during regulation due to VOUT_COMMAND or MARGIN_HIGH/LOW commands. mV/ μ s; exp = -3
VOUT_DROOP	Read/Write Word	28h	Sets the slope at which the output voltage decreases or increases with increasing or decreasing output current for use with Adaptive Voltage Positioning. 5/256 m Ω per LSB.
VOUT_MIN	Read/Write Word	2Bh	Sets a lower limit on the output voltage the unit can command. Commands below this level are clamped to the setting. Format according to VOUT_MODE.
MFR_OVP_RELATIVE_THRESH	Read/Write Word	2Ch	Sets the over-voltage fault relative to the set point in mV from 50 to 400. Q = 50 mV
MFR_UVP_RELATIVE_THRESH	Read/Write Word	2Dh	Sets the under-voltage fault relative to the set point in mV from -50 to -400. Q = 50 mV
FREQUENCY_SWITCH	Read/Write Word	33h	Sets the switching frequency in kHz from 200 to 3000; Exp = 0, 1
POWER_MODE	Read/Write Word	34h	In PMBus mode, sets the device power state for Max Efficiency (Auto PS Mode), Max Power (PS0), or MFR Specific power state PS1 (1 or 2 phases) or PS2 (Diode Emulation).
VIN_ON	Read/Write Word	35h	Sets the value of the input voltage at which the unit should begin power conversion. Exp = -2.
VIN_OFF	Read/Write	36h	Sets the value of the input voltage that the unit,

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
	Word		once operation has started, should stop power conversion. Exp = -2.
IOUT_CAL_GAIN	Read/Write Word	38h	Set the current sense gain for Total Iout. 2's complement with exp = -7.
IOUT_CAL_OFFSET	Read/Write Word	39h	Used to null out any offsets in the output current sensing circuitry for Total Iout. 2's complement Exp = -3.
VOUT_OV_FAULT_LIMIT	Read/Write Word	40h	Sets the value of the output voltage measured at the sense or output pins that causes an output over-voltage fault. Format according to VOUT_MODE.
VOUT_OV_FAULT_RESPONSE	Read/Write Byte	41h	Instructs the device on what action to take in response to an output over-voltage fault. Only shutdown and ignore are supported.
VOUT_OV_WARN_LIMIT	Read/Write Word	42h	Sets the value of the output voltage, measured at the sense or output pins, that causes an output over-voltage warning. Format as determined by VOUT_MODE.
VOUT_UV_WARN_LIMIT	Read/Write Word	43h	Sets the value of the output voltage, measured at the sense or output pins, that causes an output voltage low warning. Format as determined by VOUT_MODE.
VOUT_UV_FAULT_LIMIT	Read/Write Word	44h	Sets the value of the output voltage, measured at the sense or output pins, that causes an output under-voltage fault. Format according to VOUT_MODE.
VOUT_UV_FAULT_RESPONSE	Read/Write Byte	45h	Instructs the device on what action to take in response to an output under-voltage fault. Only shutdown and ignore are supported.
IOUT_OC_FAULT_LIMIT	Read/Write Word	46h	Sets the value of the output current, in amperes, that causes the over-current detector to indicate an over-current fault condition. Exp = 1, 2.
IOUT_OC_FAULT_RESPONSE	Read/Write Byte	47h	Instructs the device on what action to take in response to an output over-current fault. Shutdown immediately no retry, and retry 1 to 6 times and continuously are supported.
IOUT_OC_LV_FAULT_LIMIT	Read/Write Word	48h	Sets the minimum operating output voltage when the response to an over-current condition is set to constant current mode. Format according to VOUT_MODE
IOUT_OC_LV_FAULT_RESPONSE	Read/Write Word	49h	Instructs the device on what action to take in response to an output over-current low voltage fault. Shutdown immediately no retry, and retry 1 to 6 times and continuously are supported.
IOUT_OC_WARN_LIMIT	Read/Write	4Ah	Sets the value of the output current that causes

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
	Word		an output over-current warning. Exp = 1, 2.
OT_FAULT_LIMIT	Read/Write Word	4Fh	Sets the temperature, in degrees Celsius, of the unit at which it should indicate an over-temperature fault. Exp = 0.
OT_FAULT_RESPONSE	Read/Write Byte	50h	Instructs the device on what action to take in response to an over-temperature fault. Shutdown, resume when fault is no longer present, and Ignore are supported.
OT_WARN_LIMIT	Read/Write Word	51h	Sets the temperature, in degrees Celsius, of the unit at which it should indicate an Over-temperature Warning alarm. Exp = 0.
VIN_OV_FAULT_LIMIT	Read/Write Word	55h	Sets the value of the input voltage that causes an input over-voltage fault. Exp = -4.
VIN_OV_FAULT_RESPONSE	Read/Write Byte	56h	Instructs the device on what action to take in response to an input over-voltage fault. Only shutdown and ignore are supported.
VIN_OV_WARN_LIMIT	Read/Write Byte	57h	Sets the value of the input voltage that causes an input over-voltage warning. Exp = -4.
VIN_UV_WARN_LIMIT	Read/Write Word	58h	Sets the value of the input voltage that causes an input voltage low warning. Exp = -4.
IIN_OC_FAULT_LIMIT	Read/Write Word	5Bh	Sets the value of the input current, in Amperes, that causes an Input Overcurrent Fault. Exp = -3.
IIN_OC_FAULT_RESPONSE	Read/Write Word	5Ch	Instructs the device on what action to take in response to an input over-current fault. Shutdown immediately no retry, and retry 1 to 6 times and continuously are supported.
IIN_OC_WARN_LIMIT	Read/Write Word	5Dh	Sets the value of the input current, in amperes, That causes a warning that the input current is high. Exp = -3.
POWER_GOOD_ON	Read/Write Word	5Eh	Sets the output voltage at which an optional POWER_GOOD signal should be asserted. Format according to VOUT_MODE.
POWER_GOOD_OFF	Read/Write Word	5Fh	Sets the output voltage at which an optional POWER_GOOD signal should be negated. Format according to VOUT_MODE.
TON_DELAY	Read/Write Word	60h	Sets the time, in milliseconds, from when a start condition is received (as programmed by the ON_OFF_CONFIG command) until the output voltage starts to rise. Exp = 0, -1, -2.
TON_RISE	Read/Write Word	61h	Sets the time, in milliseconds, from when the output starts to rise until the voltage has entered the regulation band. Exp = 0, -1, -2.
TON_MAX_FAULT_LIMIT	Read/Write	62h	Sets an upper limit, in milliseconds, on how long the unit can attempt to power up the output. Exp

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
	Word		= 0, -1, -2.
TON_MAX_FAULT_RESPONSE	Read/Write Byte	63h	Instructs the device on what action to take in response to a TON_MAX fault. Only shutdown and ignore are supported.
TOFF_DELAY	Read/Write Word	64h	Sets the time, in milliseconds, from when a stop condition is received (as programmed by the ON_OFF_CONFIG command) until the unit stops transferring energy to the output. Exp = 0, -1, -2.
TOFF_FALL	Read/Write Word	65h	Sets the time, in milliseconds, from the end of the turn-off delay time until the voltage is commanded to zero. Exp = 0, -1, -2.
TOFF_MAX_WARN_LIMIT	Read/Write Word	66h	Sets an upper limit, in milliseconds, on how long the unit can attempt to power down the output without reaching 12.5% of the output voltage programmed at the time the unit is turned off. Exp = 0, -1, -2.
POUT_OP_WARN_LIMIT	Read/Write Word	6Ah	Sets the value of the output power, in watts, that causes a warning that the output power is high. Exp = 0, 1, 2.
PIN_OP_WARN_LIMIT	Read/Write Word	6Bh	Sets the value of the input power, in watts, that causes a warning that the input power is high. Exp = 0, 1, 2.
STATUS_BYTE	Read/Write Byte	78h	Returns 1 byte where the bit meanings are: Bit <7> Busy Bit <6> Output off (due to fault or enable) Bit <5> Output over-voltage fault Bit <4> Output over-current fault Bit <3> Input Under-voltage fault Bit <2> Temperature fault Bit <1> Communication/Memory/Logic fault Bit <0>: None of the Above
STATUS_WORD	Read/Write Word	79h	Returns 2 bytes where the Low byte is the same as the STATUS_BYTE data. The High byte has bit meanings are: Bit <7> VOUT fault Bit <6> IOUT/POUT fault Bit <5> INPUT fault Bit <4> MFR_SPECIFIC Bit <3> POWER_GOOD# Bit <2> FANS Bit <1> OTHER Bit <0> UNKNOWN
STATUS_VOUT	Read/Write Byte	7Ah	Bit <7> Output Over-voltage Fault Bit <6> Output Over-voltage Warning

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
			Bit <5> Output Under-voltage Warning Bit <4> Output Under-voltage Fault Bit <3> VOUT_MAX_MIN Warning Bit <2> TON_MAX_FAULT Bit <1> TOFF_MAX_WARNING Bit <0> Reserved
STATUS_IOUT	Read/Write Byte	7Bh	Bit <7> Output Over-current Fault Bit <6> Output Over-current & Low Voltage Fault Bit <5> Output Over-current Warning Bit <4:1> Reserved Bit <0> Output Power Warning
STATUS_INPUT	Read/Write Byte	7Ch	Bit <7> Input Over-voltage Fault Bit <6> Input Over-voltage Warning Bit <5> Input Under-voltage Warning Bit <4> Reserved Bit <3> Unit Off for Insufficient Input Voltage Bit <2> Input Over-current Fault Bit <1> Input Over-current Warning Bit <0> Input Over-power Warning
STATUS_TEMPERATURE	Read/Write Byte	7Dh	Bit <7> Over-Temperature Fault Bit <6> Over-Temperature Warning Bit <5:0> Reserved
STATUS_CML	Read/Write Byte	7Eh	Returns 1 byte where the bit meanings are: Bit <7> Invalid or unsupported command Bit <6> Invalid or unsupported data Bit <5> Packet Error Check (PEC) fault Bit <4> Memory Fault Bit <3> Processor Fault Bit <2> Reserved Bit <1> Other communication fault not listed here Bit <0> Reserved
STATUS_OTHER	Read/Write Byte	7Fh	Returns 1 byte where the bit meanings are: Bit <7:1> Reserved Bit <0> First to assert SMBALERT#
STATUS_MFR_SPECIFIC	Read/Write Byte	80h	Returns 1 byte where the bit meanings are: <u>No power stage Fault ID</u> Bit <7:6> Reserved Bit <5> Common Fault (see C7h and C9h) Bit <4:2> Reserved Bit <1> TSEN High Fault Bit <0> Reserved

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
			<u>With power stage Fault ID</u> Bit <7> PS VDRV UV Fault Bit <6> Reserved Bit <5> Common Fault (see C7h and C9h) Bit <4> Reserved Bit <3> PS Over-temperature Fault Bit <2> PS High-side FET short Fault Bit <1> PS Over-current Fault Bit <0> PS Boot UV Fault
STATUS_MFR_FANS_1_2	Read/Write Byte	81h	Reserved
READ_EIN	Block Read Cnt = 6	86h	Returns 6 bytes consisting of Energy Count (2 bytes), Rollover Count (1 byte), and Sample Count (3 bytes)
READ_EOUT	Block Read Cnt = 6	87h	Reserved
READ_VIN	Read Word	88h	Returns the input voltage in Volts. Exp = -5
READ_IIN	Read Word	89h	Returns the input current in amperes. Loop1 Exp = -3. Loop2 Exp = -4
READ_VOUT	Read Word	8Bh	Returns the output voltage in the format set by VOUT_MODE
READ_IOUT	Read Word	8Ch	Returns the output current in amperes. Exp = 2, 1, 0, -1, -2
READ_TEMPERATURE_1	Read Word	8Dh	Returns sensor 1 temperature in degrees Celsius. Sensor input configured with MFR_SELECT_TEMPERATURE_SENSOR. Exp = 0.
READ_TEMPERATURE_2	Read Word	8Eh	Returns sensor 2 temperature in degrees Celsius. Sensor input configured with MFR_SELECT_TEMPERATURE_SENSOR. Exp = 0.
READ_DUTY_CYCLE	Read Word	94h	Returns the duty cycle of the PMBus device's main power converter in percent. Exp = -3
READ_POUT	Read Word	96h	Returns the output power in watts. Exp = -2, 0, 1, 2
READ_PIN	Read Word	97h	Returns the input power in watts. Loop1 Exp = 1, 0, or -1. Loop2 Exp 0 or -1
PMBus_REVISION	Read Byte	98h	Reports PMBus Part I, Part II & Part III Rev 1.4 (55h)
MFR_ID	Block Read/Write Cnt = 2	99h	The MFR_ID command is used to either set or read the manufacturer's ID (IF - ASCII 0x4649). MFR_ID is typically only set once, at the time of manufacture.
MFR_MODEL	Block Read/Write	9Ah	The MFR_MODEL command is used to either set or read the manufacturer's model number.

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
	Cnt = 2		MFR_MODEL is typically set once, at the time of manufacture.
MFR_REVISION	Block Read, Cnt = 2	9Bh	The MFR_REVISION command is used to either set or read the manufacturer's revision number. Each manufacturer uses the format of their choice for the revision number. MFR_REVISION is typically set at the time of manufacture or if the device is updated to a later revision.
MFR_DATE	Block Read/Write Cnt = 2	9Dh	The MFR_DATE command is used to either set or read the date the device was manufactured. While each manufacturer uses the format of their choice for the revision number, the recommended MFR_DATE format is YYMMDD where Y, M and D are integer values from 0 to 9, inclusive. MFR_DATE is typically only set once, at the time of manufacture
MFR_PIN_MAX	Read Word	A3h	Returns the maximum rated value, in watts, of the input power. Exp = 3
MFR_VOUT_MIN	Read Word	A4h	Returns the minimum rated value, in Volts, to which the output voltage may be set. Exp = -8
MFR_VOUT_MAX	Read Word	A5h	Returns the maximum rated value, in Volts, to which the output voltage may be set. Exp = -8
MFR_IOUT_MAX	Read Word	A6h	Returns the maximum rated value, in Amperes, to which the output may be loaded. Exp = 1
MFR_POUT_MAX	Read Word	A7h	Returns the maximum rated output power, in watts, that the unit is rated to supply. Exp = 3
MFR_TAMBIENT_MAX	Read Word	A8h	Returns the maximum rated ambient temperature, in degrees Celsius, in which the unit may be operated. Exp = 0
IC_DEVICE_ID	Block Read Cnt = 2	ADh	Returns a 1 byte code with the following values: 9Eh = XDPE1A2G5
IC_DEVICE_REV	Block Read Cnt = 2	AEh	The IC revision that is stored inside the IC
MFR_IOUT_MIN	Read Word	AFh	Returns the minimum rated value, in Amperes, to which the output may be loaded. Exp = 1
USER_DATA_00	Block Read/Write Cnt = 2	B0h	General Purpose USER Scratchpad
USER_DATA_01	Block Read/Write Cnt = 2	B1h	General Purpose USER Scratchpad
FW_CONFIG_REGULATION	Block Read/Write Cnt = 14	C5h	Read or Write 14 bytes for regulation controls: Bit <111:96> MFR_RDROOP_IDELTA_SEG3 Bit <95:80> MFR_RDROOP_ITHR_SEG2 Bit <79:72> FATR_PID_KD_UV

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COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
			Bit <71:64> FATR_PID_KP_UV Bit <63:48> FATR_PID_KD_UV Bit <47:32> MFR_RDROOP_RLL_SEG3 Bit <31:24> MFR_RDROOP_RLL_SEG2 Bit <23:16> CURRENT_LIMIT_AT_STARTUP Bit <15:12> Reserved Bit <11> EN_RAMP_SWITCHING_FREQ Bit <10> Reserved Bit <9> EN_IOUT_APC_TEMP_COMP Bit <8:6> Reserved Bit <5> EN_ILIM_STARTUP Bit <4> EN_C_ESTIMATE Bit <3> Reserved Bit <2:1> FSW_MAX_PERIOD Bit <0> Reserved
FW_CONFIG_TELEMETRY	Block Read/Write Cnt = 4	C6h	Read or Write 4 bytes for telemetry controls: Bit <31:24> Reserved Bit <23:20> READ_DUTY_EXP Bit <19:16> READ_IIN_EXP Bit <15:12> READ_IOUT_EXP Bit <11:8> READ_VIN_EXP Bit <7:5> Reserved Bit <4> DISABLE_AUTO_EXPONENT Bit <3> FW_FILTER_EN Bit <2:0> READ_POWER_EXP
MFR_COMMON_FAULT_STATUS1	Block Read Cnt = 4	C7h	Returns additional status information when the STATUS_MFR_SPECIFIC Bit 5 is set: Bit <31:24> Reserved Bit <23:8> FAULTED_PHASE Bit <7> TSEN_HI Fault Bit <6> P2CL, NCL Bit <5> FAST_VMODE_EVENT Bit <4> CURRENT_LIMITED Bit <3> Autophase Detect Short or VDRV UVLO Bit <2> Autophase Detect Open Bit <1> MIN_TMON_CHK_NOT_OK_AFTER_TON_DELAY Bit <0> VAUX_UV
FW_CONFIG_FAULTS	Read/Write Byte	C8h	Sets the Delay time unit for the fault retry response: Bit <7:6> VOUT_DELAY_UNIT Bit <5:4> VIN_DELAY_UNIT Bit <3:2> IOUT_DELAY_UNIT Bit <1:0> TEMP_DELAY_UNIT

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
MFR_COMMON_FAULT_STATUS2	Block Read Cnt = 4	C9h	Returns additional status information when the STATUS_MFR_SPECIFIC Bit 5 is set: Bit <31:29> Reserved Bit <28> OTHER_LOOP_FAULT Bit <27:5> Reserved Bit <4> VDD UVLO Bit <3> TSEN_PI_CAL_FAULT Bit <2> SLOT_ASSIGN_FAULT Bit <1> PWR_IN_ALERT Bit <0> Reserved
MFR_SETUP_PASSWORD	Block Read/Write Cnt = 4	CAh	Password setup command. This can only be modified when disable security once has the same matching setup password.
MFR_DISABLE_SECURITY_ONCE	Block Read/Write Cnt = 4	CBh	Password try location. If correct value is used the setup password command is unlocked. Command Masking is set in D2h and D3h. Returns the access state: 0 = unlocked, 1 = locked, 2 = deadlocked (requires reset to unlock)
MFR_SELECT_TEMPERATURE_SENSOR	Read/Write Word	CCh	This sets the temperature source used by Iout Temp Comp, Fault and Telemetry. Bit <5> Iout Temp Comp Select: 0 = READ_TEMPERATURE_2, 1 = READ_TEMPERATURE_1 Bit <4:3> Fault Source Select 0 = TSEN1, 1 = TSEN2, 2 = TSEN Internal Bit <2:0> READ Temperature 1 and 2 select: 0 = TSEN1, TSEN2 1 = TSEN1, TSEN Internal 2 = TSEN2, TSEN1 3 = TSEN2, TSEN Internal 4 = TSEN Internal, TSEN1 5 = TSEN Internal, TSEN2
MFR_GAMER	Block Read/Write Cnt = 4	CDh	Gamer Mode Commands
MFR_AHB_ADDRESS	Block Read/Write Cnt = 4	CEh	Used in conjunction with MFR_REG_WRITE and MFR_REG_READ PMBus commands as the address pointer, will auto increment after each write/read
MFR_DEBUG_BUFF	Block Read Cnt = 8	CFh	Provides internal access to the microprocessor Debug Buffer. Used to determine what type of processor faults are occurring.
MFR_SECURITY_BIT_MASK_LOW	Block Read/Write	D2h	Mask bits for Command Codes 0-127, a 1 in the bit position is protected, and a 0 in the bit

Application information

COMMAND	PMBus PROTOCOL	COMMAND CODE	DESCRIPTION
	Cnt = 16		position is unprotected.
MFR_SECURITY_BIT_MASK_HIGH	Block Read/Write Cnt = 16	D3h	Mask bits for Command Codes 128-255, a 1 in the bit position is protected, and a 0 in the bit position is unprotected.
MFR_REG_WRITE	Block Write Cnt = 4	DEh	Used in conjunction with address provided in MFR_AHB_ADDRESS for AHB write
MFR_REG_READ	Block Read Cnt = 4	DFh	Used in conjunction with address provided in MFR_AHB_ADDRESS for AHB read
MFR_PID	Block Read/Write Cnt = 31	F2h	PID settings registers for 3 states: soft start, steady state, high frequency load oscillation.
MFR_COUT_CONFIG	Read/Write Word	F4h	Reserved
MFR_READ_COUT	Read Byte	F5h	Reserved
MFR_FIRMWARE_COMMAND_DATA	Block Read/Write Cnt = 4	FDh	MFR_FIRMWARE_COMMAND protocol always involves a block write to MFR_FIRMWARE_COMMAND_DATA (0xFD) then a byte write to MFR_FIRMWARE_COMMAND (0xFE) then a block read of MFR_FIRMWARE_COMMAND_DATA (0xFE).
MFR_FIRMWARE_COMMAND	Write Byte	FEh	Used in conjunction with MFR_FIRMWARE_COMMAND_DATA when required

8.16.2 11-bit Linear Format

Monitored parameters use the Linear Data Format (Figure 34) encoding into 1 Word (2 bytes), where:

$$Value = Y \times 2^N$$

Note N and Y are “signed” values. All commands, other than those using the format described by VOUT_MODE, use this format unless otherwise noted.

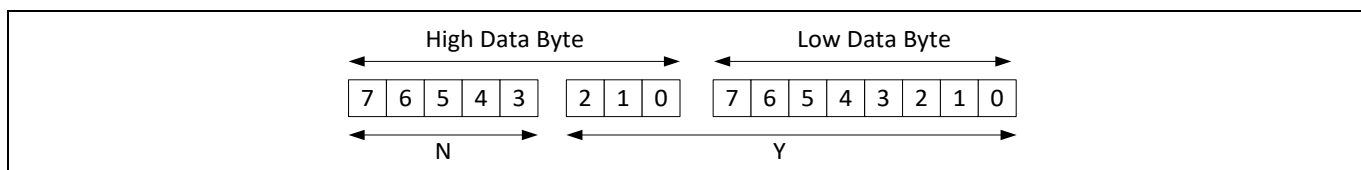


Figure 34 11-bit Data format

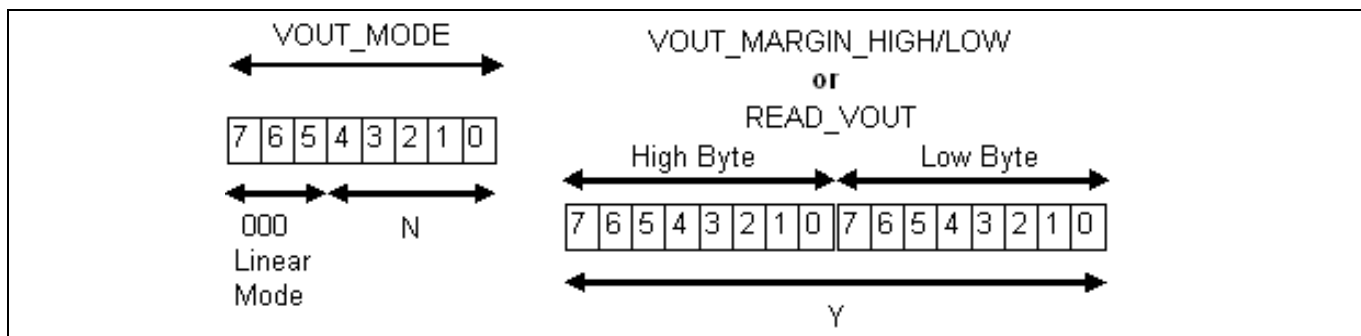
8.16.3 16-bit Linear Format

This format is only used for VOUT related commands (i.e. READ_VOUT, VOUT_COMMAND, VOUT_MARGIN_HIGH, VOUT_MARGIN_LOW, POWER_GOOD_ON, and POWER_GOOD_OFF):

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$$Value = Y \times 2^N$$

Note N and Y are “signed” values. If VOUT is set to linear format (by VOUT_MODE), then N is set by the VOUT_MODE command and only Y is returned in the data-field as a 16-bit unsigned number.



16-bit Data format

8.17 Configuration management

The XDPE1A2G5 provides non-volatile memory (NVM) for storage of most of the programmable registers in the device. This includes all trim and configuration registers, and patch space for firmware updates. Thus, the controller can startup and operate autonomously after power-on reset of the device, without user or system intervention.

The memory macro is a one-time programmable (OTP) device that consists of 64 Kbytes of memory space. The memory is split into 5 partitions as shown in Figure 35. Partition 0 stores trim and user configuration space. Partition 1 stores the patch errata that is used to update the firmware stored in ROM. Partition 2 can be used to store a customer specific patch. Partition 4 stores up to ten black box data packets. Partitioning and Patch Loading tools are available in the Infineon XDP™ Designer GUI and Gang Programmer Software.

Partitioning		
Partition0	32 Kbytes	- Trim - Config User - PMBus User - AVS User
Partition1	26 Kbytes	Patch Errata
Partition2	4 Kbytes	Custom Patch
Partition3	0 Kbytes	
Partition4	2 Kbytes	Black Box Data

Figure 35 OTP partitioning

The memory macro is inherently a one-time programmable device. However, multiple-time programming is provided by dividing the macro into multiple “Base” or “Partial” configurations. Table 38 shows the amount of memory space required for the first base configuration plus trim, and also the size of all subsequent base config

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updates. If memory was programmed with all 3 User sections every time an update was made, the number of re-writes would be limited to 22.

Table 38 Size of first base configuration plus trim and maximum re-writes (Note 19)

Section #	Section Type	Size (bytes)
1	Trim (one-time write)	340
2	Config User	748
3	PMBus User	576
4	AVS User	48
Memory size for first write (Section 1+2+3+4)		1,712
Memory size for all subsequent based configs (Section 2+3+4)		1,372
Number re-writes to memory (Base Configs Only)		22 times

Note:

19. These sizes are based on a typical configuration. Use the OTP tools provided in the XDP™ Designer GUI to validate sizes for each design.

Memory space can be conserved and thus allow for more re-writes, if updates are limited to 1 User Section, or if “Partial” configurations are used. A Partial configuration can be used for each User Section to only store the differences from the base configuration. The Partial storage size is approximately 10 to 20 bytes per difference depending on which Partial section is being stored (Config = 10 bytes per, PMBus = 20 bytes per). This would allow for 1000's of re-writes after the initial programming of the base configuration plus trim.

The memory space can be configured to store multiple configurations that are selectable by the external resistor connected to the XADDR pin. This capability is referred to as “multi-config”. This allows customers to order one part number that can support multiple types of rails & applications. The XDPE1A2G5 controller is capable of storing up to 16 configurations. Table 39 shows an example of 4 multi-configs programmed into the device; one example using all Base Configs and one using Partial Configs. Both types of programming are valid and programming files can be configured using the OTP Tools provided in the XDP™ Designer GUI.

Table 39 4x Multi-config – Base Config versus Partial

R _{addr}	Xvalent	Config (Note 20)	Size (Bytes)	Config (Note 21)	Size (Bytes)
49.9 kΩ	XV0	Base0 + Trim	1,712	Base0 + Trim	1,712
38.3 kΩ	XV1	Base1	1,372	Partial Config1	80
22.6 kΩ	XV2	Base2	1,372	Partial Config2	80
29.4 kΩ	XV3	Base3	1,372	Partial Config3	80
Total Size:			5,828	Total Size:	1,952
Size Remaining:			26,940	Size Remaining:	30,816

Note:

20. Each Base Config consists of 2x User Sections (Config, and PMBus)

21. Each Partial Config has 8 register differences totaling 80 bytes.

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For additional and more in depth information on configuration management, refer to application note AN001-XDPE1x2xx.

8.18 Security

8.18.1 Write Protect

The WRITE_PROTECT (10h) command is used to control writing to the PMBus device. The intent of this command is to provide protection against accidental changes. The command settings of 0x20, 0x40, and 0x80 are not intended to provide protection against deliberate or malicious changes to a device's configuration or operation. The command setting of 0x01 will provide an additional level of protection against deliberate or malicious changes because once set it cannot be changed until device reset. Available settings are shown in Table 40 below. STATUS_CML (7Eh) [6] "Invalid/Unsupported Data" is asserted when a protected command is written to.

Table 40 WRITE_PROTECT (10h)

Value	Description
0x00	No Write Protection
0x01	Write Protect All. Will lock once written until Reset
0x20	Write Protection except for WRITE_PROTECT, OPERATION, PAGE, ON_OFF_CONFIG, and VOUT_COMMAND commands
0x40	Write Protection except for WRITE_PROTECT, OPERATION, and PAGE commands
0x80	Write Protection except for WRITE_PROTECT

Note: The WRITE_PROTECT command is not valid if password protection is used. Read PMBUS command 0xCB. If 0xCB = 1, password protection is enabled

8.18.2 Password Protect

The XDPE controller provides a PMBus password protection feature, which prevents selected PMBus commands from being accidentally modified by unauthorized users. A password can be set up with the MFR_SETUP_PASSWORD (CAh) to protect PMBus commands selected in the MFR_SECURITY_BIT_MASK_LOW (D2h) and MFR_SECURITY_BIT_MASK_HIGH (D3h) commands. The length of the password is 4-bytes. A read of the MFR_SETUP_PASSWORD will return a hashed value of the set password. The password protection will remain disabled as long as there is no write to the MFR_SETUP_PASSWORD command and storage to OTP. The MFR_DISABLE_SECURITY_ONCE (CBh) command is used to disable the protection once the protection is enabled with a write to the MFR_SETUP_PASSWORD command. Entering the correct password at this command will unlock the protection. Reading MFR_DISABLE_SECURITY_ONCE command will provide the status of the security: 0 - Security Unlocked, 1 - Security Locked, 2 - Security Deadlocked (incorrect password enter 4x, reset to try again). STATUS_CML (7Eh) [6] "Invalid/Unsupported Data" is asserted when a protected command is written to.

Table 41 PMBus password protection commands

Command Code	Command Name	Description
CAh	MFR_SETUP_PASSWORD	A Write to this command sets up the password protection. A Read of the command returns a Hash

Application information

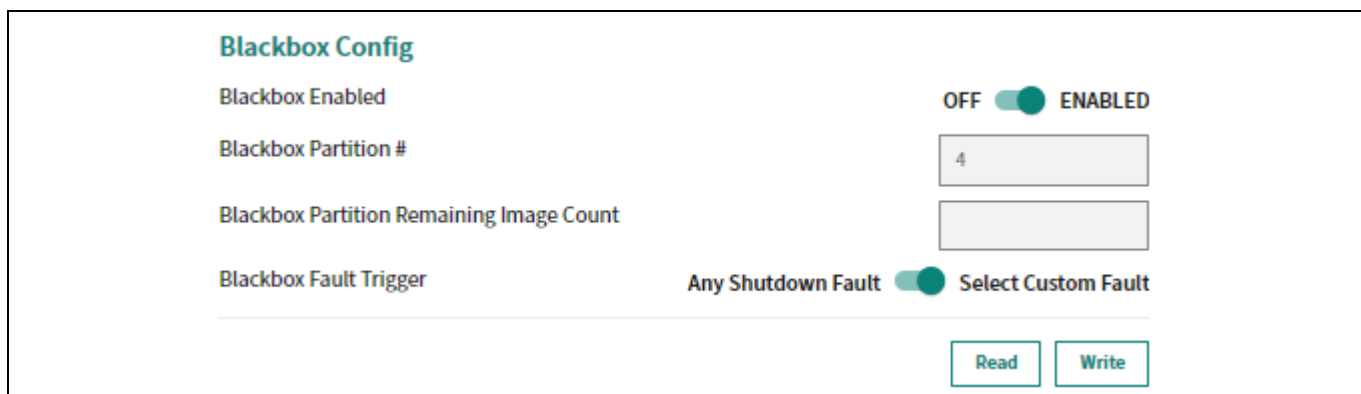
Command Code	Command Name	Description
		Code of the set password.
CBh	MFR_DISABLE_SECURITY_ONCE	Writing the correct password to this command unlocks the password protection. A Read of this command provides password protection status. 0 - no protection 1 - protection enabled 2 - too many incorrect passwords entered (4x). Must reset to try again.
D2h	MFR_SECURITY_BIT_MASK_LOW	Selects command codes 0x00 to 0x7F to be protected.
D3h	MFR_SECURITY_BIT_MASK_HIGH	Selects command codes 0x80 to 0xFF to be protected.

8.19 Blackbox feature

The XDPE1A2G5C allows the user to save key telemetry data and fault status registers during a fault off shutdown event.

8.19.1 User configuration

The user needs to enable the blackbox feature using the Infineon XDP™ Designer GUI. The blackbox feature is disabled in the default configuration file programmed at the factory



Blackbox Config

Blackbox Enabled OFF ☒ ON

Blackbox Partition #

Blackbox Partition Remaining Image Count

Blackbox Fault Trigger Any Shutdown Fault ☒ Select Custom Fault

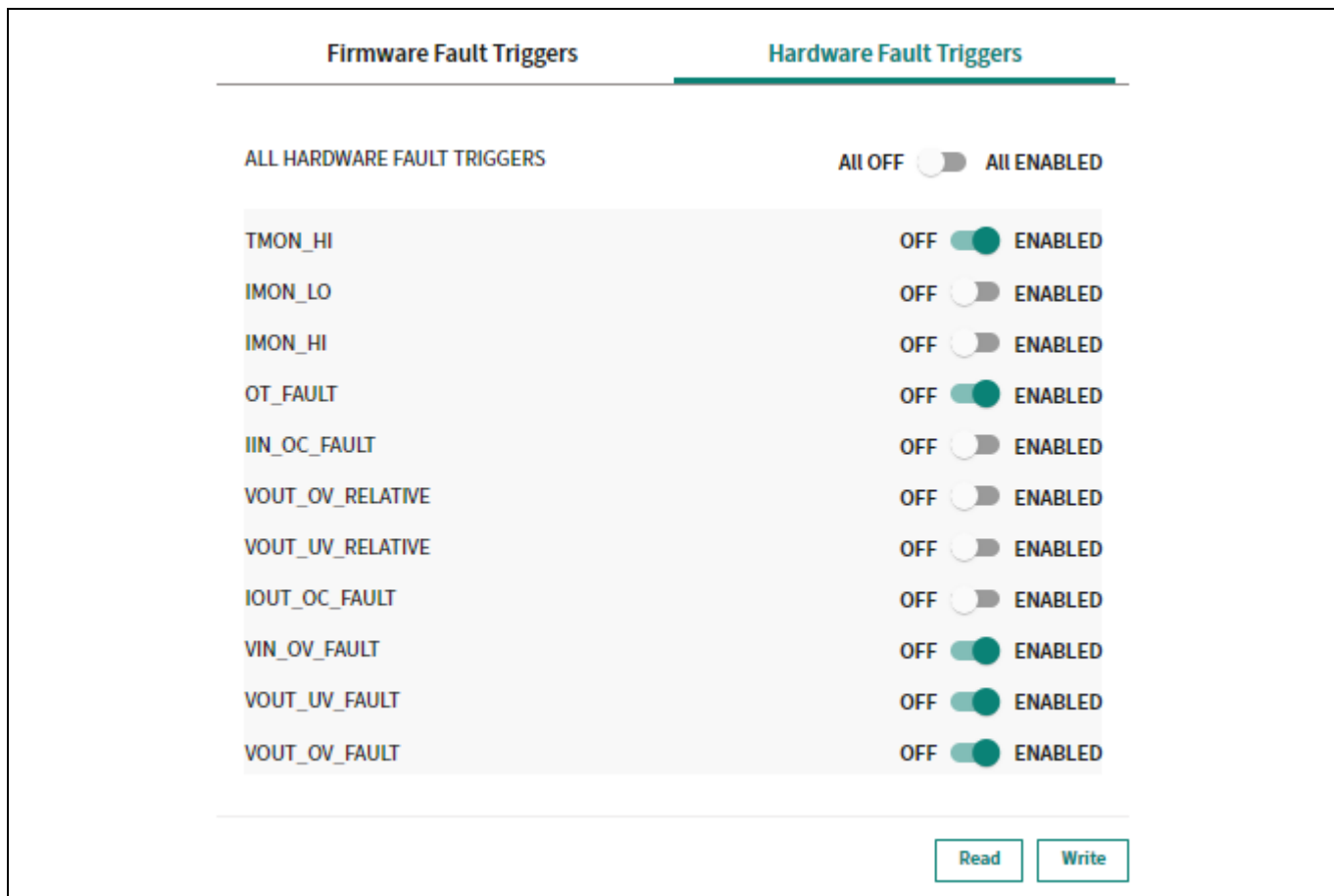
Figure 36 Blackbox enable

The user needs to set the faults that will cause the blackbox data to be written to OTP. The following faults can be set to force an OTP upload;

- PWM Short
- TON_MAX_FAULT (PMBUS)
- Power stage fault
- Over temperature fault
- Input over current

Application information

- Relative output over voltage
- Relative output under voltage
- Output over current
- Input over voltage
- Fixed output under voltage
- Fixed output over voltage



Firmware Fault Triggers	Hardware Fault Triggers
ALL HARDWARE FAULT TRIGGERS	
All OFF ☐ All ENABLED ☒	
TMON_HI	OFF ☒ ENABLED
IMON_LO	OFF ☐ ENABLED
IMON_HI	OFF ☐ ENABLED
OT_FAULT	OFF ☒ ENABLED
IIN_OC_FAULT	OFF ☐ ENABLED
VOUT_OV_RELATIVE	OFF ☐ ENABLED
VOUT_UV_RELATIVE	OFF ☐ ENABLED
IOUT_OC_FAULT	OFF ☐ ENABLED
VIN_OV_FAULT	OFF ☒ ENABLED
VOUT_UV_FAULT	OFF ☒ ENABLED
VOUT_OV_FAULT	OFF ☒ ENABLED

Read Write

Figure 37 Configuration settings to trigger a blackbox OTP upload event

These blackbox configuration bits are all part of the config file that gets stored in OTP

8.19.2 Stored blackbox data

The blackbox data is stored in 200 byte increments. There is enough storage to allow for the data from 10 shutdown events to be stored in OTP. The user can look at the data from any individual event by using the Infineon XDP™ Designer GUI. The Infineon XDP™ Designer GUI also allows the user to store the OTP contents to a file and view the contents of a stored file. The blackbox function requires that the 3.3V VCC voltage be available for at least 20ms after a fault to store the data to OTP

RAM	OTP Online	Decode Blackbox File
Blackbox Image Size:		0xC8 (200 bytes)
Selected OTP Blackbox Image Address:		0x1002F800
OTP Valid Image Count:		3
OTP Offset Store:		0x258 (600 bytes)
OTP Remaining Image Slots:		7
Select OTP Blackbox Image:		2 ▼ 0 1 2

Refresh OTP Data

Manually Trigg

Figure 38 OTP blackbox image selection drop down

The blackbox stores;

- Periodic telemetry (updated every 1ms)
- The fault that cause the OTP upload
- Telemetry immediately after the fault
- Timestamp that indicates the length of time from 3.3V POR to when the fault occurred.

The timestamp range is 0 to 20 years in 1ms increments.

Blackbox Image Decoded Data Display - Decoded File Valid Image #0

Display selection: All Blackbox Image Sections Advanced Mode

Name	Size	Address	Hex	Value	Unit
Snapshot Header					
OTP file system command type	2	0x1002F990	0x0014	20	
Option	2	0x1002F992	0x0000	0	
Size	2	0x1002F994	0x00C8	200	
Size Redundant	2	0x1002F996	0x00C8	200	
Header CRC	4	0x1002F998	0x2D097B17	755596055	
LoopA (telemetry data after fault event) (high priority)					
Duty Cycle	2	0x1002F99C	0x0000	0	%
Iout	2	0x1002F99E	0xFF84	-7.75	A
Vout	2	0x1002F9A0	0x131F	0.5975	V
Iin	2	0x1002F9A2	0x7FFD	-0.375	A
LoopB (telemetry data after fault event) (high priority)					
Duty Cycle	2	0x1002F9A4	0x0117	6.8115	%
Iout	2	0x1002F9A6	0xFFF7	-0.5625	A
Vout	2	0x1002F9A8	0x198F	0.7987	V
Iin	2	0x1002F9AA	0x0000	0	A

Figure 39 Sample of decoded blackbox OTP data

8.20 TLVR secondary open and short detection

The controller has a detection mechanism to determine if the output inductor is open or short during the manufacturing process and before systems are shipped to the field. Based upon the measured output voltage and phase current during the detection process, it can be determined if the output inductors are good or bad. More precisely, it can be determined if there is an open or short, and also determine which phase has been shorted.

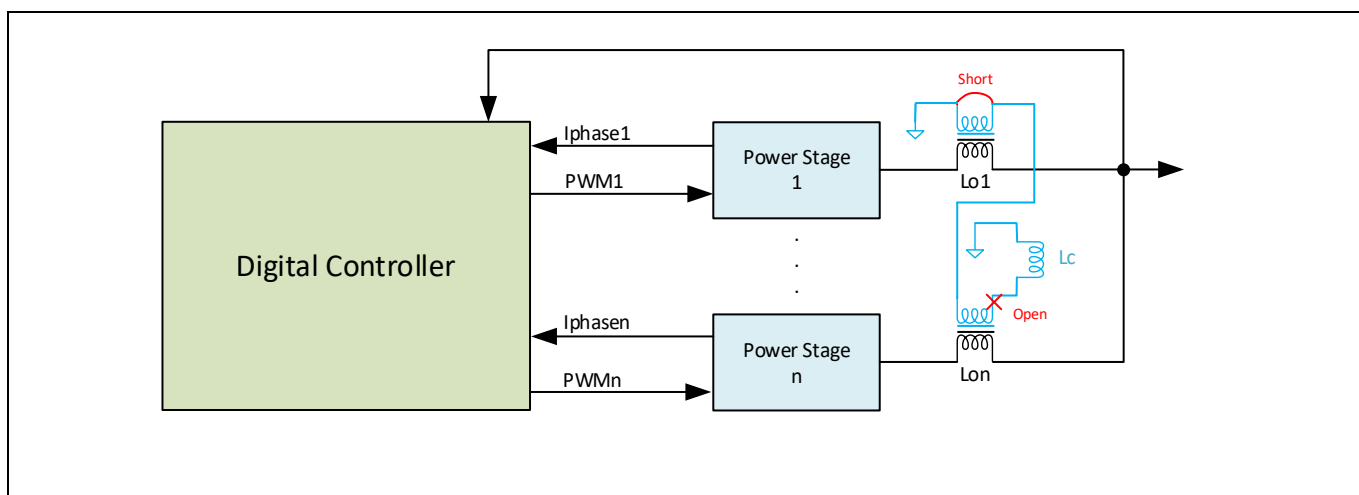


Figure 40 Shorted or open secondary on TLVR inductors

Application information

The TLVR Open and Short Commands are only allowed when the controller is in the inactive state (VREN is de-asserted). The user will setup the Open and Short commands based upon the design and the expected “Good” Vout and Iout thresholds. The setup parameters will be written to the MFR_FIRMWARE_COMMAND_DATA before the Open or Short command is executed with the MFR_FIRMWARE_COMMAND. Required parameters for each command are shown in Table 42 and Table 43.

Table 42 Open Command – MFR_FIRMWARE_COMMAND = 0x33/0x50 for bank 0/1

Setup Parameters	Bit Location	Description
Pulse Count	Bit [3:0]	Sets the number of pulses fired during the test. 0 to 15 pulses
Pulse Width	Bit [7:4]	Sets the pulse width of the fired pulse. 50 to 275 ns, 15 ns resolution
Vout Threshold	Bit [15:8]	Sets the output voltage threshold level. 0 to 2 V, 5 mV resolution
Iout Threshold	Bit [23:9]	Sets the output current threshold level. 0 to 510 A, 2 A resolution
PWM Firing Map	Bit [27:24]	Sets the PWM firing location. 0 = PWM1, 1 = PWM2... 15 = PWM16
Reserved	Bit [31:28]	

Table 43 Short Command – MFR_FIRMWARE_COMMAND = 0x34/0x51 for bank 0/1

Setup Parameters	Bit Location	Description
Pulse Count	Bit [3:0]	Sets the number of pulses fired during the test. 0 to 15 pulses
Pulse Width	Bit [7:4]	Sets the pulse width of the fired pulse. 50 to 275 ns, 15 ns resolution
Vout Threshold	Bit [15:8]	Sets the output voltage threshold level. 0 to 2 V, 5 mV resolution
Iout Threshold	Bit [23:16]	Sets the output current threshold level. 0 to 510 A, 2 A resolution
PWM Firing Map	Bit [27:24]	Firing Map is ignored and automatically one phase at one time based on total phase map at the rate of about 20KHz
Auto Complete	Bit [28]	Set to 1 to ignore set PWM firing location and auto detect each phase in the phase map

Following the execution of each command, the pass/fail, and measured Vout and Iout results can be returned by performing a read on the MFR_FIRMWARE_COMMAND_DATA. Configuration of the returned data is shown in Table 44 below.

Table 44 TLVR Open, short and missing decoding

TLVR Open test results decoding			
Bit location	FDh returned value	Bit count	Definition
Bit [0]	Pass/ fail result	1	0: Pass 1: Fail Note: If Vout and Iout are greater than Vout and Iout threshold, then FDh returns 0 (Pass); otherwise returns 1(Fail)
Bit [15:1]	Iout measurement	15	Reports peak total Iout Range 0 ~ 2048A with LSB= 1/8 (125 mA)
Bit [26:16]	Delta Vout measurement	10	Range 0 ~ 8V with LSB = 1/128V (8 mV)
Bit [31:27]	Not used	15	
	Total bit count	32	
TLVR short/ missing test results decoding			
Bit location	FDh returned value	Bit count	Definition
Bit [0]	Pass/ fail result	1	0: Pass

Application information

TLVR Open test results decoding

			1: Fail Note: Short: If Vout and Iout are greater than Vout and Iout threshold, then FDh returns 1 (Fail); otherwise returns 0(Pass) Missing phase: If Vout and Iout are both smaller than Vout and Iout threshold, then FDh returns 1 (Fail); otherwise returns 0(Pass)
Bit [15:1]	Iph measurement	15	Short: Reports the highest phase peak current Missing: Reports the lowest phase peak current Range 0~2048A with LSB = 1/8(125mA)
Bit [31:16]	Shorted phase location	16	Shorted/missing phase location if auto detect is enabled 0~15: Phase 1~phase 16 is short (note: same definition as phase map. The associated bit will be set for faulted phase)
	Total bit count	32	

9 Physical characteristics (56-lead QFN package, 7 mm x 7 mm)

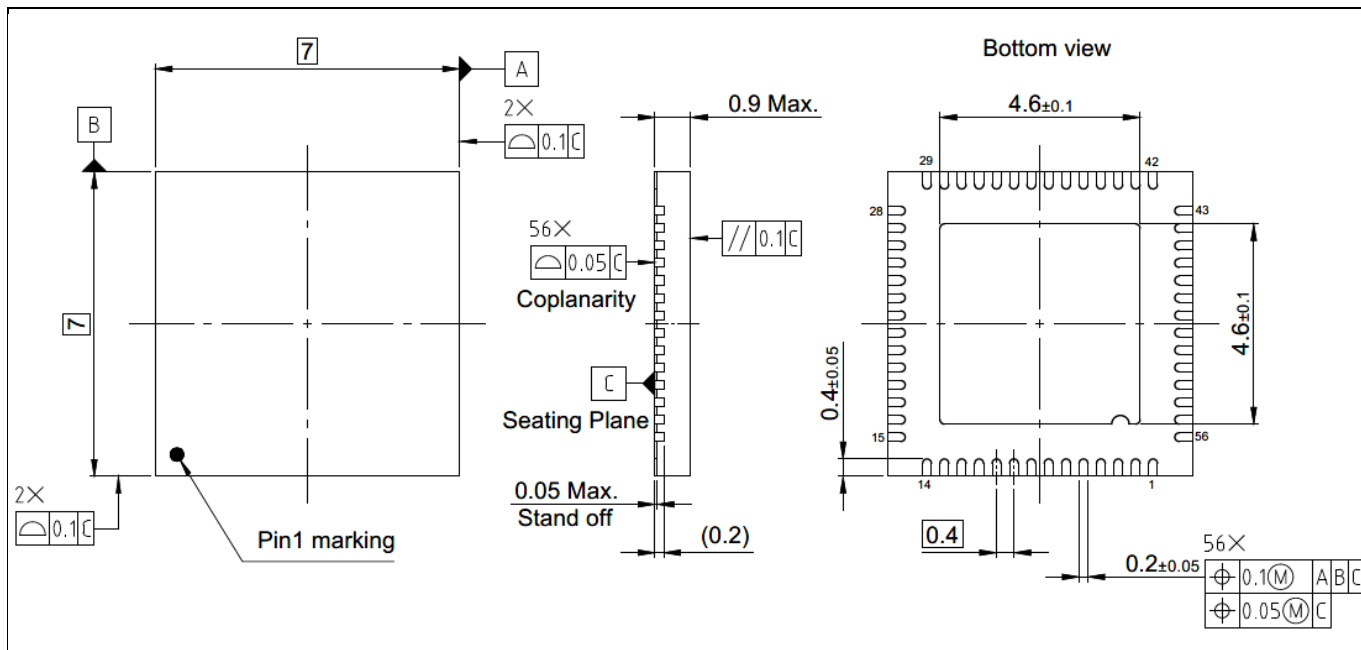


Figure 41 Package

10 Package layout guidelines

The XDPE1A2G5C is built in a 56-lead QFN package which uses a solid ring frame. The thermal pad design on the board should be based on the exposed paddle area as shown below.

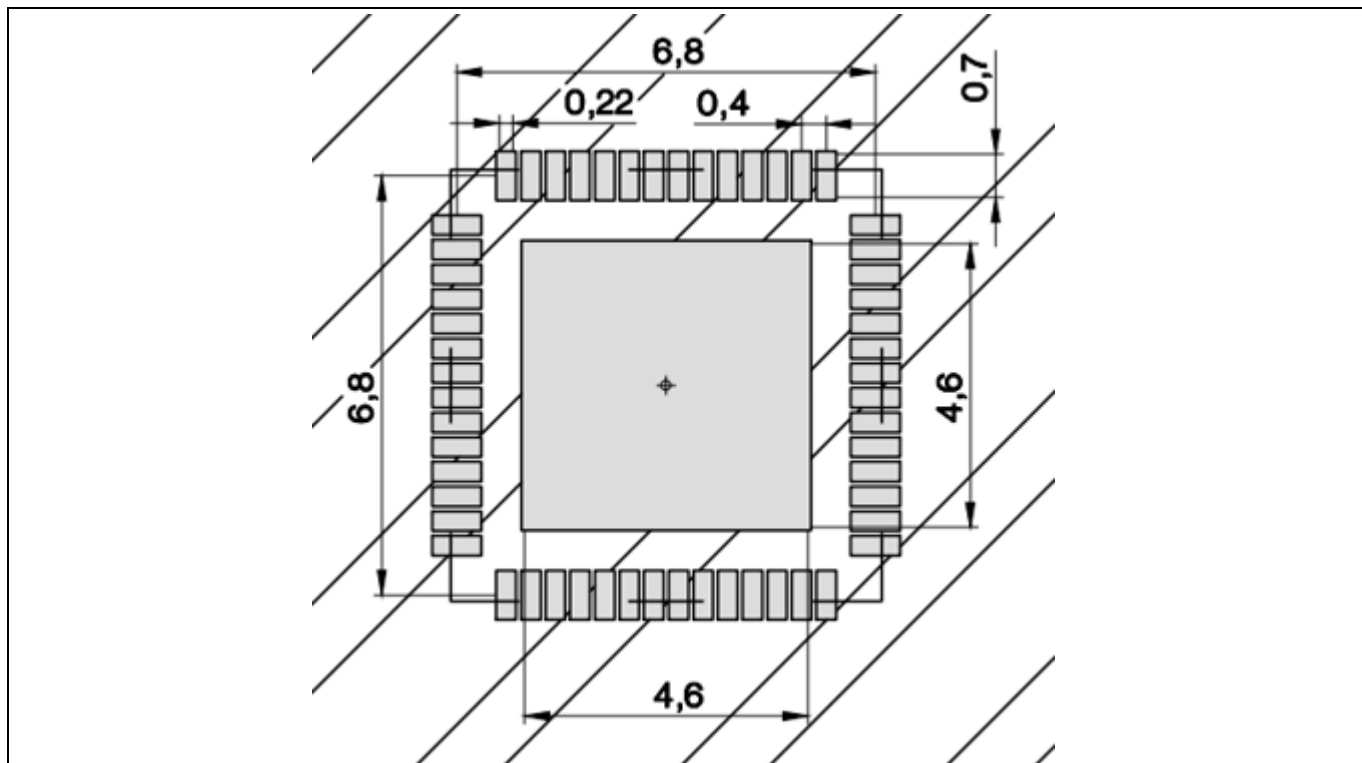


Figure 42 PCB PAD Dimensions

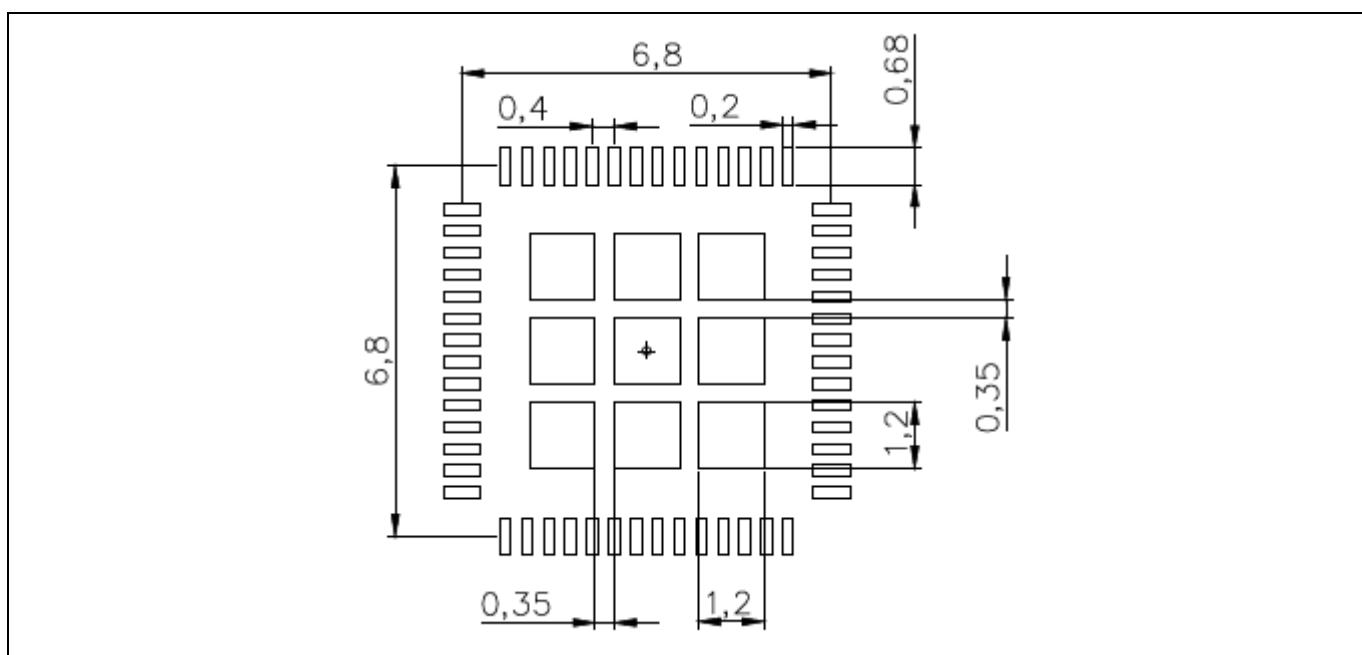


Figure 43 Paste Mask Dimensions

11 Layout considerations

In order to optimize voltage regulator performance, it is important to minimize the effects of printed circuit board (PCB) parasitic impedances on the digital controller and power stages. The following layout techniques highlight important practices which should be incorporated during design to optimize the heat-dissipating capabilities of the PCB. For additional layout guidelines and component placement tips, refer to AN002-XDPE122xx.

11.1 Component placement

Within the allotted implementation area, orient the switching components first. The switching components are the most critical because they carry large amounts of energy and tend to generate high levels of noise. The power dissipation of those components should also be taken into consideration during PCB placement. Align the output inductors and power stages such that space between the components is minimized. It is recommended that the same placement is duplicated for all phases.

The XDPE1A2G5C controller can be placed off to one side or centered relative to the individual phase switching components. Critical small signal components including the ISEN resistors, TSEN capacitors and voltage feedback RC filters should be placed near the controller.

11.2 Routing of AVS signals

AVS signal lines are very sensitive to noise sources and should be kept away from switching components. AVS signals need to be ground referenced and routed away from all noise sources such as power stages and switch nodes.

12 Package marking

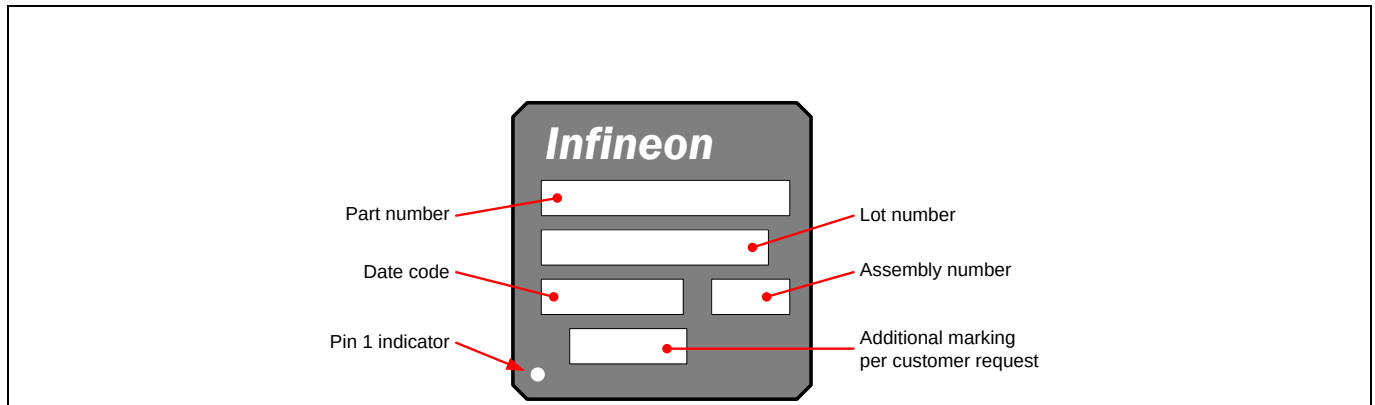


Figure 44 Package marking

13 Environmental qualifications

Table 45

Qualification Level		Industrial	
Moisture Sensitivity		QFN Package	MSL3
ESD	Human Body Model	JS-001, Class 2	
	Charged Device Model	JS-002, Class C3	
	Latch-up	JESD78, Class 2	
RoHS Compliant		Yes	

Table 46 History

rev	date	Description
1.0	02/25/2026	Page 15, Add 333 Ω accuracy to EC table Page 24, section 8.3, add tables 9 & 10 for phase current max/min Page 24, add section 8.3.1 to show support for currents above 2048 A Page 32, add table of MFR specific AVS commands Page 33, add tables for AVS telemetry resolution Page 39, update IOUT max in Table 28 Page 50, section 8.13, update load line LSB description Page 51, update table 35 pin 51 description from VAUX to GPIO Page 52, add +1 stride values to table 36 Page 54, update PMBUS 1.3.1 to 1.4 Page 62, update command 0x98h to 0x55 (rev 1.4) Page 67, update section 8.17 with new size values Page 70, add section 8.19, Black Box feature Page 73, add section 8.20, TLVR fault detection